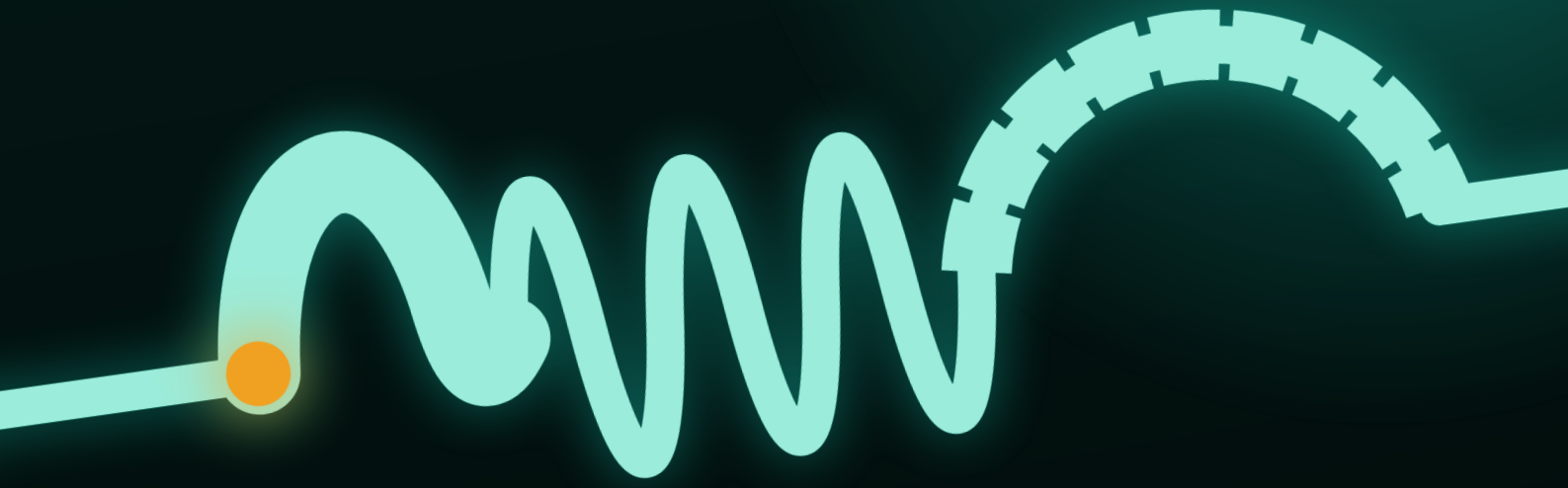
Clinical *Conditions*

REVIEW ARTICLES
LAST UPDATED 11 NOVEMBER 2026

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Esophagus
Stomach
Liver
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FROM THE ESOPHAGUS TO THE APPENDIX

Gastrointestinal

Follow food from the mouth to the rectum, with the liver, biliary tree, pancreas, and appendix as side branches.

For educational use only. This is a study aid, not a clinical reference.

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Embryology of the gut and its defects: separation, rotation, migration, and closure

The gut tube forms in week 3 from endoderm, which lines it and buds the liver, gallbladder, and pancreas, wrapped in mesoderm, which forms its wall and muscle, and colonized by neural crest, which forms the ENS. The foregut, midgut, and hindgut take their blood from the celiac artery, SMA, and IMA. The midgut herniates, rotates 270° around the SMA, and returns by week 10. Most congenital defects follow one failed step: separation of trachea and esophagus, recanalization of the duodenum, rotation of the midgut or ventral pancreas, obliteration of the vitelline duct, migration of neural crest, or closure of the abdominal wall. Here, we review these steps and their defects.

Key points

- Endoderm forms the epithelium, mesoderm the wall and muscle, and neural crest the ENS.
- The midgut rotates 270° counterclockwise around the SMA while herniated in weeks 6–10.
- Duodenal atresia follows failed recanalization, shows a double bubble, and occurs with Down syndrome in 30–40%.
- Pyloric stenosis causes projectile, nonbilious vomiting at 3–6 weeks with hypochloremic, hypokalemic metabolic alkalosis.
- Gastroschisis is uncovered and beside the umbilicus; omphalocele is covered by a sac and linked to trisomies.

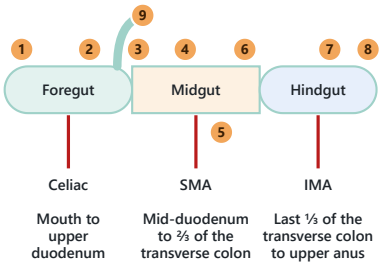


Fig. 1 | The gut tube in three regions: the foregut, from the mouth to the upper duodenum, supplied by the celiac artery; the midgut, from the mid-duodenum to the proximal two-thirds of the transverse colon, supplied by the SMA; and the hindgut, from the distal third of the transverse colon to the upper anus, supplied by the IMA. Numbered defects: 1, esophageal atresia and TEF; 2, pyloric stenosis; 3, duodenal atresia and annular pancreas; 4, malrotation and volvulus; 5, Meckel diverticulum, at the vitelline duct; 6, jejunal and ileal atresia; 7, Hirschsprung disease; 8, anorectal malformations; and 9, biliary atresia. 1, Esophageal atresia and TEF; 2, Pyloric stenosis; 3, Duodenal atresia and annular pancreas; 4, Malrotation and volvulus; 5, Meckel diverticulum; 6, Jejunal and ileal atresia; 7, Hirschsprung disease; 8, Anorectal malformations; 9, Biliary atresia.

Germ layers and regions

Endoderm forms the epithelial lining of the gut, liver, gallbladder, and pancreas, mesoderm forms the wall, connective tissue, and smooth muscle, and neural crest from the ectoderm forms the enteric neurons. The foregut runs from the mouth to the upper duodenum and is supplied by the celiac artery, the midgut from the mid-duodenum to the proximal two-thirds of the transverse colon by the SMA, and the hindgut from the distal third of the transverse colon to the upper anus by the IMA. The gut tube forms in week 3, the buccopharyngeal membrane resorbs in week 4, the vitelline duct closes in week 7, the cloacal membrane opens and villi begin in week 9, muscle layers appear by week 11 and crypts by week 12, and absorption begins by week 24 and reaches adult levels by week 32. Organs form in weeks 3–8, when teratogens matter most.

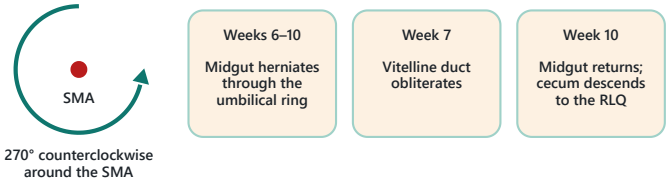


Fig. 2 | Midgut rotation: between weeks 6 and 10 the midgut herniates through the umbilical ring, rotates 270 degrees counterclockwise around the SMA, and returns at week 10, when the cecum descends to the right lower quadrant; the vitelline duct obliterates at week 7.

Rotation and the pancreas

Between weeks 6 and 10, the midgut herniates through the umbilical ring, rotates 270° counterclockwise around the SMA, and returns, after which the cecum descends to the right lower quadrant. The ventral pancreatic bud forms the lower head and uncinate process and its duct becomes the main duct of Wirsung, while the dorsal bud forms the body and tail and its duct becomes the accessory duct of Santorini; the buds rotate and fuse in

weeks 4–8. A fetus that cannot swallow amniotic fluid past an obstruction develops polyhydramnios, an amniotic fluid index over 25 cm, and GI obstruction can be diagnosed by ultrasound in the late second to early third trimester.

When it fails

Failed lateral septation of the foregut causes esophageal atresia and TEF, most often type C, a blind upper pouch with a distal fistula, in 84%, with choking and respiratory distress, polyhydramnios, a nasogastric tube that will not pass, and cardiac defects. Pyloric stenosis, acquired after birth, causes projectile, nonbilious vomiting at 3–6 weeks, an olive in 60–80%, and hypochloremic, hypokalemic metabolic alkalosis, with a wall of 3 mm or more on ultrasound, and is treated by pyloromyotomy. Duodenal atresia follows failed recanalization below the ampulla, causes vomiting within 24–38 hours of the first feed and a double bubble, and occurs with Down syndrome in 30–40%. Annular pancreas, a ventral bud that fails to rotate and encircles the duodenum, causes 30% of congenital duodenal obstruction, and pancreas divisum, in about 10%, is the most common pancreatic malformation. Biliary atresia causes jaundice beyond 2 weeks, acholic stools, dark urine, and hepatomegaly, and needs a Kasai portoenterostomy before 60 days, though many need a transplant.

Malrotation leaves fibrous bands and a mesentery that can twist into volvulus, and bilious vomiting, usually the first sign, needs urgent evaluation. Incomplete obliteration of the vitelline duct leaves a Meckel diverticulum, with painless bleeding from ectopic gastric mucosa and the rule of 2s, while a fully patent duct drains meconium from the umbilicus. Jejunal and ileal atresia follow mesenteric vessel disruption, meconium ileus calls for testing for CF, Hirschsprung disease follows failed neural crest migration, affecting the rectosigmoid in about 80%, and anorectal malformations affect 1 in 5000 births, about 70% of affected boys having a recto-urethral fistula. Gastroschisis is an uncovered defect beside, usually right of, the umbilicus in young mothers, with survival over 90%, whereas omphalocele herniates through the cord base inside a peritoneal sac and is linked with Beckwith-Wiedemann syndrome and trisomies 13, 18, and 21.

Relationships to other topics

Enteric nervous system. Enteric neurons come from neural crest, so failed migration leaves the aganglionic colon of Hirschsprung disease.

Gastric secretion. Projectile vomiting in pyloric stenosis loses HCl, causing hypochloremic metabolic alkalosis.

Gastric secretion: neural, endocrine, and local control of acid and its protection

Food triggers gastric secretion in three overlapping phases, which start it, sustain it, and then slow it down. The cephalic phase contributes about 30% of HCl, the gastric phase about 60%, and the intestinal phase about 10%. Secretion depends on seven gastric cell types and on signals that converge on the parietal cell, while a two-layer mucosal barrier protects the mucosa and D cells apply negative feedback. Here, we review this control, how it fails, and how drugs suppress acid or protect the mucosa, and consider how it relates to acid-related conditions.

Key points

- Higher-order neurons drive the DMV, and efferent CN X stimulates the stomach; thoracolumbar sympathetic fibers inhibit it.
- Gastrin promotes ECL, parietal, and chief cells, and histamine from ECL cells promotes parietal cells.
- Excess protons drive D cells, whose somatostatin inhibits G, ECL, parietal, and chief cells.
- Low pH is the brake on gastrin, so losing HCl raises gastrin.
- The intestinal phase mostly slows secretion, giving the duodenum time to neutralize and digest chyme.

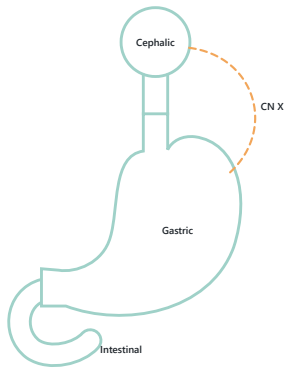


Fig. 1 | A head, stomach, and duodenum, showing that the cephalic phase starts in the head and reaches the stomach through the vagus nerve, the gastric phase happens in the stomach, and the intestinal phase happens in the duodenum.

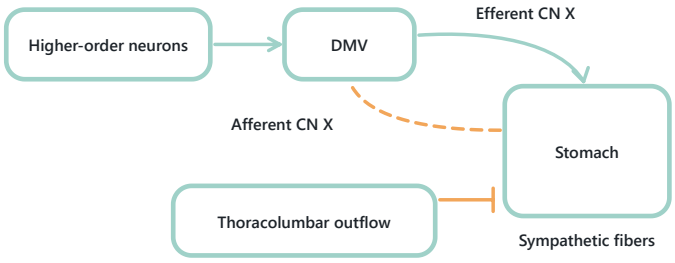


Fig. 2 | Higher-order neurons drive the dorsal motor nucleus of the vagus, whose efferent vagus fibers stimulate the stomach. Afferent vagus fibers return from the stomach to the nucleus. Thoracolumbar sympathetic fibers inhibit the stomach.

Cephalic phase

Sight, smell, taste, and thought of food act through higher-order neurons on the DMV, also called DMX. Efferent CN X synapses at the stomach and stimulates it, so secretion starts before food arrives. Afferent CN X carries vagovagal feedback from the stomach to the brainstem, whereas sympathetic fibers from the thoracolumbar outflow inhibit stomach activity. This phase depends on CN X, so vagotomy abolishes it.

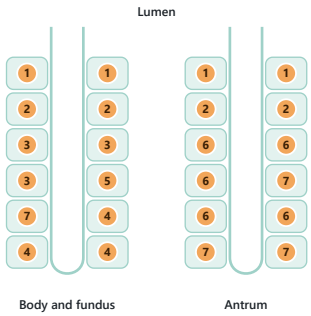


Fig. 3 | Two gastric glands, numbered from the surface down. The body and fundus gland holds foveolar, mucous neck, parietal, enterochromaffin-like, D, and chief cells. The antral gland holds foveolar, mucous neck, G, and D cells. 1, Foveolar cells; 2, Mucous neck cells; 3, Parietal cells; 4, Chief cells; 5, ECL cells; 6, G cells; 7, D cells.

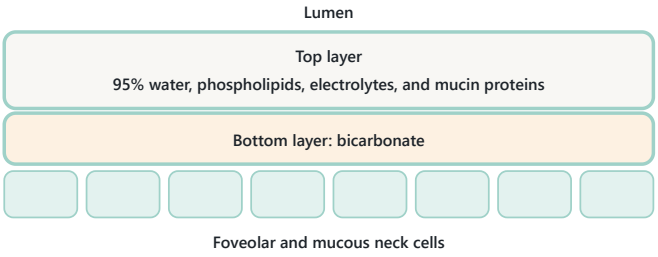


Fig. 4 | The mucosal barrier. A top layer of water, phospholipids, electrolytes, and mucin proteins lies over a bottom layer of bicarbonate, above foveolar and mucous neck cells.

Gastric cells and the mucosal barrier

Foveolar cells, at the surface and in gastric pits, and mucous neck cells, in the gland neck, secrete mucus. Parietal cells in the body and fundus secrete protons and IF, and chief cells at the gland base secrete pepsinogen and gastric lipase. ECL cells in the body and fundus secrete histamine, G cells in the antrum secrete gastrin, and D cells in the body, fundus, and antrum secrete somatostatin. Foveolar and mucous neck cells form the mucosal barrier, which has a top layer of 95% water, phospholipids, electrolytes, and mucin proteins, and a bottom layer of bicarbonate.

Gastric phase

Local and vagal stimulation act on G cells. Gastrin then promotes ECL, parietal, and chief cells, and histamine from ECL cells promotes parietal cells, which secrete protons through the H+/K+ ATPase in exchange for K+. Vagal ACh promotes chief cells, and the pepsinogen they secrete is activated to pepsin at pH under 3.5. D cells are negative regulators: excess protons lead to somatostatin, which inhibits G, ECL, parietal, and chief cells. Inside the parietal cell, carbonic anhydrase makes H+ and HCO_3^-; the H+ is pumped into the lumen by the H+/K+ ATPase, while the HCO_3^- leaves into the blood through the basolateral Cl^-/HCO_3^- exchanger (AE2), so blood HCO_3^- rises briefly after meals, the alkaline tide. Vomiting or NG suction loses HCl and leaves its HCO_3^- in the blood, causing hypochloremic metabolic alkalosis, and the volume loss activates RAAS, causing hypokalemia.

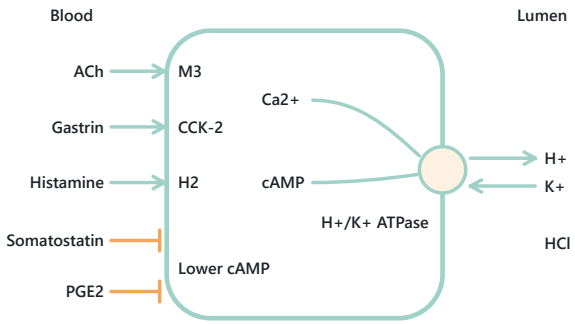


Fig. 5 | A parietal cell. Acetylcholine, gastrin, and histamine stimulate it through M3, CCK-2, and H2 receptors. Somatostatin and prostaglandin E2 inhibit it. The H+/K+ ATPase pumps H+ into the lumen in exchange for K+.

Receptors and intracellular pathways

On parietal cells, ACh acts through M3 and gastrin through CCK-2, both using Gq and increased Ca2+, whereas histamine acts through H2, using Gs and increased cAMP. Somatostatin through SSTR2 and PGE2 through EP3 inhibit parietal cells, using Gi and decreased cAMP. On ECL cells, gastrin stimulates through CCK-2, and somatostatin inhibits through SSTR2. On chief cells, ACh through M3 and gastrin through CCK-2 stimulate, and somatostatin through SSTR2 inhibits. On G cells, GRP through BB2 and amino acids through CaSR stimulate, using Gq and increased Ca2+, and somatostatin through SSTR2 inhibits.

Intestinal phase

Chyme in the duodenum mostly slows secretion. Peptides and amino acids release gastrin from G cells, causing a brief rise in parietal cell activity, and enteroxyntin, which stimulates parietal cells. Chyme and distension release 5-HT from enterochromaffin cells, increasing motility and secretion. Acid releases secretin from S cells, which decreases gastrin and HCl and increases pancreatic HCO3−. Fat and amino acids release CCK from I cells, which decreases HCl, slows gastric emptying, and causes gallbladder contraction and pancreatic enzyme release. Fat and glucose release GIP from K cells, which decreases HCl and increases insulin. Chyme reaching the ileum releases PYY and GLP-1 from L cells, which inhibit parietal cells and slow gastric emptying. Distension or hyperosmolar chyme triggers the enterogastric reflex, which decreases HCl and slows gastric emptying.

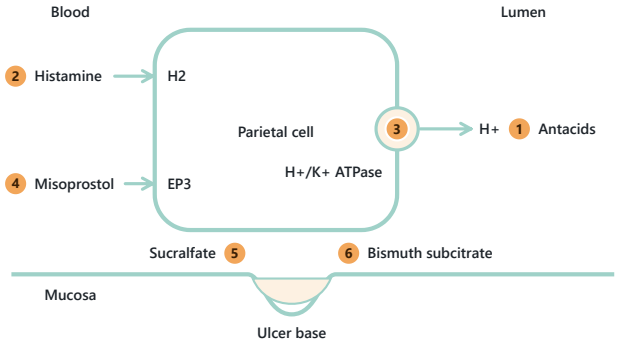


Fig. 6 | Six numbered drug sites. Antacids neutralize protons in the lumen, H2 receptor antagonists block H2 receptors, PPIs block the H+/K+ ATPase, misoprostol acts on EP3 receptors, and sucralfate and bismuth subcitrate coat the ulcer base. 1, Antacids; 2, H2R antagonists; 3, PPIs; 4, Misoprostol; 5, Sucralfate; 6, Bismuth subcitrate.

Pharmacology

Antacids (calcium carbonate, sodium bicarbonate, magnesium salts, and aluminum hydroxide) neutralize HCl in the lumen, increasing gastric pH and giving fast, short-acting relief. H2R antagonists (famotidine and cimetidine) competitively block histamine at H2 receptors on parietal cells, decreasing cAMP and H+/K+ ATPase activity, and so decrease HCl, especially basal and nocturnal secretion. PPIs (omeprazole and lansoprazole) are prodrugs activated in parietal cell canaliculi that irreversibly bind the pump, causing a profound decrease in basal and stimulated HCl. Misoprostol is a PG analog acting on EP receptors on parietal cells; it decreases HCl and increases mucus and HCO3−, giving gastroprotection in NSAID users. Sucralfate polymerizes at pH under 4 into a paste that binds ulcer base proteins and increases local PG and HCO3−, forming a barrier against acid, pepsin, and bile; because it needs acid, it works poorly with acid-suppressing drugs. Bismuth subcitrate coats the ulcer base, kills *H. pylori*, and increases PG and HCO3−.

Relationships to other conditions

Peptic ulcer disease. *H. pylori* increases gastrin and so HCl, and NSAIDs and alcohol inhibit COX, decreasing PG, which increases HCl and decreases mucus. Both therefore damage the balance between acid and the mucosal barrier.

Zollinger-Ellison syndrome *and* ***autoimmune gastritis***. Both raise gastrin, but for opposite reasons. In Zollinger-Ellison syndrome, gastrin from a tumor ignores feedback, so HCl is high; in autoimmune gastritis, parietal cell loss decreases HCl and IF, so feedback is lost. High HCl favors the former, and low HCl with B12 deficiency favors the latter. Secretin normally inhibits gastrin release but paradoxically increases it from gastrinoma cells.

Cushing ulcers *and* ***NSAID injury***. Both increase HCl. In Cushing ulcers, increased ICP increases CN X activity, whereas NSAIDs act through COX inhibition and also decrease mucus.

GERD. Here the problem is the location of HCl, which enters the esophagus, so the drugs that suppress acid apply here as well.

Digestion and absorption. Secretion links directly to digestion: HCl converts pepsinogen to pepsin, IF is needed for B12 absorption, and the intestinal phase slows the stomach so that the duodenum can neutralize and digest chyme.

The emetic center: inputs, the vomiting reflex, and antiemetic targets

Vomiting is coordinated not by a single center but by loosely organized neuron pools in the medulla, which a central pattern generator fires in sequence. Four inputs converge on the NTS, the start of the final common pathway: the CTZ in the area postrema, which samples blood and CSF through leaky capillaries; vagal afferents from the gut; the vestibular system; and the cortex and thalamus. Each input carries characteristic receptors, D2, 5-HT3, NK1, μ-opioid, H1, and muscarinic, which explain both the triggers of vomiting and the targets of antiemetics. Here, we review the inputs, the reflex, its failures, and the drugs that block it.

Key points

- The CTZ lies in the area postrema, outside the BBB, and carries D2, 5-HT3, NK1, and μ-opioid receptors.
- All inputs converge on the NTS, which drives a central pattern generator in the medullary reticular formation.
- Chemotherapy releases 5-HT from the gut wall onto 5-HT3 receptors; delayed CINV on days 2–5 depends on substance P.
- Motion sickness comes from vestibular, visual, and proprioceptive mismatch and is blocked by H1 antagonists and scopolamine.
- 5-HT3 antagonists prevent acute CINV, NK1 antagonists delayed CINV, and D2 antagonists act on the CTZ.

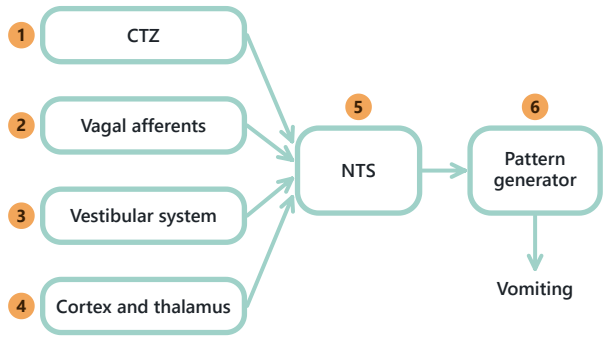


Fig. 1 | Four inputs to vomiting: 1, the chemoreceptor trigger zone; 2, vagal afferents from the gut; 3, the vestibular system; and 4, the cortex and thalamus. All converge on 5, the nucleus tractus solitarius, which drives 6, the central pattern generator, to cause vomiting. 1, CTZ; 2, Vagal afferents; 3, Vestibular system; 4, Cortex and thalamus; 5, NTS; 6, Central pattern generator.

Inputs

The CTZ lies in the area postrema on the floor of the fourth ventricle, a circumventricular organ whose capillaries lack tight junctions, so it detects toxins in both blood and CSF. Its receptors include D2, 5-HT3, NK1, and μ-opioid receptors, which respond to chemotherapy, opioids, and antiparkinsonian drugs. Gastric irritation releases 5-HT from the gut wall onto 5-HT3 receptors on vagal afferents, which end in the NTS. The vestibular system, with H1 and muscarinic receptors, responds to motion that the vestibular, visual, and proprioceptive systems report differently, and higher centers in the cortex and thalamus form the fourth input.

The area postrema, NTS, and dorsal motor nucleus of the vagus form the dorsal vagal complex. The area postrema receives afferents from CN IX and CN X and the hypothalamus and projects to the NTS, which carries NK1 and μ-opioid receptors and begins the final common pathway. Lesioning the area postrema stops vomiting from most emetic drugs but not from motion or vagal afferents. The NTS drives the central pattern generator in the retrofacial nucleus of the reticular formation, which sequences the respiratory and abdominal muscles, so there is no single vomiting center.

The vomiting reflex

Outputs from the NTS coordinate swallowing, salivation, breathing, the circulation, and the gut. The diaphragm and abdominal muscles contract, the external intercostals contract while the internal intercostals relax, the esophagus contracts while the LES relaxes, and the stomach and duodenum show gastric dysrhythmia and duodenal retroperistalsis. Repeated vomiting loses HCl, leaving its HCO_3^- in the blood and causing hypochloremic metabolic alkalosis.

When it fails

Chemotherapy and radiation release 5-HT from the gut wall onto 5-HT3 receptors, causing acute CINV within 24 hours, while delayed CINV on days 2–5 depends on substance P. Opioids stimulate D2 and μ receptors in the CTZ, but high doses cross the BBB to μ receptors in the NTS, which suppress vomiting. PONV is most likely in young to middle-aged nonsmoking women with previous PONV or motion sickness and rises with volatile anesthetics, nitrous oxide, long surgery, and postoperative opioids. Antiparkinsonian drugs act on the D2-dense area postrema and cause nausea.

Motion sickness arises from vestibular, visual, and proprioceptive mismatch, and people with total loss of labyrinthine function are immune. In NMOSD, attacks on the AQP4-rich area postrema cause intractable nausea, vomiting, and hiccups. Cannabinoid hyperemesis syndrome follows years of cannabis use, with cyclical vomiting relieved by hot showers that resolves after stopping. Cyclic vomiting syndrome causes stereotyped episodes separated by baseline health, is linked with migraine, and is diagnosed on Rome criteria after excluding mimics. Hyperemesis gravidarum starts at 4–6 weeks with weight loss of 5% or more, ketonuria, and electrolyte imbalance, and thiamine prevents Wernicke encephalopathy. On the condition pages, appendicitis causes periumbilical pain moving to the right lower quadrant followed by anorexia and vomiting, acute pancreatitis causes severe epigastric pain radiating to the back with nausea and vomiting, and Bouveret syndrome, a gallstone causing gastric outlet obstruction, causes vomiting and epigastric pain.

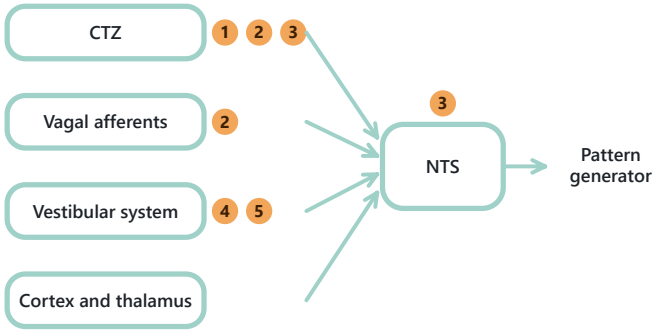


Fig. 2 | Five numbered drug sites. In the chemoreceptor trigger zone, 1, D2 antagonists, 2, 5-HT3 antagonists, and 3, NK1 antagonists act; 5-HT3 antagonists also act on vagal afferents; NK1 antagonists also act in the nucleus tractus solitarius; and 4, H1 antagonists, and 5, antimuscarinics, act on the vestibular input. 1, D2 antagonists; 2, 5-HT3 antagonists; 3, NK1 antagonists; 4, H1 antagonists; 5, Antimuscarinics.

Pharmacology

D2 antagonists, metoclopramide and prochlorperazine, block D2 receptors in the CTZ, and metoclopramide also stimulates 5-HT4 receptors, releasing ACh in the gut and speeding gastric emptying. 5-HT3 antagonists such as ondansetron and granisetron block 5-HT on vagal afferents and in the CTZ, for acute CINV, radiation, and PONV. NK1 antagonists such as aprepitant and netupitant block substance P in the area postrema and NTS, preventing delayed CINV.

H1 antagonists such as promethazine, dimenhydrinate, and diphenhydramine block histamine and acetylcholine in the brain, reducing vestibular input, especially in motion sickness, while second-generation antihistamines, which act peripherally, do not prevent it. Scopolamine, given as a transdermal patch, blocks muscarinic receptors and inhibits input to the vestibular nuclei, preventing motion sickness.

Relationships to other topics

Gastric secretion. Repeated vomiting loses HCl, and the HCO_3^- left in the blood causes hypochloremic metabolic alkalosis.

Liver function: structure, metabolism, bile, and detoxification

The liver is organized around portal triads and central veins, with blood flowing through sinusoids from zone 1 to zone 3 and bile flowing the opposite way. Its hepatocytes run the core pathways of carbohydrate, amino acid, and lipid metabolism, synthesize most plasma proteins, and store vitamins. They also make bile salts, conjugate bilirubin, metabolize xenobiotics in three phases, and move molecules through dedicated channels. Here, we review these functions, how they fail, and the drugs that target hepatitis viruses and bile salts, and consider how they relate to clinical conditions.

Key points

- Blood reaches zone 3 last, so zone 3 is first to suffer from low perfusion.
- During tissue damage, stellate cells become myofibroblasts that secrete collagen I, causing fibrosis.
- CYP7A1 is the key first step of bile salt synthesis, and CYP8B1 decides between cholic and chenodeoxycholic acid.
- Only conjugated bilirubin reaches the urine, so dark urine points to conjugated hyperbilirubinemia.
- Phase I modifies a molecule, phase II conjugates it, and phase III pumps it out.

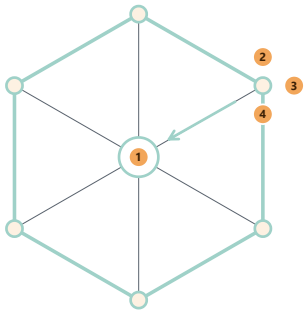


Fig. 1 | A hexagonal liver lobule with a central vein in the middle and a portal triad at each corner. The labeled triad holds a hepatic artery, portal vein, and bile duct. Blood flows from the corner toward the center. 1, Central vein; 2, Hepatic artery; 3, Portal vein; 4, Bile duct.

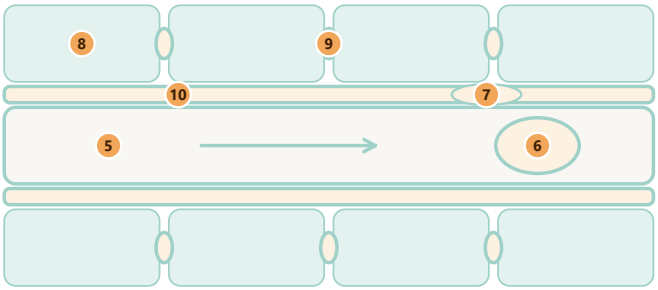


Fig. 2 | A sinusoid between two rows of hepatocytes, showing the sinusoidal capillary, a Kupffer cell inside it, the space of Disse on each side holding a stellate cell, and bile canaliculi between hepatocytes. 5, Sinusoidal capillaries; 6, Kupffer cells; 7, Stellate cells; 8, Hepatocytes; 9, Bile canaliculi; 10, Space of Disse.

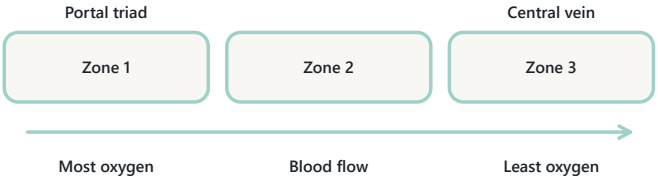


Fig. 3 | Three zones in a row from the portal triad to the central vein. Blood flows from zone 1 to zone 3, so oxygen falls along the way.

Anatomy and histology

Each portal triad holds a hepatic artery carrying oxygenated blood, a portal vein carrying blood from the gut, and a bile duct, and the central vein drains the sinusoids of each lobule. Sinusoidal capillaries carry mixed arterial and portal blood past hepatocytes, the main functional cells. Kupffer cells sense gram-negative LPS, parasites, and bacteria, produce cytokines, and convert old RBCs into heme and globin. Stellate cells store vitamin A during quiescence and, during tissue

damage, become myofibroblasts that secrete collagen I. The space of Disse lies between sinusoids and hepatocytes, and bile canaliculi collect bile between hepatocytes. Bile contains bile salts, bilirubin, phospholipid, cholesterol, water, and electrolytes. In the acinus, periportal zone 1 receives the most oxygen and handles gluconeogenesis, beta oxidation, and urea synthesis; zone 2 is midzonal; and centrilobular zone 3 receives the least oxygen and handles glycolysis, lipogenesis, and CYP metabolism.

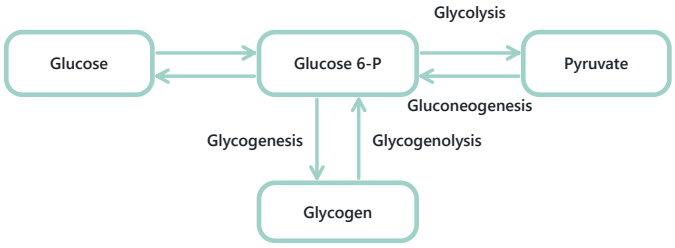


Fig. 4 | Glucose 6-phosphate sits at the center. Glucokinase converts glucose to it and glucose 6-phosphatase reverses this. Glycolysis leads to pyruvate and gluconeogenesis reverses this. Glycogenesis leads to glycogen and glycogenolysis reverses this.

Carbohydrate metabolism

In glycolysis, glucokinase forms glucose 6-P, PFK-1 forms fructose 1,6-bisphosphate, and pyruvate kinase forms pyruvate. In gluconeogenesis, pyruvate carboxylase forms oxaloacetate, PEPCK forms PEP, fructose 1,6-bisphosphatase forms fructose 6-P, and glucose 6-phosphatase forms glucose. In glycogenesis, glucose 6-P forms glucose 1-P, UDP-glucose, and then glycogen through glycogen synthase. In glycogenolysis, glycogen phosphorylase forms glucose 1-P, which becomes glucose 6-P and then glucose. Glucose 6-phosphatase lets the liver release free glucose into the blood.

Amino acids and protein synthesis

ALT and AST perform transamination, converting an amino acid and α-ketoglutarate to a keto acid and glutamate, and glutamate dehydrogenase performs deamination, releasing ammonia. In the urea cycle, carbamoyl phosphate synthetase I forms carbamoyl phosphate from ammonia and CO_2, ornithine transcarbamylase forms citrulline, argininosuccinate synthetase adds aspartate, argininosuccinate lyase forms arginine and releases fumarate, and arginase forms urea and releases ornithine. The liver makes albumin; the alpha and beta globulins transferrin, ceruloplasmin, haptoglobin, retinol-binding protein, transcortin, and thyroxine-binding protein; factors I, II, V, VII, IX, and X; AT-III and proteins C and S; complement proteins; and TPO, IGF-1, angiotensinogen, and 25-OH cholecalciferol.

Lipids and vitamins

In lipogenesis, acetyl-CoA from excess glucose forms malonyl-CoA through acetyl-CoA carboxylase, fatty acids through fatty acid synthase, and triglycerides with glycerol 3-phosphate. Fatty acids enter mitochondria through CPT1 for beta oxidation to acetyl-CoA, which can form HMG-CoA through HMG-CoA synthase and then the ketone bodies acetoacetate and β-hydroxybutyrate, used by other tissues. Hepatocytes package triglycerides with ApoB-100 as VLDL, which loses triglycerides in the blood to form IDL and LDL, and the liver makes HDL, which returns cholesterol to it. HMG-CoA reductase forms mevalonate, the rate-limiting step of cholesterol synthesis, and statins block it; cholesterol leaves in VLDL, into bile, or as bile salts. Vitamin A is stored in stellate cells and vitamin E in the liver. Cholecalciferol from the skin is stored and converted to 25-OH cholecalciferol in the liver, then to calcitriol in the kidney. Reduced vitamin K is needed for gamma-carboxylation of factors II, VII, IX, and X and proteins C and S, which forms vitamin K epoxide; VKOR restores it, and warfarin blocks VKOR.

Bile salt synthesis

Cholesterol passes through CYP7A1, the key first step, and forms cholic acid with CYP8B1 and chenodeoxycholic acid without it. The liver conjugates these with glycine or taurine, which makes them more water soluble, forming glycocholic, taurocholic, glycochenodeoxycholic, and taurochenodeoxycholic acid. Micelles digest fat in the small intestine, and most bile salts are reabsorbed in the terminal ileum. In the colon, bacteria deconjugate the rest and form the secondary bile salts deoxycholic and lithocholic acid, some of which are absorbed and return through the portal vein, where the liver reconjugates them.

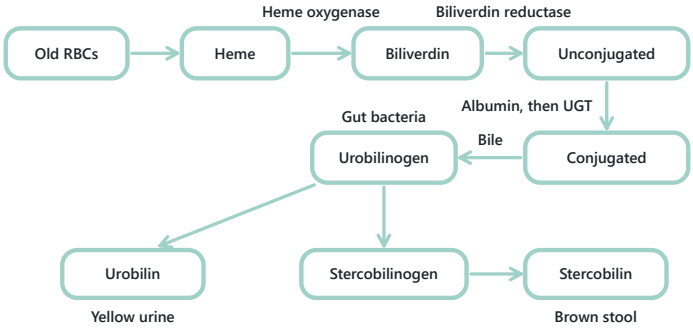


Fig. 5 | Old red cells yield heme, heme oxygenase forms biliverdin, and biliverdin reductase forms unconjugated bilirubin. Albumin carries it to the liver, where UGT forms conjugated bilirubin. Gut bacteria form urobilinogen, which leads to urobilin in urine and stercobilinogen then stercobilin in stool.

Bilirubin conjugation

Kupffer cells form heme and globin from old RBCs, and globin is broken down to amino acids. Heme oxygenase forms iron and biliverdin, and biliverdin reductase forms unconjugated bilirubin, which albumin carries to the liver. UGT forms conjugated bilirubin in hepatocytes. Gut bacteria form urobilinogen; reabsorbed urobilinogen reaches the kidney and colors urine yellow as urobilin, and urobilinogen that stays in the gut forms stercobilinogen, then stercobilin, which colors stool brown. Unconjugated hyperbilirubinemia leaves urine and stool normal in color, because bilirubin is water insoluble and bound to albumin, whereas conjugated hyperbilirubinemia causes dark urine, and pale stool if bile is obstructed.

Xenobiotic metabolism

In phase I, CYP enzymes oxidize, reduce, or hydrolyze; in phase II, conjugation makes the molecule water soluble; and in phase III, channels pump it into bile or blood. UGT adds glucuronic acid to bilirubin and paracetamol, GST adds glutathione to NAPQI, SULT adds sulfate to paracetamol, and NAT adds an acetyl group to isoniazid. CYP3A4 handles statins, calcium channel blockers, benzodiazepines, and tacrolimus; CYP2D6 handles codeine, tramadol, metoprolol, and tamoxifen; CYP2C9 handles warfarin, phenytoin, and NSAIDs; CYP2C19 handles clopidogrel, PPIs, and diazepam; CYP1A2 handles caffeine, theophylline, and clozapine; and CYP2E1 handles alcohol, paracetamol, and isoniazid. In CYP2D6 poor metabolizers, active drugs build up, so the dose is lowered, and prodrugs such as codeine work poorly; in ultrarapid metabolizers, active drugs clear fast, so the dose is raised, and codeine forms too much morphine. Alcohol is metabolized by alcohol dehydrogenase in the cytosol, the main pathway, by CYP2E1, which chronic drinking induces, and by catalase in peroxisomes. Acetaldehyde then forms acetate through aldehyde dehydrogenase, and the raised NADH:NAD+ ratio increases fatty acid synthesis and decreases fatty acid breakdown, causing fatty liver. GSH neutralizes peroxides through glutathione peroxidase, forming GSSG, which glutathione reductase and NADPH restore, and GST joins GSH to toxins such as NAPQI. NAC replenishes glutathione.

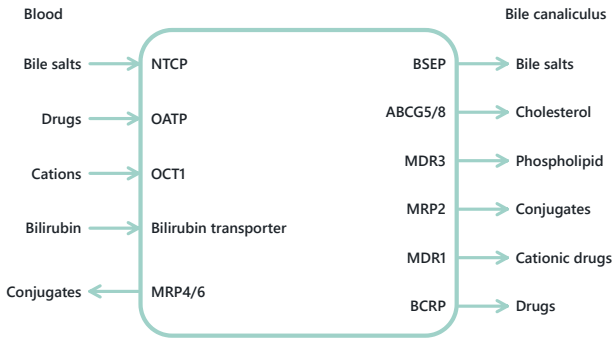


Fig. 6 | A hepatocyte with blood on the left and a bile canaliculus on the right. NTCP, OATP, OCT1, and a bilirubin transporter bring molecules in from the blood. MRP4 and MRP6 return molecules to the blood. BSEP, ABCG5 and ABCG8, MDR3, MRP2, MDR1, and BCRP send molecules into bile.

Channels

NTCP brings conjugated bile salts into the liver with Na+, OATP brings unconjugated bile salts and drugs such as statins, OCT1 brings organic cations such as metformin, and a bilirubin transporter brings unconjugated bilirubin. Into bile, BSEP sends bile salts, ABCG5/8 cholesterol, MDR3 phospholipid, MRP2 conjugated bilirubin and other conjugates, MDR1 cationic drugs, and BCRP drugs and sulfate conjugates. MRP4/6 send bile salts and other conjugates into blood.

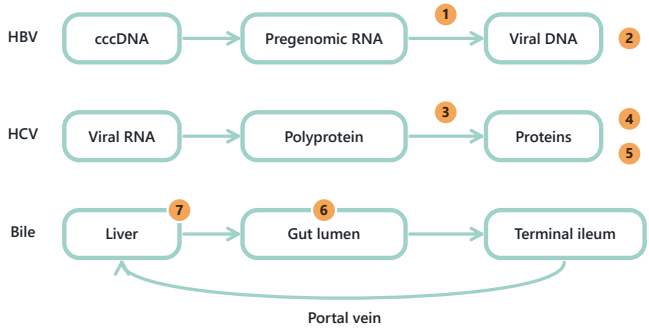


Fig. 7 | Seven numbered drug sites across three rows. In HBV, drugs block conversion of pregenomic RNA to viral DNA, and peginterferon acts on the host. In HCV, drugs block polyprotein cleavage and replication. In bile salt circulation, cholestyramine acts in the gut lumen, and ursodeoxycholic acid acts in the liver. 1, Nucleotide analogs; 2, Peginterferon-α; 3, NS3/4A inhibitors; 4, NS5A inhibitors; 5, NS5B inhibitors; 6, Cholestyramine; 7, Ursodeoxycholic acid.

Pharmacology

Tenofovir and entecavir are nucleotide analogs that compete at HBV reverse transcriptase, blocking reverse transcription of pregenomic RNA and suppressing HBV DNA, although cccDNA persists, so HBV is hard to cure. Peginterferon-α acts on interferon receptors, inducing antiviral proteins and boosting the immune response. In HCV, NS3/4A inhibitors such as glecaprevir and grazoprevir block cleavage of the viral polyprotein; NS5A inhibitors such as ledipasvir, velpatasvir, and pibrentasvir bind NS5A, impairing RNA replication and virion assembly; and sofosbuvir, an NS5B inhibitor, is a nucleotide prodrug incorporated into HCV RNA that ends the chain. Combinations cure over 95% of HCV infections. Cholestyramine is a non-absorbed resin that binds bile salts in the intestine and blocks their reabsorption, so the liver converts more cholesterol to bile salts; it decreases cholestatic pruritus, bile acid diarrhea, and LDL, and it binds other drugs too, so it is given apart from them, including ursodeoxycholic acid. Ursodeoxycholic acid is a hydrophilic bile salt that displaces toxic hydrophobic bile salts, increases bile flow, and decreases biliary cholesterol saturation, reducing bile salt toxicity, improving LFTs, slowing progression in PBC, and dissolving cholesterol gallstones.

Relationships to other conditions

Acute liver failure and cirrhosis. Lost synthesis and clearance explain the features: decreased albumin causes ascites and edema, decreased clotting factors cause coagulopathy with increased INR, decreased ammonia clearance causes hepatic encephalopathy, and decreased gluconeogenesis causes hypoglycemia. In cirrhosis, stellate cells drive the fibrosis.

Ischemic hepatitis *and* ***paracetamol overdose***. Both cause zone 3 necrosis. Decreased perfusion causes zone 3 hypoxia in ischemic hepatitis, whereas in paracetamol overdose, increased NAPQI uses up glutathione. Shock favors the former, and a paracetamol history favors the latter.

Unconjugated and conjugated hyperbilirubinemia. Normal urine color favors an unconjugated cause, such as hemolysis, whereas dark urine and pale stools favor conjugated hyperbilirubinemia from obstruction, as in choledocholithiasis.

Inherited hyperbilirubinemias. Gilbert syndrome and Crigler-Najjar syndrome are both unconjugated, from decreased UGT activity. Gilbert syndrome is benign, with mild jaundice with fasting, stress, or illness; Crigler-Najjar syndrome type I has absent activity, with severe neonatal jaundice and kernicterus; and type II is milder and responds to phenobarbital. Dubin-Johnson syndrome and Rotor syndrome are both conjugated and benign: defective MRP2 with a black liver favors Dubin-Johnson syndrome, and defective OATP reuptake with no black liver favors Rotor syndrome.

PBC and PSC. Cholestasis increases bile salts in blood and conjugated bilirubin, which is why cholestyramine and ursodeoxycholic acid both apply.

Gallstones. Cholesterol stones follow increased cholesterol, or decreased bile salts or phospholipids, whereas black pigment stones follow chronic hemolysis and increased unconjugated bilirubin.

Viral hepatitis. HBV and HCV differ in their targets: HBV drugs block reverse transcriptase, whereas HCV drugs target NS3/4A, NS5A, and NS5B, and only HCV is usually cured.

Digestion and absorption. Bile salts form the micelles that fat digestion needs, and the terminal ileum returns them to the liver, while ATP7B handles the copper that the gut absorbs.

PHYSIOLOGY

Digestion and absorption: enzyme activation, nutrient transport, and the sites that fail

Enzymes break food down in the lumen and at the brush border, and enterocytes absorb the products, mainly in the duodenum and jejunum. Five sites contribute: the mouth, stomach, pancreas, bile, and brush border. Many enzymes are secreted active, whereas pepsin and the pancreatic proteases are secreted inactive, so they cannot digest the cells that make them. Carbohydrates and proteins reach the portal vein, most fats leave in chylomicrons, and each micronutrient has its own site and route. Here, we review enzyme activation, how each nutrient is digested and absorbed, and how these steps fail, and consider how they relate to clinical conditions.

Key points

- Enterokinase activates trypsinogen, and trypsin then activates the other pancreatic zymogens and more trypsinogen.
- Pepsinogen is activated by HCl at pH under 3.5.
- Glucose and galactose enter through SGLT1 with Na+, fructose through GLUT5, and all three leave through GLUT2.
- Bile salts emulsify fat and form micelles, and chylomicrons leave into a lacteal.
- Iron, calcium, and copper are absorbed in the duodenum, folate in the jejunum, and B12 in the terminal ileum.

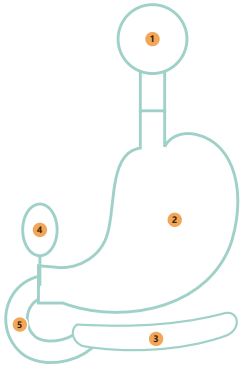


Fig. 1 | The digestive tract from mouth to duodenum, with five numbered sites: the mouth, stomach, pancreas, bile from the gallbladder, and the brush border of the duodenum. 1, Mouth; 2, Stomach; 3, Pancreas; 4, Bile; 5, Brush border.

Enzyme activation

Salivary amylase, lingual lipase, gastric lipase, pancreatic amylase, pancreatic lipase, and cholesterol esterase are secreted active, although pancreatic lipase needs colipase and bile salts, and cholesterol esterase needs bile salts, to work. Brush border enterokinase, disaccharidases, and peptidases are active and stay on the enterocyte. Pepsin is secreted as pepsinogen, which HCl activates at pH under 3.5. Trypsin is secreted as trypsinogen, which enterokinase activates, and trypsin then activates more trypsinogen. Chymotrypsinogen, proelastase, procarboxypeptidases, prophospholipase A2, and procolipase are all activated by trypsin. Enterokinase sits in the duodenum, so trypsin normally stays inactive until it leaves the pancreas.

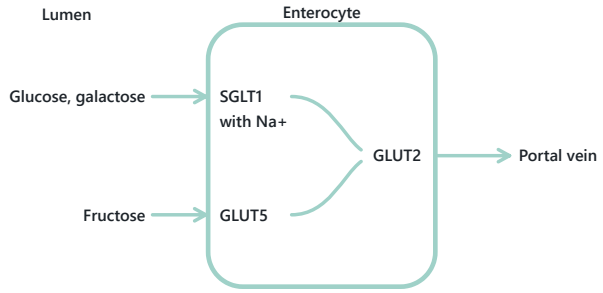


Fig. 2 | An enterocyte. Glucose and galactose enter through SGLT1 with sodium, fructose enters through GLUT5, and all three leave through GLUT2 into the portal vein.

Carbohydrates

Salivary amylase starts on starch until stomach acid stops it. Pancreatic amylase forms maltose, maltotriose, and α-limit dextrins, and brush border disaccharidases form monosaccharides: lactase converts lactose to glucose and galactose, sucrase converts sucrose to glucose and fructose, and maltase converts maltose to glucose. Glucose and galactose enter enterocytes through SGLT1 with Na+, fructose enters through GLUT5, and all three leave through GLUT2 into the portal vein. Fiber is not digested, so colonic bacteria ferment it.

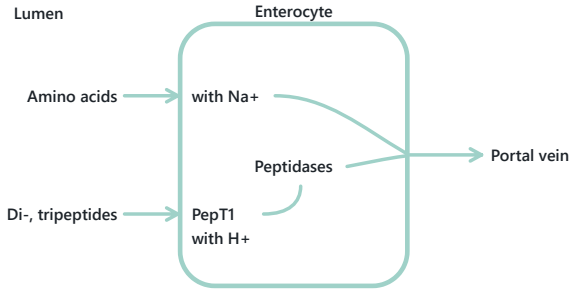


Fig. 3 | An enterocyte. Amino acids enter with sodium, di- and tripeptides enter through PepT1 with hydrogen ions and are split by peptidases, and amino acids leave into the portal vein.

Proteins

In the stomach, HCl converts pepsinogen to pepsin, which forms peptides. Trypsin, chymotrypsin, elastase, and carboxypeptidases form oligopeptides and amino acids, and brush border peptidases form amino acids, dipeptides, and tripeptides. Amino acids enter enterocytes with Na+, and dipeptides and tripeptides enter through PepT1 with H+, after which peptidases form amino acids that leave into the portal vein.

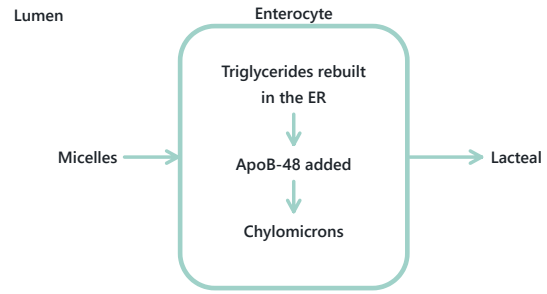


Fig. 4 | An enterocyte. Micelles deliver fatty acids and monoglycerides, which enter, are rebuilt into triglycerides, packaged with ApoB-48 into chylomicrons, and leave into a lacteal.

Fats

Lingual and gastric lipase have a minor role in adults. Bile salts emulsify fat into small droplets, and lipase and colipase form monoglycerides and fatty acids, with cholesterol esterase and phospholipase A2 also acting. Micelles of bile salts deliver the products to enterocytes, where triglycerides are rebuilt in the ER, ApoB-48 is added, and chylomicrons leave into a lacteal. Short- and medium-chain fatty acids skip chylomicrons and enter

the portal vein directly. Bile salts in micelles in the jejunum are reabsorbed in the terminal ileum and return to the liver through the portal vein.

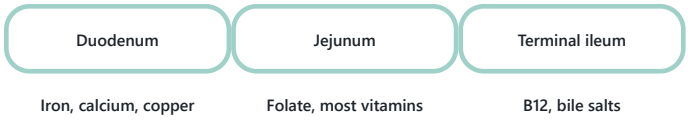


Fig. 5 | The small intestine in three parts. Iron, calcium, and copper are absorbed in the duodenum, folate and most vitamins in the jejunum, and B12 and bile salts in the terminal ileum.

Micronutrients

Iron, calcium, and copper are absorbed in the duodenum, folate and most vitamins in the jejunum, and B12 and bile salts in the terminal ileum. Iron is reduced from Fe3+ to Fe2+, helped by HCl and vitamin C, enters through DMT1, and reaches the blood through ferroportin, carried by transferrin. Hepcidin is the brake on iron: it binds ferroportin and causes it to be degraded. B12 binds haptocorrin from saliva in the stomach, proteases free it in the duodenum, it binds IF from parietal cells, the complex is absorbed in the terminal ileum, and transcobalamin carries it in blood. Copper enters through CTR1, which zinc blocks, reaches the blood through ATP7A, and in the liver ATP7B moves it into bile and onto ceruloplasmin. Vitamins A, D, E, and K travel in micelles and leave in chylomicrons, water-soluble vitamins are absorbed by transporters, mostly in the jejunum, and vitamin D increases calcium absorption. In SIBO, bacteria deconjugate bile salts and cause fat malabsorption, and B12 falls while folate stays normal or high. In abetalipoproteinemia, no apo B lipoproteins form, so enterocytes cannot export chylomicrons, and fat and vitamin E stay behind.

Relationships to other conditions

Maldigestion and malabsorption. Maldigestion leaves food undigested, as in lactose intolerance, pancreatic insufficiency, and bile salt deficiency. Malabsorption reflects mucosal injury, as in celiac disease, where blunted villi decrease absorption, with iron and folate deficiency. Normal enzymes with injured mucosa favor malabsorption, and altered enzyme function without mucosal injury favors maldigestion.

Acute pancreatitis *and pancreatic insufficiency*. Both involve pancreatic enzymes in opposite ways. In acute pancreatitis, trypsin is activated inside the pancreas, causing autodigestion, whereas in pancreatic insufficiency, decreased enzymes leave fat, protein, and carbohydrate undigested.

Autoimmune gastritis *and* ***ileal Crohn disease***. Both cause B12 deficiency. Autoimmune gastritis decreases IF, whereas ileal Crohn disease decreases ileal absorption and also causes bile salt loss.

Zollinger-Ellison syndrome. Excess acid inactivates pancreatic enzymes, causing diarrhea, which links gastric secretion to maldigestion.

Hemochromatosis and Wilson disease. Both are metal overload states. In hemochromatosis, decreased hepcidin increases iron absorption, whereas in Wilson disease, decreased ATP7B function decreases copper excretion into bile, so copper builds up in the liver.

Gastric secretion. HCl activates pepsinogen and helps iron reduction, IF comes from parietal cells, and the intestinal phase slows the stomach so that the duodenum can neutralize and digest chyme.

The immune system of the gut: sampling, secretory IgA, and tolerance

GALT must defend a vast surface exposed to pathogens while tolerating food and normal flora. M cells over Peyer patches sample luminal antigens and pass them to dendritic cells and macrophages, which prime lymphocytes that travel to the mesenteric lymph nodes. B cells switch to IgA-secreting plasma cells, and enterocytes carry dimeric IgA into the lumen on pIgR as SIgA, the first-line mucosal barrier. Here, we review this system, its tolerance of food, the pathogens and diseases that exploit or misdirect it, and the drugs that suppress it.

Key points

- Goblet cells make mucus, Paneth cells secrete defensins and lysozyme, and tuft cells start type 2 immunity against helminths through IL-25.
- M cells over Peyer patches, almost half of them in the distal ileum, carry antigen to dendritic cells and macrophages by transcytosis.
- Dimeric IgA crosses enterocytes on pIgR and enters the lumen as SIgA, binding antigens before they reach the submucosa.
- Peyer patch T cells secrete TGF-β and IL-10 toward food; loss of tolerance causes food allergy.
- Selective IgA deficiency, about 1 in 700, causes sinopulmonary infections and giardiasis and is linked to celiac disease.
- Vedolizumab blocks gut-homing lymphocytes; anti-TNF, IL-12/23, and JAK drugs block cytokines.

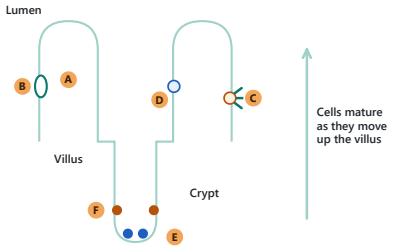


Fig. 1 | The epithelium of a villus and crypt: A, enterocytes line the villus; B, goblet cells secrete mucus; C, tuft cells sense parasites; D, enteroendocrine cells sense luminal contents; E, Paneth cells at the crypt base secrete defensins and lysozyme; and F, stem cells at the crypt base give rise to all of them, maturing as they move up the villus. A, Enterocytes; B, Goblet cells; C, Tuft cells; D, Enteroendocrine cells; E, Paneth cells; F, Stem cells.

Epithelial cells

Stem cells, LGR5-positive and at the base of the crypts, give rise to every epithelial cell, which matures as it moves up the villus. Enterocytes, about 70% of cells in the ileum, absorb nutrients. Goblet cells secrete mucin into a mucus layer that, in the colon, keeps bacteria off the epithelium, help keep tight junctions, and can import antigen. Enteroendocrine cells sense luminal contents and release incretins such as GLP-1 and GLP-2, which tune absorption, and GLP-1 strengthens the barrier. Paneth cells, at the base of small intestinal crypts beside the stem cells, secrete α-defensins 5 and 6 and lysozyme, keeping the crypt sterile, and support the stem cells with Wnt3, EGF, and Notch signals. Tuft cells sense parasites and release IL-25, which activates type 2 innate lymphoid cells to release IL-13; this promotes more tuft and goblet cells, more mucus, and helminth expulsion, the "weep and sweep" response.

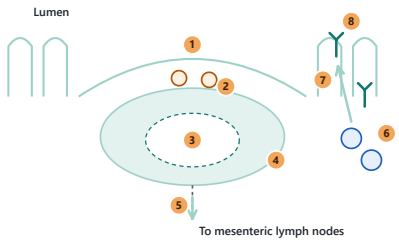


Fig. 2 | A Peyer patch in the distal small intestine: 1, M cells in the epithelium over the dome take up antigen; 2, dendritic cells and macrophages in the M cell pocket present it; 3, B cells mature in the germinal center of the follicle; 4, T cells sit in the parafollicular zone; 5, activated lymphocytes travel to the mesenteric lymph nodes; 6, IgA plasma cells sit in the lamina propria; 7, enterocytes carry dimeric IgA across on pIgR; and 8, secretory IgA enters the lumen. 1, M cells; 2, Dendritic cells and macrophages; 3, Germinal center; 4, Parafollicular zone; 5, Mesenteric lymph nodes; 6, IgA plasma cells; 7, Enterocytes; 8, SIgA.

Peyer patches

Peyer patches are organized lymphoid follicles in the lamina propria and submucosa of the jejunum and ileum, and sometimes the duodenum, with almost 50% in the distal ileum; they are most numerous at 15–25 years. They contain mainly B cells, with T cells, plasma cells, mast cells, eosinophils, macrophages, and basophils. The follicle-associated epithelium over each dome has few villi, thin mucus, and M cells, which have few microvilli and a thin glycocalyx and carry antigen across by transcytosis to dendritic cells and macrophages waiting in a basolateral pocket. B cells mature in the germinal center, T cells sit in the parafollicular zone, and activated lymphocytes travel to the mesenteric lymph nodes, where the response is amplified.

Secretory IgA

IgA is made in greater amounts than any other antibody. Plasma cells in the lamina propria secrete monomeric IgA, a J chain joins two monomers, and the dimer binds pIgR at the basolateral surface of enterocytes, crosses the cell, and is released when pIgR is cleaved at the apical surface, leaving the secretory component. SIgA binds antigens in the lumen and keeps them out of the submucosa and circulation, and it is also anti-inflammatory, helping induce tolerance through regulatory T cells. IgA1, about 80% of the total, lines mucosal surfaces and IgA2 is found mainly in the colon; class switching is promoted by IL-5, IL-6, and retinoic acid from vitamin A. IgG crosses the placenta but IgA does not, and IgA in breast milk protects the newborn from enteric infection.

Tolerance and allergy

Food proteins and normal flora are tolerated, and Peyer patch T cells secrete TGF-β and IL-10 toward common food allergens. Cesarean delivery, early antibiotics, and changes in the gut microbiome affect tolerance and barrier integrity. When tolerance is lost, food allergy follows, either IgE-mediated or not; in IgE-mediated allergy, dendritic cells present food antigen to Th2 cells, which drive sensitization, and IL-4 variants exaggerate the Th2 response.

When it fails

Salmonella targets and destroys M cells, and in typhoid, Peyer patch necrosis perforates the distal ileum, most often in the third week of fever. Shigella, Listeria, and Yersinia enter through M cells, Yersinia using them as its main route, HIV spreads through them to mucosal T cells, and prions are taken up by M cells and replicate in Peyer patches. Peyer patches enlarged by new infections in infancy can lead the bowel into intussusception.

Selective IgA deficiency, the most common primary immunodeficiency at about 1 in 700, is IgA below 7 mg/dL after age 4 with normal IgG and IgM; it is often silent but causes recurrent sinopulmonary infections and giardiasis, is linked most often with celiac disease and with autoimmune disease in 20–30%, and anti-IgA antibodies can cause anaphylaxis after blood products containing IgA.

Misdirected immunity underlies several conditions: in celiac disease, linked to HLA DQ2/8, gliadin activates T helper cells and B cells make tTG, DGP, and EMA antibodies; in IBD, Th1 predominates in Crohn disease, with NOD-2 mutations and defective Paneth cell defensins in ileal disease, and Th2 in ulcerative colitis; in autoimmune gastritis, autoantibodies against parietal cells reduce HCl and intrinsic factor; chronic H. pylori gastritis can cause gastric MALT lymphoma, treated by eradication; in eosinophilic esophagitis, Th2 cytokines such as IL-5 and IL-13 recruit 15 or more eosinophils per

high-power field; and in Whipple disease, Tropheryma whipplei infects macrophages.

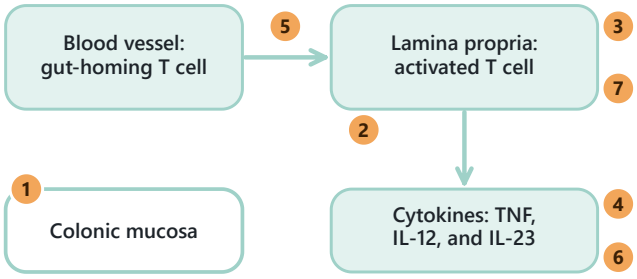


Fig. 3 | Seven numbered drug sites. 1, 5-ASA acts on the colonic mucosa. 2, glucocorticoids and 3, thiopurines and methotrexate act on activated lymphocytes. 4, anti-TNF drugs and 6, ustekinumab block cytokines. 5, vedolizumab blocks gut-homing T cells from leaving the blood. 7, JAK inhibitors block cytokine signaling inside the cell. 1, 5-ASA; 2, Glucocorticoids; 3, Thiopurines and methotrexate; 4, Anti-TNF drugs; 5, Vedolizumab; 6, Ustekinumab; 7, JAK inhibitors.

Pharmacology

5-ASA drugs, mesalazine and sulfasalazine, act topically on the colonic mucosa, inhibiting COX, lipoxygenase, and NF-κB. Glucocorticoids such as budesonide and prednisone act through the cytoplasmic glucocorticoid receptor to reduce inflammatory gene transcription, and budesonide's high first pass limits systemic effects. Azathioprine becomes thioguanine nucleotides that block purine synthesis, and methotrexate inhibits dihydrofolate reductase, both reducing lymphocyte proliferation.

Anti-TNF antibodies, infliximab and adalimumab, bind TNF-α. Vedolizumab blocks the α4β7 integrin from binding MAdCAM-1, a gut-selective block on lymphocytes entering the mucosa. Ustekinumab binds the p40 subunit of IL-12 and IL-23, reducing Th1 and Th17 responses, and JAK inhibitors such as tofacitinib and upadacitinib block signaling by many cytokines.

Relationships to other topics

Digestion and absorption. The distal ileum that reabsorbs most bile salts also holds almost half of the Peyer patches.

The enteric nervous system: plexuses, pacemakers, and motility

The gut has its own nervous system. The submucosal plexus governs secretion and blood flow, and the myenteric plexus, running from the upper esophagus to the internal anal sphincter, governs motility, while ICC generate the slow waves on which contractions ride. Parasympathetic fibers in the vagus and pelvic nerves are mostly excitatory and sympathetic fibers from T8–L2 inhibitory. Here, we review the structure of the ENS, the peristaltic reflex and regional motility patterns, the disorders that follow missing or damaged neurons, and the drugs that speed or slow the gut.

Key points

- The myenteric plexus controls motility and the submucosal plexus secretion and blood flow; ICC set the rhythm.
- Distension makes the gut contract above, through ACh, and relax below, through NO and VIP.
- The fasting MMC sweeps the small intestine every 90–120 minutes.
- Hirschsprung disease is failed neural crest migration, with no ganglion cells in the rectosigmoid in about 80%.
- Prucalopride and metoclopramide stimulate 5-HT4; loperamide acts on μ receptors to slow transit.

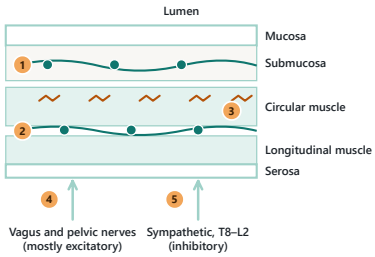


Fig. 1 | The gut wall in layers, with the enteric nervous system: 1, the submucosal (Meissner) plexus in the submucosa; 2, the myenteric (Auerbach) plexus between the circular and longitudinal muscle; 3, interstitial cells of Cajal in the muscle; and the extrinsic nerves, 4, parasympathetic fibers in the vagus and pelvic nerves, mostly excitatory, and 5, sympathetic fibers from T8 to L2, inhibitory. 1, Submucosal (Meissner) plexus; 2, Myenteric (Auerbach) plexus; 3, ICC; 4, Parasympathetic; 5, Sympathetic.

Structure

The submucosal (Meissner) plexus, present in the small and large intestine but not the stomach or esophagus, senses the mucosa and controls secretion and blood flow. The myenteric (Auerbach) plexus lies between the circular and longitudinal muscle as a continuous network from the upper esophagus to the internal anal sphincter, controls motility, and about 30% of its neurons are sensory. ICC, between nerve terminals and smooth muscle, generate slow waves, but contraction needs a spike on top of the slow wave. Parasympathetic fibers, through the vagus to the esophagus, stomach, pancreas, and upper colon and through pelvic nerves to the lower colon, rectum, and anus, are mostly excitatory, with ACh exciting and NO inhibiting, while sympathetic fibers from T8–L2 inhibit through prevertebral ganglia. ENS neurons come from neural crest cells.

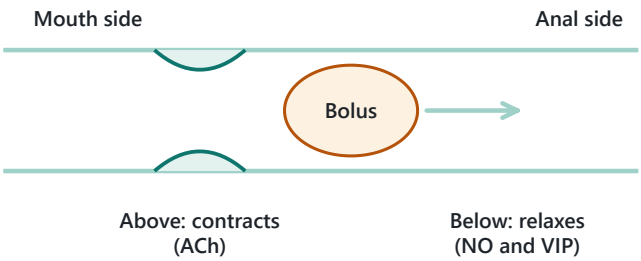


Fig. 2 | The peristaltic reflex: distension by a bolus makes the gut contract above it, through acetylcholine, and relax below it, through nitric oxide and VIP, pushing the bolus toward the anus.

Motility patterns

By the law of the intestine, distension is sensed by mechanoreceptors, and the gut contracts above and relaxes below. Mucosal 5-HT activates afferent nerves through 5-HT3 receptors and enhances ACh release through 5-HT4 receptors. In the esophagus, swallowing starts vagal inhibition followed by a contraction wave as the LES relaxes. The stomach relaxes to receive food through inhibitory vagal fibers, then mixes it under NANC, sympathetic, and vagal control, contracting 3–8 times a minute. The small intestine mixes its contents,

and between meals the MMC, quiet for about 70 minutes, intermittent for 10–20, then propulsive for 5, recurs every 90–120 minutes; a meal reaches the colon in 90–120 minutes. The colon mixes, segments, and pushes stool by mass movements over 8–15 hours.

After a meal, gastric stretch increases colonic motility through the gastrocolic reflex, with possible mediators CCK, serotonin, neurotensin, and gastrin. Motilin, like gastrin, is released directly by neural stimulation. In defecation, a mass movement fills the rectum, the internal sphincter relaxes involuntarily, the external sphincter and puborectalis relax voluntarily through the pudendal nerves, and Valsalva helps expel stool.

When it fails

In Hirschsprung disease, neural crest cells fail to reach the distal colon, so both plexuses lack ganglion cells, in the rectosigmoid in about 80%; it affects 1 in 5000 births, 4 boys for each girl, is linked to RET mutations, delays meconium, and is diagnosed by rectal suction biopsy with calretinin staining. In achalasia, myenteric ganglion cells are lost, so the LES cannot relax. In chronic Chagas disease, about 30% develop heart disease or megaesophagus and megacolon. Gastroparesis delays gastric emptying without obstruction, causing nausea, vomiting, and early satiety; it is most often idiopathic, then diabetic, in 25% and worse in type 1, postsurgical, or postinfectious, and is confirmed by 4-hour scintigraphy.

In diffuse esophageal spasm, the esophagus contracts instead of relaxing when distended. In ileus, slow waves continue without spikes under continuous inhibitory input. Acute colonic pseudo-obstruction affects sick inpatients through autonomic dysregulation, and CIPO follows disorders of enteric nerves, smooth muscle, or ICC. In infectious diarrhea, rapid, powerful peristalsis leaves no time to absorb water.

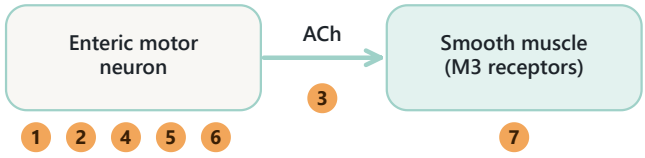


Fig. 3 | Seven numbered drug sites on an enteric motor neuron releasing acetylcholine onto smooth muscle. 1, metoclopramide and 2, prucalopride stimulate 5-HT4 receptors to enhance release, and 4, bisacodyl stimulates enteric neurons. 5, loperamide acts on μ-opioid receptors and 6, octreotide on somatostatin receptors to reduce it. 3, neostigmine blocks acetylcholine breakdown. 7, hyoscine butylbromide blocks M3 receptors on the muscle. 1, Metoclopramide; 2, Prucalopride; 3, Neostigmine; 4, Bisacodyl; 5, Loperamide; 6, Octreotide; 7, Hyoscine butylbromide.

Pharmacology

Metoclopramide blocks D2 receptors and stimulates 5-HT4, speeding gastric emptying, and with erythromycin, a motilin agonist, is first line in gastroparesis. Prucalopride, a selective 5-HT4 agonist, increases colonic peristalsis in laxative-refractory constipation, neostigmine blocks AChE and is the first-line drug for acute colonic pseudo-obstruction, and bisacodyl stimulates enteric neurons.

Loperamide acts on μ-opioid receptors on enteric neurons, reducing ACh release and slowing transit with little effect on the brain. Octreotide, through somatostatin receptors, inhibits secretion of hormones including VIP and serotonin, reducing secretion and motility, and hyoscine butylbromide blocks M3 receptors on smooth muscle as an antispasmodic.

Relationships to other topics

Gastric secretion. The vagus that relaxes and mixes the stomach also drives the cephalic phase of acid secretion.

GI embryology and defects. Enteric neurons come from migrating neural crest cells, and Hirschsprung disease is their failure to reach the distal colon.

Esophagitis: routes of mucosal injury and relationships to other foregut disease

Esophagitis is inflammation of the esophagus that follows mucosal damage from pills, caustics, allergens, infection, or reflux. The route of injury largely determines management, and endoscopy with biopsy identifies the cause by its pattern and histology. Presentation is dominated by dysphagia, odynophagia, heartburn, chest pain, and, in eosinophilic disease, food impaction, and the principal threats are bleeding, perforation, strictures, and progression through Barrett esophagus to adenocarcinoma. Here, we review the etiology, diagnosis, complications, and management of esophagitis, and consider how it relates to GERD, esophageal motility disorders, gastritis, and PUD.

Key points

- Esophagitis is a common endpoint of mucosal injury, and identifying the cause determines treatment.
- Pills lodge in the mid-esophagus, alkali penetrates deeper than acid, and infection follows impaired cell-mediated immunity.
- EGD with biopsy separates the causes: white plaques in Candida, small deep ulcers in HSV, large shallow ulcers in CMV, and 15 or more eosinophils per high-power field in eosinophilic esophagitis.
- PPIs are given in all cases, while caustic ingestion requires EGD within 24 hours and no induced vomiting.
- Odynophagia favors esophagitis over GERD and motility disorders.

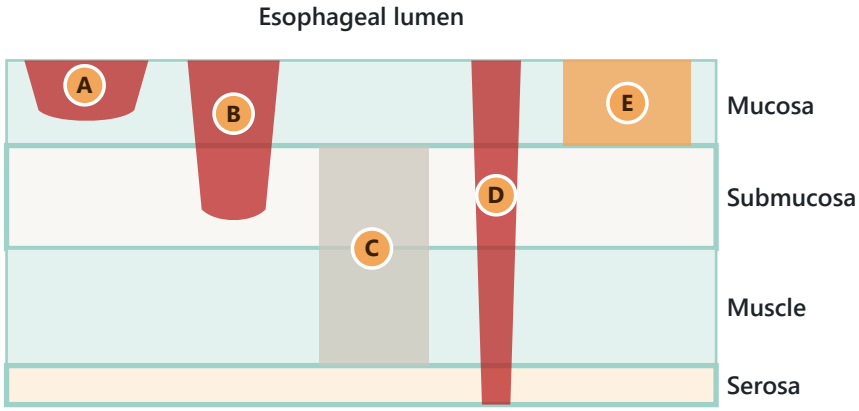


Fig. 1 | A wall of the esophagus in section, showing shallow mucosal injury, an ulcer into the submucosa, scar in the wall, a full-thickness perforation, and a patch of metaplasia. A, Mucosal inflammation; B, Ulcers; C, Strictures; D, Perforation; E, Barrett esophagus.

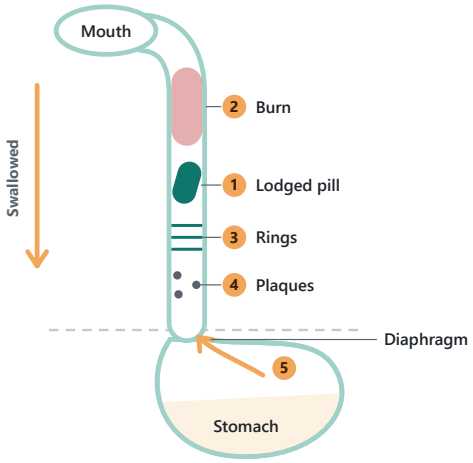


Fig. 2 | Diagram of the esophagus from the mouth to the stomach: pills, caustic agents, and food allergens are swallowed from above, infection affects the wall, and acid refluxes up from the stomach. 1, Pill-induced; 2, Caustic; 3, Eosinophilic; 4, Infectious; 5, Reflux-associated.

Pathophysiology

Pills, caustics, allergens, infection, and reflux all injure the esophageal mucosa, and the resulting inflammation causes odynophagia, heartburn, and chest pain. Deeper injury forms ulcers that bleed, with hematemesis and melena, and healing can narrow the lumen, causing strictures with dysphagia and obstruction. Full-thickness injury perforates, causing pneumomediastinum and mediastinitis, and chronic injury can progress through intestinal metaplasia and dysplasia to esophageal adenocarcinoma.

Etiology and mechanisms

Causes are best grouped by how injury reaches the mucosa. Pills, including NSAIDs, KCl, doxycycline, and bisphosphonates, lodge in the mid-esophagus, where the aortic arch or an enlarged left atrium narrows it, especially when taken lying down with little water. Strong alkali causes liquefactive necrosis that penetrates deeply, whereas strong acid causes coagulative necrosis that is more superficial, with skip lesions. Eosinophilic esophagitis is an allergic reaction to food and air antigens, in patients with asthma, allergic rhinitis, or atopic dermatitis. Infection follows impaired cell-mediated immunity, as with HIV, chemotherapy, or corticosteroids, and is due to Candida in 88%, HSV in 10%, and CMV in 2%. Reflux-associated disease arises from GERD, including conditions nonresponsive to PPIs.

Clinical presentation and complications

Patients present with dysphagia, odynophagia, heartburn, and chest pain, and eosinophilic esophagitis often presents with food impaction. Complications include bleeding (hematemesis, melena), perforation (pneumomediastinum, mediastinitis), strictures and ulcers, obstruction, and aspiration pneumonia. Barrett esophagus may also develop, in which intestinal metaplasia progresses through dysplasia to esophageal adenocarcinoma.

Diagnosis

EGD with biopsies of the proximal, mid, and distal esophagus identifies the cause by its endoscopic pattern, and histology confirms it. Pill injury produces an ulcer in the middle third, at the aortic arch or left atrium, with symptoms within hours to 10 days of the pill. In caustic ingestion, EGD within 24 hours grades the injury and predicts strictures, whereas after 96 hours friable tissue risks perforation. Eosinophilic esophagitis shows rings, longitudinal furrows, white exudates, and strictures, with 15 or more eosinophils per high-power field, and biopsy after an 8-week PPI trial separates it from GERD; endoscopy is normal in up to 10%, so biopsies are taken even when the mucosa looks normal.

Candida produces white plaques that do not wash off, HSV small, deep ulcers with multinucleated giant cells and Cowdry type A inclusions, and CMV large, shallow ulcers with intranuclear and cytoplasmic inclusions. Reflux-associated disease is worked up as GERD. Bleeding is evaluated by EGD with biopsy, and perforation by contrast esophagram and CXR, followed by surgery.

Management

PPIs are given in all cases to prevent further mucosal damage, followed by treatment of the cause. Offending pills are stopped, and pills are taken with at least 6 oz of water, staying upright for 30 minutes. Caustic ingestion requires securing the airway, NPO, EGD within 24 hours to grade the injury, avoidance of induced vomiting, and surgery for perforation, and strictures are the main late complication. Eosinophilic esophagitis is treated first with a PPI, then with elimination of trigger foods (cow's milk, wheat, peanuts and tree nuts, egg, soy, and seafood) or swallowed topical fluticasone or budesonide for 8 weeks, and strictures are dilated once it is treated. Candida esophagitis is treated with oral fluconazole for 14–21 days, always systemic rather than topical, and

HSV and CMV with antivirals. Reflux-associated disease is treated as GERD.

Relationships to other conditions

GERD. Reflux is one cause of esophagitis, and the two share dysphagia, heartburn, chest pain, bleeding, perforation, aspiration, and Barrett esophagus. Odynophagia favors esophagitis and dyspepsia favors GERD, whereas rings, furrows, atopy, food impaction, and persistent eosinophils after a PPI trial favor eosinophilic esophagitis.

Motility disorders. These share dysphagia and aspiration but are neuromuscular rather than mucosal. Dysphagia to solids and liquids from onset favors a motility disorder, for which manometry is the gold standard. Scleroderma links the two, as a weak LES promotes reflux.

Gastritis *and PUD*. NSAIDs cause all three by different mechanisms: a lodged pill injures the esophagus directly, whereas COX inhibition injures the stomach. Pain on swallowing favors esophagitis, and epigastric pain that changes with meals favors PUD. Perforation produces pneumomediastinum in esophagitis, assessed by contrast esophagram and CXR, and pneumoperitoneum in PUD, assessed by upright CXR. Intestinal metaplasia is common to both, but Barrett esophagus leads to esophageal adenocarcinoma, whereas chronic H. pylori infection leads to gastric cancer.

Cardiac disease. As chest pain overlaps with cardiac causes, an ECG should be obtained first.

Viral hepatitis. CMV and HSV also cause hepatitis in immunocompromised patients.

Esophageal motility disorders: neuromuscular failure of the LES and esophageal body

Esophageal motility disorders are neuromuscular dysfunctions of the esophagus that cause dysphagia. Achalasia, diffuse esophageal spasm, and esophageal scleroderma are separated by the state of the LES and the pattern of mid-distal motility, seen on barium swallow and defined by high-resolution manometry, the gold standard. Endoscopy excludes pseudoachalasia, and structural causes of dysphagia, including Zenker diverticulum, Schatzki ring, Plummer-Vinson syndrome, and eosinophilic esophagitis, must be distinguished. Complications include aspiration pneumonia and cancer. Here, we review their mechanisms, diagnosis, and treatment, and consider how they relate to esophagitis and GERD.

Key points

- LES tone separates the disorders: increased in achalasia and decreased in esophageal scleroderma.
- Achalasia causes progressive dysphagia to solids and liquids, a bird's beak on barium swallow, and an IRP over 15 mm Hg with absent peristalsis on manometry.
- Diffuse esophageal spasm causes episodic chest pain and a corkscrew esophagus.
- EGD excludes pseudoachalasia from a tumor at the gastroesophageal junction.
- Achalasia is treated by disrupting the LES, whereas scleroderma is treated with PPIs and Nissen fundoplication is avoided.

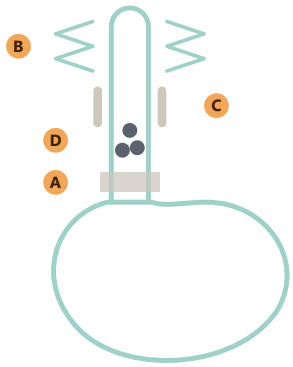


Fig. 1 | An esophagus above a stomach. A tight sphincter marks raised LES tone, disorganized contractions affect the wall, a thinned section shows fibrosis, and food sits above the sphincter. A, Failed LES relaxation; B, Disorganized contractions; C, Atrophy and fibrosis; D, Retained food.

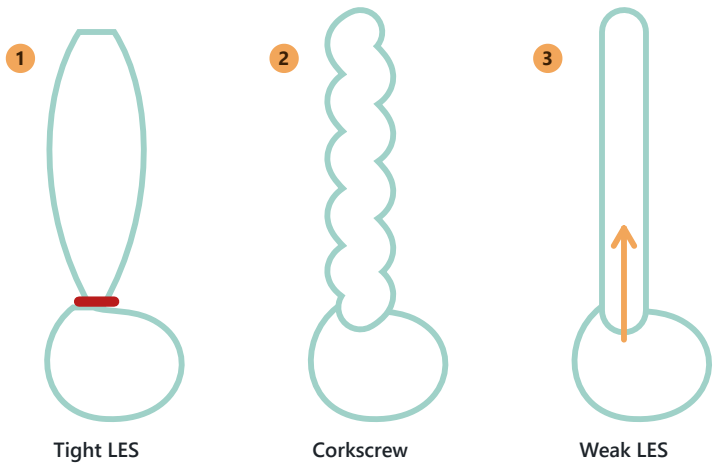


Fig. 2 | Three esophagi: achalasia with a dilated esophagus and tight LES, diffuse esophageal spasm with a corkscrew shape, and scleroderma with a weak LES that lets acid rise. 1, Achalasia; 2, Diffuse esophageal spasm; 3, Esophageal scleroderma.

Pathophysiology

Neuromuscular dysfunction in the LES or mid-distal esophagus stops food passing normally, causing dysphagia. In achalasia, decreased NO and VIP raise LES tone; in diffuse esophageal spasm, mid-distal contractions become uncoordinated; and in scleroderma, atrophy and fibrosis weaken smooth muscle, decreasing LES tone and motility. Food that stays in the esophagus predisposes to aspiration pneumonia and cancer.

Etiology and mechanisms

Achalasia is idiopathic or due to Trypanosoma cruzi. Decreased myenteric plexus activity lowers NO and VIP neurotransmitter activity, which increases LES tone and decreases mid-distal esophageal motility. In diffuse esophageal spasm, a trigger, possibly increased ACh release, produces simultaneous, premature contractions and disorganized mid-distal motility. In esophageal scleroderma, systemic sclerosis, often the limited form (CREST), causes atrophy and fibrosis of the distal smooth muscle, decreasing LES tone and mid-distal motility while proximal motility is preserved.

Clinical presentation and complications

Symptoms are dysphagia to solids and liquids, regurgitation, chest pain, weight loss, and heartburn. Achalasia causes progressive dysphagia to solids and liquids, regurgitation, and weight loss, with heartburn in 27–42% that is often mistreated as GERD, so reflux symptoms that persist despite PPIs should raise suspicion of achalasia. Diffuse esophageal spasm causes episodic chest pain and intermittent dysphagia, set off by eating quickly or by hot or cold drinks, and the pain can mimic angina. Esophageal scleroderma causes severe reflux, within CREST: calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia. Complications are aspiration pneumonia and cancer, squamous cell carcinoma or adenocarcinoma in achalasia.

Diagnosis

Barium swallow is the first test, EGD excludes pseudoachalasia from cancer, and high-resolution manometry, interpreted by the Chicago classification, is the gold standard. Achalasia shows a bird's beak with a dilated esophagus on barium, and an IRP over 15 mm Hg with absent peristalsis on manometry: type I without pressurization, type II with panesophageal pressurization, and type III with premature, spastic contractions. Diffuse esophageal spasm shows a corkscrew or rosary-bead esophagus, with premature contractions in 20% or more of swallows, and esophageal scleroderma weak distal peristalsis and low LES pressure, with proximal motility preserved.

Other causes of dysphagia must be distinguished. Pseudoachalasia reflects a tumor at the gastroesophageal junction or fundus, found by EGD. Zenker diverticulum affects older men, with regurgitation of food, halitosis, aspiration, and food sticking in the throat, and is shown by barium swallow, as manometry is unhelpful; it is a false diverticulum herniating through the Killian triangle between the cricopharyngeus and the inferior pharyngeal constrictor. Schatzki ring causes intermittent dysphagia to solids or food impaction, shown by barium swallow and EGD. Plummer-Vinson syndrome affects middle-aged women with iron-deficiency anemia and postcricoid dysphagia, investigated with iron studies and EGD for an upper esophageal web, and eosinophilic esophagitis presents with atopy and food impaction, diagnosed by EGD with biopsy.

Management

Achalasia is treated with pneumatic dilation, first line for most types, or myotomy, Heller with a partial fundoplication or POEM, which is preferred for type III; CCB, nitrates, or botulinum injection are used if patients are unfit for these, and type II responds best and type III least. Diffuse esophageal spasm is treated with CCB or nitrates, with pneumatic dilation or botulinum injection as other options. Esophageal scleroderma is treated with PPIs, and Nissen fundoplication is avoided. Zenker diverticulum is treated with cricopharyngeal myotomy and diverticulectomy, open or endoscopic, Schatzki ring with dilation followed by a PPI to prevent recurrence, and Plummer-Vinson syndrome with iron replacement and dilation of the web if needed, with surveillance for squamous cell carcinoma.

Relationships to other conditions

Esophagitis. Both cause dysphagia and aspiration, but motility disorders are neuromuscular rather than mucosal. Dysphagia to solids and liquids from onset favors a motility disorder, whereas odynophagia favors esophagitis, and food impaction with atopy favors eosinophilic esophagitis.

GERD. Achalasia and scleroderma are opposites at the LES, and scleroderma behaves like severe GERD because low LES tone permits reflux. Treatments are therefore opposed: Heller myotomy lowers LES pressure and Nissen fundoplication tightens it, so each would worsen the other condition, and achalasia should be suspected when reflux symptoms persist despite PPIs.

Autoimmune liver disease. Scleroderma (CREST) is associated with PBC.

Gastroesophageal reflux disease: failure of the antireflux barrier and its consequences

GERD occurs when HCl enters the esophagus and causes mucosal damage. Reflux is favored by decreased LES tone, including transient LES relaxations, hiatal hernia, increased HCl production, and increased intragastric pressure. Symptoms are heartburn, regurgitation, dysphagia, dyspepsia, and chest pain, with extraesophageal symptoms such as cough, asthma, and laryngitis, and the main long-term concern is Barrett esophagus. Here, we review its causes, stepwise diagnosis, and treatment, and consider how it relates to esophagitis, motility disorders, and PUD.

Key points

- Four mechanisms drive reflux: decreased LES tone, hiatal hernia, increased HCl production, and increased intragastric pressure.
- Chest pain requires an ECG, and alarm features require EGD before a PPI trial.
- Failure of an 8-week PPI trial leads to EGD and ambulatory pH monitoring, the only test of abnormal acid exposure.
- Barrett esophagus is salmon-colored mucosa 1 cm or more above the gastroesophageal junction with goblet cells, surveyed every 3–5 years without dysplasia.

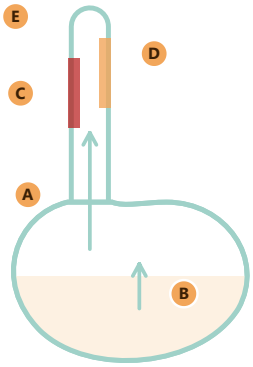


Fig. 1 | A stomach full of acid with a loose sphincter. Arrows show acid rising into the esophagus, where the lining is damaged and a patch has changed. A, Failed barrier; B, Raised pressure; C, Mucosal injury; D, Barrett esophagus; E, Aspiration.

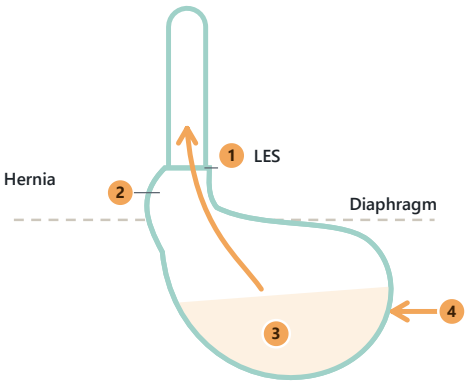


Fig. 2 | Esophagus and stomach showing acid rising through a weak LES, with a hiatal hernia above the diaphragm, acid in the stomach, and pressure on the stomach. 1, Decreased LES tone; 2, Hiatal hernias; 3, Increased HCl production; 4, Increased intragastric pressure.

Pathophysiology

Decreased LES tone or a hiatal hernia lets HCl pass into the esophagus, and increased intragastric pressure pushes it upward, more so when HCl production is increased. The HCl damages the esophagus, causing heartburn, dyspepsia, and chest pain. Chronic injury can progress through intestinal metaplasia and dysplasia to esophageal adenocarcinoma, and refluxed HCl that reaches the airway causes aspiration pneumonia.

Etiology and mechanisms

LES tone is decreased by transient LES relaxations or a low resting pressure, by smoking, alcohol, and caffeine, and by drugs such as nitrates, CCB, anticholinergics, theophylline, and morphine. Hiatal hernias promote reflux; sliding hernias (type I) make up over 95%, and paraesophageal hernias (types II–IV) can also cause volvulus and strangulation. HCl production is increased by smoking, alcohol, and Zollinger-Ellison syndrome, and aspirin and other NSAIDs are also risk factors. Intragastric pressure is increased by obesity, gastroparesis, and pregnancy.

Clinical presentation and complications

Patients present with heartburn, regurgitation, dysphagia, dyspepsia, and chest pain, and extraesophageal symptoms include chronic cough, asthma, laryngitis and hoarseness, sore throat, and dental erosions. Complications are bleeding (hematemesis, melena), perforation (pneumomediastinum, mediastinitis), strictures with progressive dysphagia, obstruction, aspiration pneumonia, and Barrett esophagus, in which intestinal metaplasia progresses through dysplasia to esophageal adenocarcinoma. Barrett esophagus is salmon-colored mucosa 1 cm or more above the gastroesophageal junction with goblet cells; its risk factors are male sex, White race, age over 50, central obesity, smoking, and a family history, screening is advised when chronic reflux comes with 3 or more, and fewer than 5% develop adenocarcinoma.

Diagnosis

Chest pain is first evaluated with an ECG. Alarm features, namely dysphagia, odynophagia, anemia, weight loss, and hematemesis or melena, call for EGD first; otherwise a PPI trial of 8 weeks, taken before the first meal, is initiated, and EGD with pH monitoring follows if there is no improvement. Ambulatory pH monitoring is the only test of abnormal acid exposure, by a wireless capsule for 48–96 hours or 24-hour pH-impedance. Manometry is performed before antireflux surgery to exclude achalasia or a scleroderma-like esophagus, and barium swallow shows strictures, hiatal hernia, and tumors but has little role in uncomplicated reflux. Bleeding is evaluated by EGD with biopsy, and perforation by contrast esophagram and CXR, followed by surgery.

Management

PPIs or H2R antagonists are always prescribed. Lifestyle measures are weight loss, raising the head of the bed, no meals 2–3 hours before bed, avoiding chocolate, caffeine, alcohol, and spicy food, and stopping smoking. A PPI is given at standard dose before the first meal for 8 weeks, and a bedtime H2RA is added for nighttime symptoms. Patients on NSAIDs stop the offending drug, refractory disease is treated with Nissen fundoplication after manometry, avoided in scleroderma, and a paraesophageal hernia is repaired surgically, as an emergency for volvulus or incarceration. In Barrett esophagus, EGD is repeated every 5 years without dysplasia if the segment is under 3 cm and every 3 years if 3 cm or more, in 6 months on twice-daily PPI for indefinite dysplasia, and low- or high-grade dysplasia is treated by endoscopic eradication, such as radiofrequency ablation.

Relationships to other conditions

Esophagitis. GERD is one cause of esophagitis, and the two share most symptoms and complications. Odynophagia favors esophagitis, and dyspepsia favors GERD.

Motility disorders. Scleroderma produces reflux through low LES tone, whereas achalasia has high LES tone and is often mistreated as GERD. Manometry confirms normal peristalsis before Nissen fundoplication, which is avoided in scleroderma.

PUD. Zollinger-Ellison syndrome links the two through increased HCl production, NSAIDs are risk factors for both, and PPIs treat both.

Cardiac disease. Chest pain is presumed cardiac until proven otherwise, particularly with cardiovascular risk factors.

Gastritis: superficial mucosal injury and the boundary with peptic ulcer disease

Gastritis is inflammation of the stomach's mucosal lining following damage to the stomach, and is associated with the antrum, shallow erosions, and shallow mucosal layers. Causes include NSAIDs and alcohol, *H. pylori*, Curling and Cushing ulcers, and autoimmune gastritis. Here, we review these mechanisms and consider how gastritis relates to PUD.

Key points

- NSAIDs and alcohol act through COX inhibition, and *H. pylori* through increased gastrin.
- Curling ulcers follow severe burns, and Cushing ulcers follow increased ICP.
- Autoimmune gastritis decreases HCl and IF, producing fundal atrophy rather than inflammation or erosions.
- Autoimmune gastritis causes pernicious anemia through IF loss, and its high gastrin can cause a type 1 gastric carcinoid.
- Chronic *H. pylori* gastritis leads to intestinal-type gastric adenocarcinoma and MALT lymphoma, so it is always eradicated.

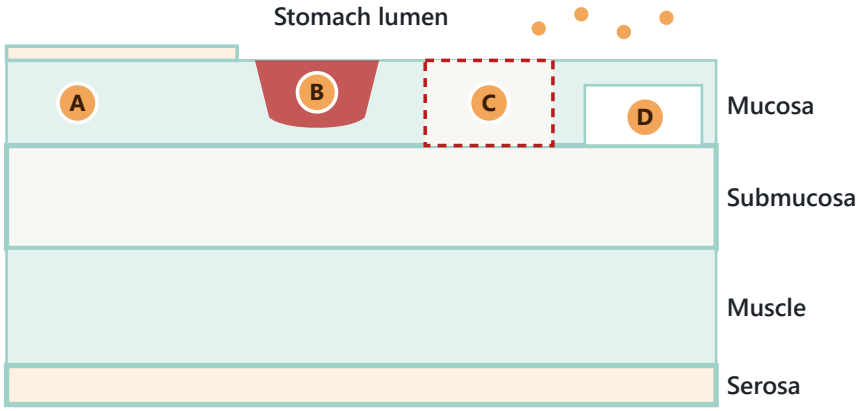


Fig. 1 | A stomach wall in section, with a thinned mucus layer, acid in the lumen, shallow erosions, a pale ischemic area, and a thinned area of lost parietal cells. A, Weakened mucus layer; B, Increased HCl; C, Mucosal ischemia; D, Parietal cell loss.

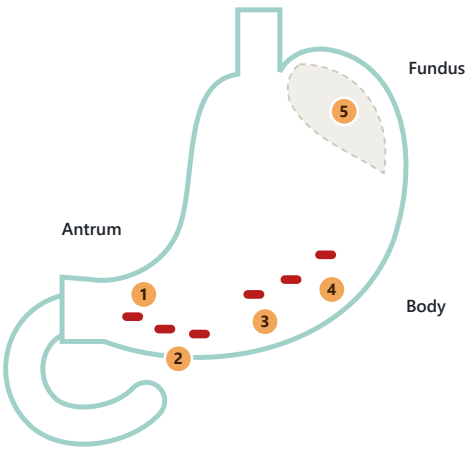


Fig. 2 | Stomach showing erosions in the antrum and body, and atrophy in the fundus. Positions are illustrative. 1, NSAIDs and alcohol; 2, H. pylori; 3, Curling ulcers; 4, Cushing ulcers; 5, Autoimmune gastritis.

Pathophysiology

Damage to the stomach inflames the mucosa and forms shallow erosions that do not pass the muscularis mucosae. COX inhibition decreases PG and so decreases mucus production, while NSAIDs, alcohol, *H. pylori*, and increased CN X activity increase HCl. Hypovolemia from severe burns causes mucosal ischemia in Curling ulcers. In autoimmune gastritis, autoantibodies against parietal cells decrease HCl and IF, causing fundal atrophy rather than erosions.

Etiology and mechanisms

NSAIDs and alcohol cause COX inhibition, which decreases PG, increasing HCl production and decreasing mucus production. *H. pylori* infection increases gastrin production and therefore HCl production. In Curling ulcers, severe burns cause hypovolemia and mucosal ischemia. In Cushing ulcers, increased ICP increases CN X activity and submucosal activity, which increases HCl production. In autoimmune gastritis, autoantibodies attack parietal cells, decreasing HCl and IF production and causing fundal atrophy rather than inflammation or erosions. *H. pylori* survives gastric acid because its urease makes ammonia. Autoimmune gastritis affects the body and fundus and spares the antrum, and often accompanies autoimmune thyroid disease.

Clinical presentation and complications

Gastritis is often asymptomatic; otherwise it causes epigastric pain, bloating, nausea, and vomiting, and erosions can bleed with hematemesis or melena. Autoimmune gastritis is usually found through anemia: parietal cell loss decreases IF and HCl, B12 malabsorption develops over 2–5 years, and pernicious anemia follows, with macrocytosis and hypersegmented neutrophils, then peripheral neuropathy and subacute combined degeneration. Low acid raises gastrin, which makes ECL cells grow and can cause a type 1 gastric carcinoid. Other complications are peptic ulcer from *H. pylori* and NSAIDs; gastric adenocarcinoma, as *H. pylori* leads to atrophy, intestinal metaplasia, and dysplasia; gastric MALT lymphoma from chronic *H. pylori* infection; iron and B12 deficiency from atrophic gastritis; and hemorrhage in stress gastritis without prophylaxis. Intestinal-type adenocarcinoma, the most common, forms glands and is linked to *H. pylori*, atrophic gastritis, pernicious anemia, smoking, and salt-preserved, smoked, or N-nitroso foods. Diffuse-type adenocarcinoma lacks cell adhesion, often with CDH1 mutations, has signet ring cells, infiltrates the wall as linitis plastica, or leather bottle stomach, and is linked to blood group A. Metastatic signs are the Virchow node (left supraclavicular), Sister Mary Joseph nodule (umbilicus), Irish node (left axilla), Krukenberg tumor (ovary), and Blumer shelf (pelvic cul-de-sac); paraneoplastic signs are acanthosis nigricans, diffuse seborrheic keratoses, and Trousseau syndrome. In Ménétrier disease, excess TGF-α activates EGFR and causes giant folds in the body and fundus, sparing the antrum, with decreased acid and protein loss that leads to hypoalbuminemia and edema. It affects mostly men, at a median age of 55, is progressive and premalignant in adults, and has a self-limited form in children after CMV.

Diagnosis

Dyspepsia is evaluated with *H. pylori* testing, by urea breath test, the most accurate noninvasive test, or stool antigen. Alarm features (weight loss, bleeding, dysphagia, a mass, persistent vomiting, and family history of gastric cancer) or age 60 or older call for EGD with biopsies from 5 sites. Autoimmune gastritis shows anti-parietal cell antibodies, which are sensitive, and anti-IF antibodies, which are specific, with body and fundus atrophy and intestinal metaplasia on biopsy, and B12 and iron are checked. In pernicious anemia, the FBC shows macrocytosis and hypersegmented neutrophils, though one-third have no macrocytosis. Ménétrier disease shows giant folds on EGD, confirmed by biopsy, which may need to be deep, with low albumin. Eradication is confirmed by a noninvasive test 4–6 weeks after finishing ABX or PPI.

Management

H. pylori gastritis is eradicated, using the regimens for PUD, and cure is confirmed. Gastritis from NSAIDs or alcohol is treated by stopping the cause and giving a PPI. Critically ill patients receive stress prophylaxis with a PPI first line or an H2R antagonist second line. *H. pylori*-negative dyspepsia under 60 is treated with an empiric PPI, then a prokinetic or tricyclic antidepressant. Autoimmune gastritis is treated with IM B12 1000 μg daily or on alternate days for 1–2 weeks, then IM or high-dose oral B12 for life, iron replacement, and EGD every 3 years if atrophy is advanced. Gastric MALT lymphoma is treated by *H. pylori* eradication if t(11;18) negative, with an 80% complete response, and otherwise by radiotherapy or systemic therapy. Ménétrier disease is treated with a high-protein diet, PPI, and albumin, treatment of CMV or *H. pylori*, and total gastrectomy if refractory.

A note on terminology

Curling and Cushing ulcers are not true ulcers because they do not erode past the muscularis mucosae.

Relationships to other conditions

PUD. Depth is the boundary: gastritis stays superficial, whereas PUD extends beyond the muscularis mucosae. NSAIDs, alcohol, and *H. pylori* cause both.

Autoimmune vs H. pylori gastritis. Autoimmune gastritis targets the body and fundus, where parietal cells live, and causes low acid with high gastrin. *H. pylori* gastritis typically starts in the antrum. Autoimmune disease also leads to B12 deficiency and raises the risk of gastric carcinoid tumors.

Other autoimmune disease. Autoimmune gastritis can accompany conditions such as Hashimoto thyroiditis and vitiligo.

Peptic ulcer disease: *H. pylori*, NSAIDs, and the balance of acid and mucosal defense

PUD is ulceration of the mucosal lining that extends beyond the muscularis mucosae. *H. pylori* is the most common cause, followed by NSAIDs and alcohol, and gastrinomas are a less common cause. Gastric and duodenal ulcers have opposite pain patterns. Bleeding, perforation, and gastric outlet obstruction are the major complications. Here, we review its etiology, diagnosis, and management, and consider its relationship to gastritis, GERD, and cancer.

Key points

- *H. pylori* is the most common cause, and first-line eradication is triple therapy.
- Gastric ulcer pain worsens with eating; duodenal ulcer pain improves with eating and worsens with fasting.
- Stop PPIs 2 weeks before *H. pylori* testing, and ABX and bismuth 4 weeks before.
- If *H. pylori* is absent, check gastrin and then a secretin stimulation test.

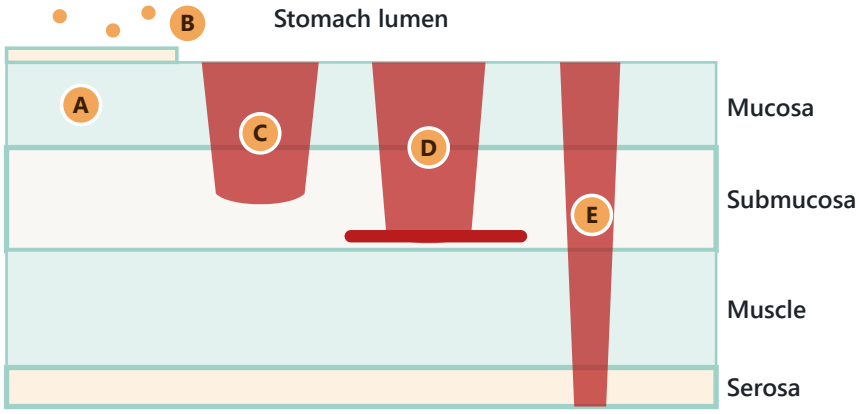


Fig. 1 | A stomach wall in section with a thinned mucus layer, acid in the lumen, a crater through the mucosa, a deeper crater reaching a vessel, and a perforation through the whole wall. A, Weakened barrier; B, Increased HCl; C, Ulcer crater; D, Vessel erosion; E, Perforation.

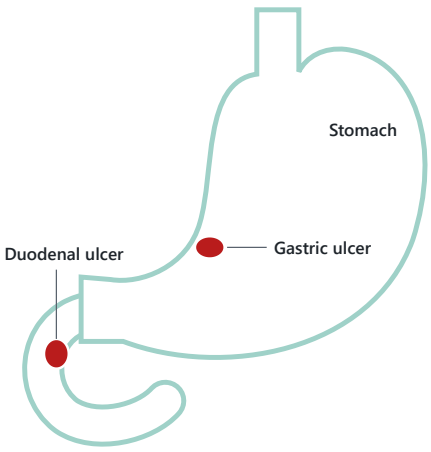


Fig. 2 | Stomach and duodenum with a gastric ulcer and a duodenal ulcer.

Pathophysiology

H. pylori, NSAIDs, and alcohol damage the mucosa and decrease mucus, and increased gastrin or decreased PG brings more acid to it. Once injury passes the muscularis mucosae, an ulcer forms, causing epigastric pain. An ulcer that reaches a vessel bleeds, with hematemesis and melena, and full-thickness injury perforates, causing pneumoperitoneum and peritonitis.

Etiology and mechanisms

H. pylori is gram-negative and urease-positive, has multiple flagella, and lives in the gastric mucosa. Its virulence factors are ureases, cytotoxins (VacA, CagA), mucinases, phospholipases, and PAF. Chronic infection leads to gastritis, then atrophy and intestinal metaplasia, then dysplasia and gastric cancer, and to gastric MALT lymphoma. NSAIDs and alcohol, the second most common cause, act through COX inhibition, decreasing PG, increasing HCl production, and decreasing mucus production. Gastrinomas, as in Zollinger-Ellison syndrome, increase gastrin production and therefore HCl production. *H. pylori* causes 90% of duodenal and 70–90% of gastric ulcers, and its urease makes ammonia, which neutralizes acid around it. Gastrinomas occur with MEN1 in 20–30%, and cause refractory or multiple ulcers, including jejunal ulcers, with diarrhea.

Clinical presentation and complications

Pain is epigastric, dull, and burning in both types; gastric ulcer pain is worse with eating and better with fasting, and duodenal ulcer pain is the reverse. Other symptoms are nausea, vomiting, heartburn, bloating, fullness, belching, and fatigue. Complications are bleeding (hematemesis, melena), perforation (pneumoperitoneum, peritonitis), gastric outlet obstruction (vomiting, pain, malnutrition, succussion splash), intestinal metaplasia, gastric cancer, and gastric MALT lymphoma. Gastric ulcers are usually on the lesser curvature, with pain 15–30 minutes after meals, and duodenal ulcers in the duodenal bulb, with pain 2–3 hours after meals and at night. A bleeding posterior duodenal ulcer erodes the gastroduodenal artery, and a lesser curvature ulcer the left gastric artery.

Diagnosis

Perforation is evaluated with an upright CXR, and bleeding with EGD with biopsy. *H. pylori* is detected by urea breath test, stool antigen test, rapid urease test, or histology. PPIs are stopped 2 weeks before testing, and ABX and bismuth 4 weeks before testing. If no *H. pylori* is detected, gastrin levels are checked, followed by a secretin stimulation test for Zollinger-Ellison syndrome. EGD is the gold standard, used for age over 50 with new dyspepsia or for alarm features at any age. A fasting gastrin over 1000 pg/mL, 10 times normal, is diagnostic of gastrinoma, after stopping PPIs for 1 week; if it is equivocal, a paradoxical rise over 120 pg/mL after secretin confirms it, and MEN1 is screened with calcium, PTH, and prolactin.

Management

Perforation is treated with IV fluids, IV PPI, IV ABX, and surgery, and bleeding with IV fluids, IV PPI, EGD, and surgery. Acid is suppressed with antacids, H2R antagonists, and PPIs, and the mucosa is protected with misoprostol, sucralfate, and bismuth. First-line *H. pylori* eradication is triple therapy, bd for 7–14 days: PPI, amoxicillin (metronidazole if allergic), and clarithromycin. Alternative regimens are PPI, amoxicillin, and levofloxacin; PPI, metronidazole, tetracycline, and bismuth; and PPI, amoxicillin, and rifabutin. NSAID ulcers are treated by stopping the NSAID or lowering the dose, with misoprostol to prevent ulcers in those who must continue, and gastrinoma with a high-dose PPI and resection if localized. A refractory ulcer, over 5 mm and not healed after 8–12 weeks of PPI, prompts a search for persistent *H. pylori*, NSAIDs, gastrinoma, or cancer, then vagotomy or partial gastrectomy. Cure is confirmed by urea breath test 4–6 weeks after treatment.

Relationships to other conditions

Gastritis. The two share causes, but PUD extends beyond the muscularis mucosae, whereas gastritis and Curling and Cushing ulcers do not.

Esophagitis *and* ***GERD***. All are treated with PPIs and can bleed. Perforation produces pneumoperitoneum here, assessed by upright CXR, and pneumomediastinum in the esophagus, assessed by contrast esophagram and CXR.

Diarrhea. Gastrinomas also cause chronic secretory diarrhea, so ulcers with diarrhea suggest Zollinger-Ellison syndrome.

Malignancy. Gastric ulcers can be malignant and are biopsied, whereas duodenal ulcers are rarely malignant. MEN1 links gastrinomas to **pancreatic neuroendocrine tumors**.

Vascular liver injury: inflow failure, outflow congestion, and zone 3 hypoxia

Decreased blood flow into or out of the liver leads to hypoxia of zone 3 (centrilobular) hepatocytes, causing hepatocellular injury. Ischemic hepatitis reflects failed arterial inflow, whereas congestive hepatopathy and Budd-Chiari syndrome reflect impaired venous outflow. Each has a distinct tempo and enzyme pattern. Here, we review their mechanisms, differentiation, and treatment, and consider how they relate to other causes of very high aminotransferases.

Key points

- Ischemic hepatitis causes a massive AST/ALT rise that falls within 7–10 days.
- JVD is present in heart failure and absent in Budd-Chiari syndrome.
- Doppler ultrasound of the hepatic veins is first line for Budd-Chiari syndrome.
- Budd-Chiari treatment steps up from anticoagulation to transplant.

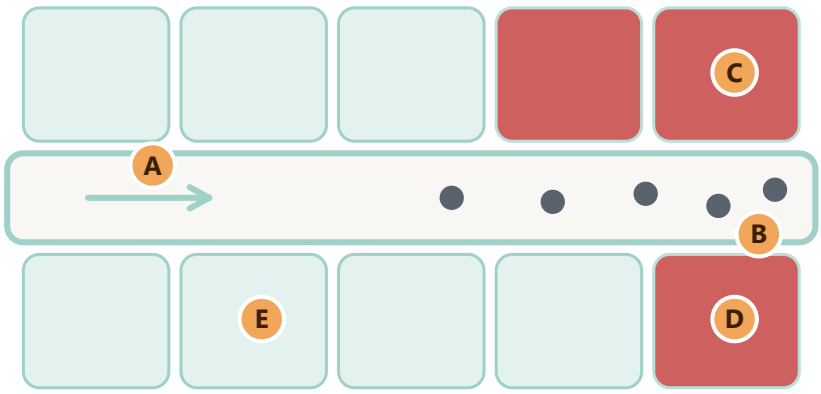


Fig. 1 | Two rows of hepatocytes along a sinusoid. Few red cells enter from the left, congestion builds on the right, and hepatocytes near the central vein have died. A, Decreased inflow; B, Blocked outflow; C, Zone 3 necrosis; D, Enzyme release; E, Chronic congestion.

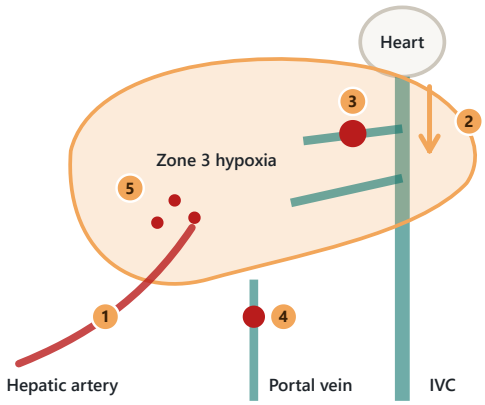


Fig. 2 | Heart, inferior vena cava, liver, hepatic veins, hepatic artery, and portal vein, with low arterial inflow, 1, back-pressure from the heart, 2, a clot in the hepatic vein, 3, a clot in the portal vein, 4, and blocked sinusoids, 5. Blood flows in by the hepatic artery and portal vein, through the sinusoids, and out by the hepatic veins. 1, Ischemic hepatitis ("shock liver"); 2, Hepatic vein congestion (congestive hepatopathy); 3, Hepatic vein clot (Budd-Chiari syndrome); 4, Portal vein thrombosis; 5, Sinusoidal obstruction syndrome (veno-occlusive disease).

Pathophysiology

Shock or cardiac arrest decreases hepatic perfusion, whereas right heart failure or a clot blocks outflow, raising sinusoidal pressure and causing sinusoidal congestion. Either way, zone 3 hepatocytes receive the least oxygen and die first, releasing AST and ALT, which often exceed 1000 U/L. Chronic congestion causes zone 3 atrophy, leading to nutmeg liver and cardiac cirrhosis.

Etiology and mechanisms

Ischemic hepatitis ("shock liver") follows shock (cardiogenic, hypovolemic, septic) or cardiac arrest: decreased hepatic perfusion causes zone 3 hypoxia and centrilobular necrosis, with increased AST/ALT, often over 1000 U/L. Congestive hepatopathy follows RHF, as well as tricuspid regurgitation and constrictive pericarditis: increased CVP and hepatic venous pressure cause sinusoidal congestion, zone 3 atrophy, and necrosis, producing nutmeg liver and, if chronic, cardiac cirrhosis. Budd-Chiari syndrome is thrombosis of the hepatic veins or IVC, most commonly from myeloproliferative neoplasms (polycythemia vera, JAK2), and also from pregnancy, OCPs, PNH, and malignancy (e.g. HCC). Blocked hepatic outflow and increased sinusoidal pressure cause hepatomegaly, ascites, and RUQ pain. Portal vein thrombosis is most often due to cirrhosis, and otherwise to myeloproliferative and other prothrombotic states or abdominal inflammation such as pancreatitis; it blocks portal inflow and causes portal hypertension, with varices and splenomegaly but near-normal liver function unless the liver is cirrhotic. Sinusoidal obstruction syndrome, or veno-occlusive disease, follows conditioning for hematopoietic cell transplant, especially with busulfan, or drugs such as cyclophosphamide and oxaliplatin, which damage the zone 3 sinusoidal endothelium and obstruct the sinusoids, causing jaundice, painful hepatomegaly, and ascites or weight gain.

Clinical presentation and complications

Ischemic hepatitis is acute, 1–3 days after hypotension, often with no symptoms or with nausea and RUQ pain; JVD depends on the cause of shock, and AST/ALT shows a massive rise, often over 1000 U/L, falling within 7–10 days. Congestive hepatopathy is usually chronic, with RUQ discomfort, hepatomegaly, and peripheral edema; JVD is present with hepatojugular reflux, and there is a mild rise, often with increased bilirubin. Budd-Chiari syndrome is acute or chronic, with a triad of RUQ pain, hepatomegaly, and ascites; JVD is absent, and the rise is variable and very high in acute disease. General symptoms are RUQ pain, hepatomegaly, ascites, jaundice, and nausea. Complications are acute liver failure (coagulopathy with increased INR, hepatic encephalopathy, hypoglycemia), cardiac cirrhosis from long-standing congestion, and portal hypertension (ascites, varices, splenomegaly).

Diagnosis

Any vascular cause is assessed with LFTs and INR. Ischemic hepatitis is supported by a history of hypotension, serial LFTs with a rapid rise then rapid fall, and markedly increased LDH. Congestive hepatopathy is evaluated by echocardiogram and BNP. Budd-Chiari syndrome is evaluated by Doppler ultrasound of the hepatic veins first, CT or MR venography to confirm, and a hypercoagulability workup (JAK2 V617F). JVD is present in heart failure and absent in Budd-Chiari syndrome. Portal vein thrombosis is shown by Doppler ultrasound, with normal or near-normal LFTs unless the liver is cirrhotic, and a chronic clot forms a portal cavernoma. Sinusoidal obstruction syndrome is diagnosed clinically by a bilirubin of 2 mg/dL or more with 2 of painful hepatomegaly, weight gain over 5%, or ascites, usually within 21 days of transplant, and Doppler shows reduced or reversed portal flow.

Management

Ischemic hepatitis is treated by restoring perfusion and treating the cause of shock (fluids, vasopressors, cardiac support); LFTs normalize within 7–10 days once blood flow is restored. Congestive hepatopathy is treated by treating heart failure (e.g. diuretics). Budd-Chiari syndrome is treated in all patients with anticoagulation and treatment of the underlying cause, then angioplasty or stenting for short venous narrowing, TIPS, and liver transplant for fulminant failure or cirrhosis. Portal vein thrombosis is treated with early anticoagulation, with management of portal hypertension and the cause, and sinusoidal obstruction syndrome with defibrotide, the only approved drug, started early.

Relationships to other conditions

Intestinal ischemia. Low flow injures vulnerable zones in both organs: zone 3 in the liver and the watershed splenic flexure in the colon.

Viral hepatitis. Both can raise aminotransferases above 1000 U/L, but ischemic hepatitis rises and falls within days with a very high LDH, whereas viral hepatitis rises and falls more slowly.

Paracetamol overdose. Both injure zone 3, so the history separates them.

Cirrhosis. Congestion and Budd-Chiari syndrome share portal hypertension with cirrhosis, and SAAG of 1.1 g/dL or more does not separate them.

Infectious liver injury: the hepatitis viruses and their clinical divergence

Viral infection of hepatocytes triggers a cytotoxic T-cell response, which destroys infected cells and causes hepatic inflammation. Hepatitis A and E are fecal-oral and acute, whereas hepatitis B, C, and D are bloodborne and can become chronic, leading to cirrhosis and HCC. CMV, HSV, and EBV also cause hepatitis. Here, we review their virology, phases, serology, and treatment, and consider how they relate to other liver injuries.

Key points

- "The vowels hit your bowels": hepatitis A and E are fecal-oral, acute only, and naked.
- HDV needs HBsAg from HBV as its envelope.
- In the window period, IgM anti-HBc is the only positive marker, and anti-HBc never comes from vaccination.
- Anti-HCV stays positive for life, so HCV RNA confirms current infection.
- Screen for HCC every 6 months in chronic HBV, or HCV with cirrhosis.

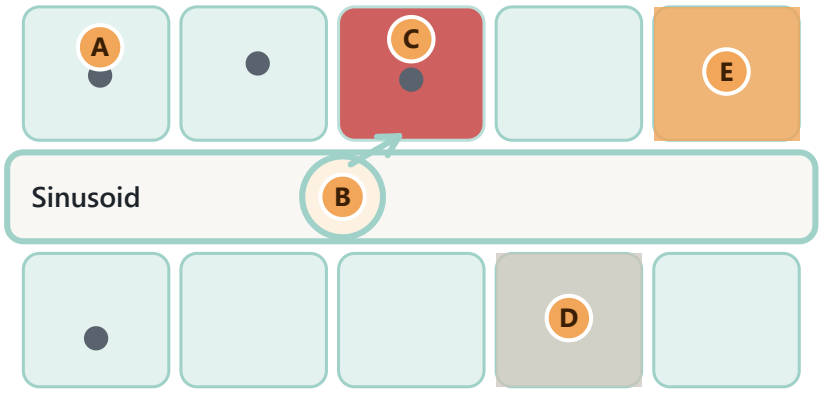


Fig. 1 | Hepatocytes containing virus with a cytotoxic T cell attacking one, a dead hepatocyte, and scar forming nearby. A, Infected hepatocyte; B, Cytotoxic T cell; C, Hepatocyte death; D, Fibrosis; E, Integration.

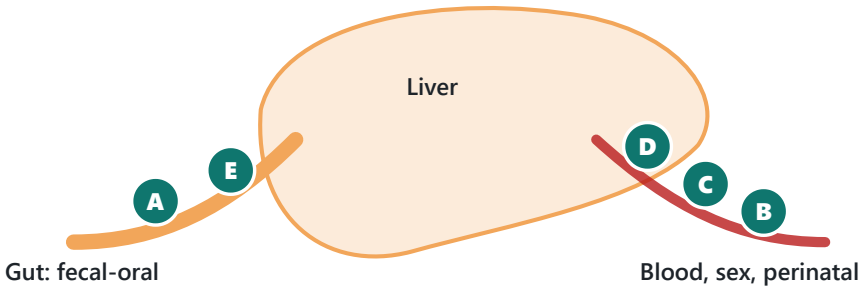


Fig. 2 | Routes into the liver: hepatitis A and E arrive from the gut by the fecal-oral route, and hepatitis B, C, and D arrive through the blood.

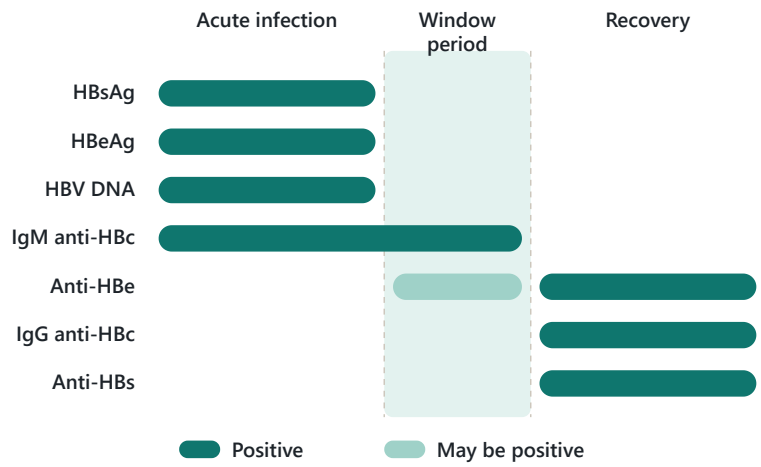


Fig. 3 | Hepatitis B markers through acute infection, the window period, and recovery. In acute infection, HBsAg, HBeAg, HBV DNA, and IgM anti-HBc are positive. In the window period, IgM anti-HBc is the only positive marker, and anti-HBe may be positive. In recovery, anti-HBe, IgG anti-HBc, and anti-HBs are positive.

Pathophysiology

Viruses replicate inside hepatocytes, and cytotoxic T cells destroy the infected cells, causing hepatic inflammation and increased AST and ALT. Chronic infection leads to fibrosis and then cirrhosis, and HBV integrates into host DNA, so it can cause HCC without cirrhosis.

Virology

HAV is a naked +ssRNA picornavirus spread by the fecal-oral route, with 2–6 weeks of incubation; it is acute only, has no carrier state, and has an inactivated vaccine. HBV is an enveloped, partially double-stranded, circular DNA (pdsDNA) hepadnavirus spread by sex, blood, and perinatally, with 1–6 months of incubation; it becomes chronic in ~90% of infants and under 5% of adults, integrates into host DNA so can cause HCC without cirrhosis, and has a recombinant HBsAg vaccine. HCV is an enveloped +ssRNA flavivirus spread by blood, mainly IV drug use, with 2 weeks–6 months of incubation; it becomes chronic in 75–85%, and high antigenic variation means no vaccine and frequent chronicity. HDV is a circular –ssRNA deltavirus with an HBsAg envelope borrowed from HBV, spread by sex, blood, and perinatally, with incubation similar to HBV; the HBV vaccine protects. HEV is a naked +ssRNA hepevirus spread by the fecal-oral route through contaminated water and undercooked pork, with 2–9 weeks of incubation; it is acute, chronic only if immunocompromised, rarely chronic overall, has no widely available vaccine, and causes fulminant hepatitis in pregnancy with high mortality.

Coinfection and superinfection

HDV only infects people who also have HBV. In coinfection, HBV and HDV are acquired at the same time; disease is usually acute and self-limited, and both viruses are usually cleared. In superinfection, HDV is acquired by someone with chronic HBV; disease is more severe, with a risk of fulminant hepatitis, and usually becomes chronic with faster progression to cirrhosis.

Viral life cycles

+ssRNA viruses (HAV, HCV, HEV) attach, enter the hepatocyte, and uncoat; +ssRNA acts directly as mRNA, is translated into a polyprotein that viral proteases cleave, and viral RdRp copies a –RNA template that makes new genomes, followed by assembly and release. They replicate in the cytoplasm, and HCV drugs target NS3/4A, NS5A, and NS5B. HBV attaches and enters via the NTCP receptor, its nucleocapsid carries pdsDNA into the nucleus, pdsDNA is repaired into cccDNA, host RNA polymerase II makes mRNAs and pregenomic RNA, viral reverse transcriptase converts pregenomic RNA into new pdsDNA, and assembly with an HBsAg envelope is followed by release. cccDNA persists, so HBV is hard to cure, and tenofovir and entecavir block reverse transcriptase. HDV replicates in the nucleus using host RNA polymerase.

Other viruses

CMV affects immunocompromised patients (transplant, AIDS), with a mono-like illness, a negative heterophile test, and usually mild hepatitis. HSV affects pregnancy (third trimester), neonates, and immunocompromised patients; it is rare, and can be fulminant and anicteric. EBV causes infectious mononucleosis in adolescents and young adults, with common mild transaminase rise and hepatosplenomegaly. Beyond viruses, pyogenic liver abscesses are usually polymicrobial, with E. coli, Klebsiella, streptococci, staphylococci, and anaerobes, spreading from the biliary tract or through the portal vein, and half of solitary abscesses lie in the right lobe. Amebic abscesses follow Entamoeba histolytica infection from contaminated food or water, most often in men

aged 18–50 who live in or travel to endemic areas. Hydatid cysts come from Echinococcus granulosus eggs shed by dogs, whose intermediate hosts are sheep and other livestock, and grow slowly, forming daughter cysts. Schistosoma mansoni and japonicum, acquired from freshwater, send eggs to the liver, where granulomas lead to Symmers pipe-stem periportal fibrosis, portal hypertension, splenomegaly, and varices.

Clinical presentation and complications

Acute viral hepatitis passes through an asymptomatic incubation phase; a prodromal phase with flu-like illness, RUQ pain, arthralgia, and increased ALT more than AST; an icteric phase with jaundice, dark urine, pale stools, and increased bilirubin; and a convalescent phase resolving over weeks to months. Many infections are anicteric, especially HAV in young children and acute HCV. Outcomes are recovery (HAV, HEV, most adult HBV), fulminant hepatitis (HBV, HDV, HEV in pregnancy), and chronic infection (HBV, HCV, HDV), in which ongoing inflammation causes fibrosis, cirrhosis, HCC, and liver failure; in chronic HBV, DNA integrates into the host genome and causes HCC even without cirrhosis. Extrahepatic manifestations, mostly from immune complexes (HBV) and cryoglobulins (HCV), are arthritis and arthralgia (serum sickness-like in HBV); polyarteritis nodosa (HBV) and mixed cryoglobulinemia vasculitis (HCV); rare myocarditis and pericarditis; glomerulonephritis, including membranous nephropathy (HBV) and MPGN (HCV and HBV); thrombocytopenia, hemolytic anemia, and neutropenia; B-cell non-Hodgkin lymphoma (HCV); and porphyria cutanea tarda and lichen planus (HCV).

Diagnosis

IgM means current or recent infection, and IgG means past infection or immunity. Acute HAV shows IgM anti-HAV, and IgG anti-HAV is also positive after vaccination. Acute HDV shows IgM anti-HDV or HDV RNA with positive HBsAg, and past infection shows IgG anti-HDV; IgM anti-HBc is positive in coinfection and negative in superinfection. Acute HEV shows IgM anti-HEV, or HEV RNA if immunocompromised, and past infection shows IgG anti-HEV; it should be considered in pregnant patients with acute hepatitis. For HBV, acute infection shows positive HBsAg, HBeAg, and HBV DNA, with IgM anti-HBc. The window period shows negative HBsAg, HBeAg, HBV DNA, and anti-HBs, anti-HBe that may be positive, and IgM anti-HBc. Recovery shows positive anti-HBs and anti-HBe, with IgG anti-HBc. Immunized patients have only anti-HBs. Chronic, replicative infection (high infectivity) shows positive HBsAg, HBeAg, and HBV DNA with IgG anti-HBc, and chronic, non-replicative infection (low infectivity) shows positive HBsAg and anti-HBe, low or undetectable HBV DNA, and IgG anti-HBc. HBsAg for 6 months or more means chronic infection. HCV is screened with anti-HCV antibody; if positive, HCV RNA (PCR) checks for active virus. RNA-negative patients have resolved past infection or a false-positive antibody, and RNA-positive patients have current infection and need fibrosis assessed before treatment using APRI (above 1.0 suggests cirrhosis), FIB-4, or elastography. CMV is evaluated by CMV PCR (blood), with "owl's eye" inclusions on biopsy; HSV by HSV PCR (blood); and EBV by heterophile antibody (Monospot) and EBV serology. A pyogenic abscess is imaged with ultrasound and then contrast CT, with guided aspiration for stains and cultures; an amebic abscess is diagnosed by imaging and serology, and aspiration, if done, yields odorless, brown anchovy paste. Hydatid cysts are staged by ultrasound, with a detached membrane, the water lily sign, or daughter cysts, often with eosinophilia, and schistosomiasis by eggs or antigen in stool and periportal fibrosis on imaging.

Management

HAV is treated supportively, with vaccine or immunoglobulin post-exposure. Acute HBV is treated supportively, with antivirals if severe; chronic HBV is treated with tenofovir or entecavir, or peginterferon-α; and post-exposure and newborns of HBsAg-positive mothers receive vaccine and HBIG. HCV is treated with direct-acting antivirals for 8–12 weeks (e.g. sofosbuvir-based), with cure over 95%. HDV is treated with peginterferon-α and treatment of the HBV. HEV is treated supportively, with ribavirin if chronic. CMV is treated with ganciclovir if immunocompromised, HSV with IV acyclovir started as soon as HSV hepatitis is suspected, and EBV supportively, avoiding contact sports while the spleen is enlarged. HCC screening with ultrasound every 6 months applies in chronic HBV, or HCV with cirrhosis. Pyogenic abscesses are drained and treated with antibiotics; amebic abscesses with metronidazole followed by a luminal agent such as paromomycin, with drainage only in select cases; hydatid cysts with albendazole, adding PAIR or surgery for larger cysts while avoiding spillage, which can cause anaphylaxis; and schistosomiasis with praziquantel.

Relationships to other conditions

Ischemic hepatitis. Viral hepatitis rises and falls more slowly, and a very high LDH favors ischemia.

AFLP. HEV and HSV both cause severe hepatitis in the third trimester, so viral causes are considered alongside AFLP.

Autoimmune liver disease. Autoimmune hepatitis also has a hepatocellular pattern, but viral serologies are negative and ANA, ASMA, and IgG are raised.

Esophagitis. CMV and HSV also cause esophagitis in immunocompromised patients.

Traumatic liver injury: hemodynamic stability as the decision point

Blunt or penetrating force causes liver lacerations and hematomas, leading to bleeding. Management turns on hemodynamic stability rather than the mechanism alone. Here, we review its causes, complications, and management, and consider how it relates to other causes of RUQ pain.

Key points

- Hemodynamic stability determines the next test and treatment.
- Unstable patients have a FAST exam, and stable patients have CT abdomen with IV contrast.
- Most stable liver injuries heal without surgery.

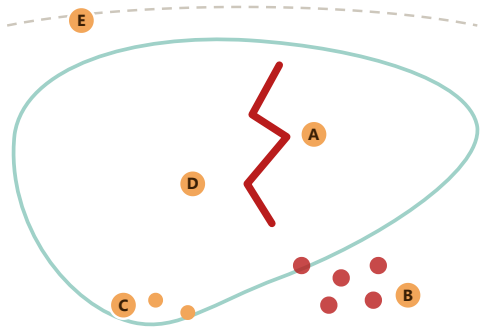


Fig. 1 | A liver with a jagged laceration, blood escaping into the abdomen, bile leaking, blood entering a bile duct, and irritation of the diaphragm above. A, Laceration; B, Hemorrhage; C, Bile leak; D, Hemobilia; E, Diaphragm irritation.

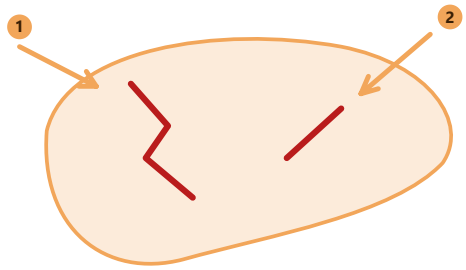


Fig. 2 | Liver with a jagged laceration from blunt force and a straight penetrating wound. 1, Blunt trauma; 2, Penetrating trauma.

Pathophysiology

Blunt or penetrating force tears liver tissue, causing RUQ pain. Torn vessels bleed, causing hemoperitoneum and hypovolemic shock, and torn ducts leak, causing biloma and bile peritonitis. Blood that enters the bile ducts causes hemobilia, with RUQ pain, jaundice, and GI bleeding, and blood under the diaphragm causes right shoulder pain.

Etiology

Blunt trauma follows motor vehicle collisions, falls, and assault. Penetrating trauma follows stab wounds and gunshot wounds.

Clinical presentation and complications

Symptoms are RUQ pain, right shoulder pain from diaphragm irritation, abdominal distention, and signs of shock. Complications are hemorrhage (hemoperitoneum, hypovolemic shock), bile leak (biloma, bile peritonitis), hemobilia (RUQ pain, jaundice, GI bleeding), and liver abscess.

Diagnosis

Unstable patients have a FAST exam to look for free fluid. Stable patients have CT abdomen with IV contrast to grade the injury. Injuries are graded on the AAST scale from I to VI, and grades I and II make up over half.

Management

Hemodynamically stable patients have nonoperative management with close monitoring, with angioembolization if actively bleeding. Unstable patients with a positive FAST have emergency laparotomy. Most stable liver injuries heal without surgery.

Relationships to other conditions

Biliary disease. Hemobilia can resemble biliary disease, with RUQ pain and jaundice, but GI bleeding after liver trauma points to it.

Ischemic hepatitis. Hypovolemic shock from hemorrhage can itself cause ischemic liver injury.

Autoimmune liver disease: bile duct injury in PBC and PSC, and autoimmune hepatitis

Immune-mediated bile duct damage leads to cholestasis, and retained bile salts cause hepatocellular injury. PBC destroys small intrahepatic ducts, whereas PSC, which is not strictly autoimmune, scars medium and large ducts. Autoimmune hepatitis and IgG4-related sclerosing cholangitis complete the group. Here, we review their mechanisms, cholestatic consequences, and treatment, and consider how they relate to IBD and cholangiocarcinoma.

Key points

- PBC: middle-aged women, AMA-positive, small intrahepatic ducts.
- PSC: young to middle-aged men, p-ANCA, beaded ducts on MRCP, and most patients have IBD.
- A cholestatic pattern is first evaluated with RUQ ultrasound, then serology.
- PSC needs surveillance for cholangiocarcinoma, gallbladder, and colorectal cancer.

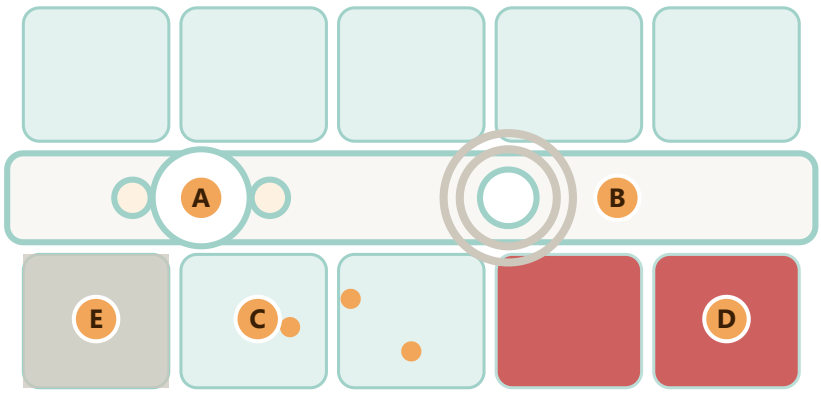


Fig. 1 | Hepatocytes around a bile duct under attack by immune cells, a second duct ringed by fibrosis, bile salts backing up, and injured hepatocytes. A, Duct destruction; B, Duct fibrosis; C, Cholestasis; D, Hepatocellular injury; E, Cirrhosis.

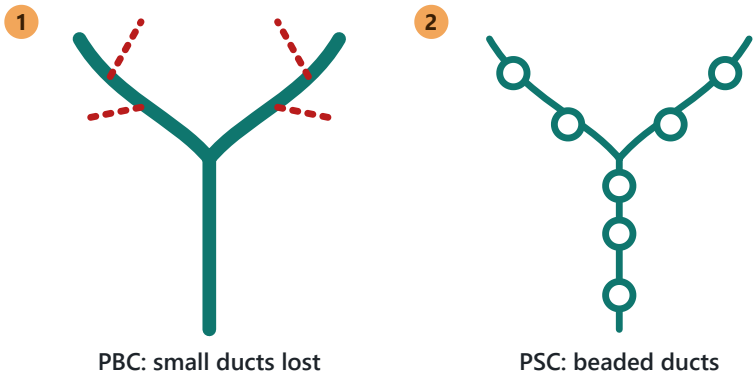


Fig. 2 | Two bile duct trees: PBC with small intrahepatic ducts lost, and PSC with beaded medium and large ducts. 1, Primary biliary cholangitis; 2, Primary sclerosing cholangitis.

Pathophysiology

T cells destroy small intrahepatic bile ducts in PBC, whereas onion-skin fibrosis causes strictures of medium and large ducts in PSC. In both, bile builds up, increasing bile salts in blood, with pruritus, and conjugated bilirubin, with jaundice. Retained bile salts then damage hepatocytes, and ongoing injury leads to fibrosis and cirrhosis.

Etiology and mechanisms

In PBC, AMA is positive in over 90%; T cells destroy small intrahepatic bile ducts, with granulomas around the ducts ("florid duct lesion"), causing loss of bile ducts (ductopenia) and cholestasis. It affects middle-aged women and is associated with other autoimmune diseases: scleroderma (CREST), Hashimoto thyroiditis, rheumatoid arthritis, and Sjögren syndrome. Cancer risk is HCC once cirrhotic. In PSC, p-ANCA is positive in many patients; inflammation and fibrosis of medium and large, intra- and extrahepatic bile ducts, with "onion-skin" fibrosis, cause strictures alternating with dilations ("beaded" ducts on MRCP) and cholestasis. It affects young to middle-aged men and is associated with other p-ANCA-associated diseases: ulcerative colitis, glomerulonephritis, and vasculitis. Cancer risks are HCC, cholangiocarcinoma, gallbladder, and colorectal cancer. Autoimmune hepatitis type 1 has ANA and ASMA, in young women with high IgG; type 2 has anti-LKM1, mostly in children. IgG4-related sclerosing cholangitis mimics PSC and is steroid-responsive.

Clinical presentation and complications

Symptoms are fatigue, pruritus (often the first symptom), jaundice, RUQ discomfort, and xanthomas. Duct destruction or stricture causes cholestasis, with increased bile salts in blood (pruritus), increased conjugated bilirubin (jaundice, dark urine), and increased cholesterol (xanthomas). Decreased bile reaching the gut causes fat malabsorption, with steatorrhea, pale stools, fat-soluble vitamin deficiencies (A, D, E, K), and osteoporosis. Toxic bile salts retained in the liver cause hepatocellular injury, fibrosis, biliary cirrhosis, HCC, portal hypertension, and liver failure. In PSC only, chronic bile duct inflammation causes cholangiocarcinoma and recurrent bacterial cholangitis, as bile stagnates behind strictures.

Diagnosis

The cholestatic LFT pattern is markedly increased ALP, increased GGT, increased conjugated bilirubin later, and mildly increased AST/ALT. RUQ ultrasound first rules out extrahepatic obstruction such as gallstones or a mass. Serology follows: AMA, p-ANCA, ANA, ASMA, anti-LKM1, and IgG4. A positive AMA means PBC, with biopsy only if AMA-negative or unclear. p-ANCA or IBD prompts MRCP, in which "beaded" bile ducts mean PSC.

Management

UDCA decreases liver injury and is first line in PBC. Cholestyramine, a bile acid sequestrant, relieves pruritus. Deficiencies are corrected with fat-soluble vitamins (A, D, E, K), and calcium and vitamin D for bone health. A dominant stricture in PSC is opened by ERCP with balloon dilation, and end-stage disease is treated by liver transplant. Autoimmune hepatitis is treated with corticosteroids with or without azathioprine, and IgG4-related disease with corticosteroids. PSC needs surveillance for cholangiocarcinoma, gallbladder, and colorectal cancer. In PBC that does not respond to UDCA, obeticholic acid, a farnesoid X receptor agonist, is added.

Relationships to other conditions

IBD. Most PSC patients have IBD, usually ulcerative colitis, so PSC prompts colonoscopy.

Cholangiocarcinoma. A new dominant stricture with rising CA 19-9 in PSC suggests cholangiocarcinoma. This risk is specific to PSC, not PBC.

Malabsorption. Bile salt deficiency causes fat malabsorption here as it does in other cholestatic disease.

Choledocholithiasis. Both give a cholestatic LFT pattern, but ultrasound finds dilated ducts with stones and non-dilated ducts in PBC.

Autoimmune pancreatitis. IgG4-related sclerosing cholangitis can occur with autoimmune pancreatitis.

Metabolic liver disease: iron, copper, misfolded protein, and fat

Buildup of iron, copper, misfolded protein, or fat in hepatocytes causes cellular stress, leading to hepatocellular injury, fibrosis, and cirrhosis. Hereditary hemochromatosis, Wilson disease, A1AT deficiency, and MASLD share this path to HCC but differ widely beyond the liver. Here, we review their inheritance, mechanisms, multisystem features, and treatment.

Key points

- All four can progress through hepatitis and fibrosis to cirrhosis and HCC.
- Hemochromatosis: decreased hepcidin and increased iron absorption.
- Wilson disease: *ATP7B* mutation, low ceruloplasmin, and Kayser-Fleischer rings.
- A1AT deficiency: protein builds up in the liver and is missing from the lungs.
- MASLD: cardiovascular disease is the leading cause of death.

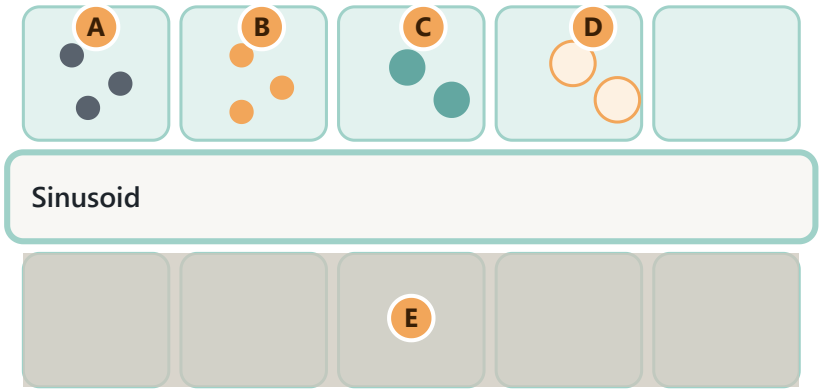


Fig. 1 | Hepatocytes containing iron, copper, trapped protein, and fat, with scar forming at the end of the row. A, Iron; B, Copper; C, Misfolded protein; D, Fat; E, Fibrosis.

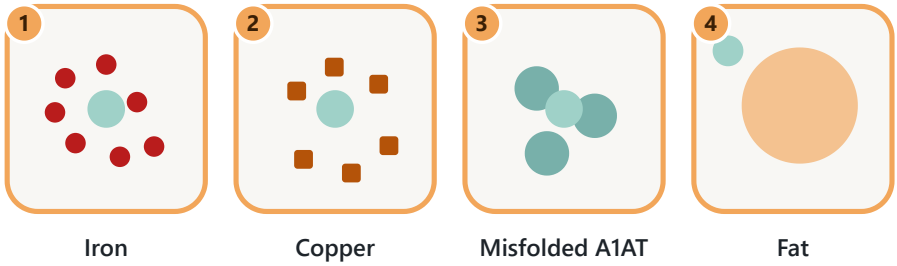


Fig. 2 | Four hepatocytes filled with iron, copper, misfolded protein, and fat. 1, Hereditary hemochromatosis; 2, Wilson disease; 3, Alpha-1 antitrypsin (A1AT) deficiency; 4, MASLD.

Pathophysiology

An inherited or acquired defect causes buildup in hepatocytes. In hemochromatosis, decreased hepcidin increases ferroportin activity, and iron causes free radical damage. In Wilson disease, an ATP7B mutation decreases copper excretion into bile, so copper builds up in the liver. In A1AT deficiency, the misfolded protein polymerizes and is trapped in the hepatocyte ER, causing hepatocellular injury. In MASLD, triglycerides accumulate, causing steatosis. Repeated cellular stress leads to fibrosis and cirrhosis.

Hereditary hemochromatosis

Inheritance is autosomal recessive, except ferroportin disease, which is autosomal dominant. Type 1 (*HFE*, C282Y homozygosity) is most common; type 2A (*HJV*, hemojuvelin) and type 2B (*HAMP*, hepcidin) are juvenile, severe, and early in onset; type 3 (*TFR2*, transferrin receptor 2) is rare; and type 4 (*SLC40A1*, FPN, ferroportin) resists hepcidin or fails to export iron. *HFE*, *HJV*, *HAMP*, and *TFR2* mutations decrease hepcidin, and *FPN* mutation makes ferroportin resist hepcidin. Increased ferroportin activity increases duodenal iron absorption; iron deposits in the liver, heart, pancreas, and skin, causing free radical damage by the Fenton reaction. Hepcidin normally blocks ferroportin, the only cellular iron exporter.

Wilson disease

Inheritance is autosomal recessive. *ATP7B* mutation on chromosome 13 decreases copper excretion into bile and loading onto ceruloplasmin; copper builds up in the liver, and free copper in the blood reaches the brain, cornea, kidneys, and RBCs.

A1AT deficiency

Inheritance is autosomal codominant. *SERPINA1* mutation (PiZZ genotype) makes misfolded A1AT polymerize. In the liver, it is trapped in the hepatocyte ER (PAS-positive globules), causing hepatocellular injury and cirrhosis. In the lungs, decreased A1AT in blood leaves neutrophil elastase unopposed, causing alveolar wall destruction.

MASLD

MASLD is acquired, and is steatosis plus at least one cardiometabolic risk factor, without significant alcohol use (formerly NAFLD). Metabolic syndrome is the main risk factor: central obesity, T2DM, insulin resistance, dyslipidemia (increased TG, decreased HDL), and hypertension. Increased fatty acids reaching the liver and increased fat production cause triglycerides to build up in hepatocytes (steatosis), then lipotoxicity, oxidative stress, and MASH (steatohepatitis).

Clinical presentation and complications

All four can progress from hepatotoxicity to hepatomegaly and hepatitis, fibrosis, cirrhosis, and HCC, with risk especially high in hemochromatosis. Wilson disease can present as acute liver failure with Coombs-negative hemolytic anemia. Beyond the liver, hemochromatosis causes restrictive cardiomyopathy (which can become dilated) and arrhythmias, diabetes mellitus from pancreatic iron, hypogonadism from pituitary iron, hyperpigmentation ("bronze diabetes"), arthropathy of the 2nd and 3rd MCP joints, and infections (*Vibrio vulnificus*, *Yersinia*). Wilson disease causes Kayser-Fleischer rings, sunflower cataract, basal ganglia (putamen) disease with tremor, parkinsonism, and psychiatric symptoms, Fanconi syndrome, and Coombs-negative hemolytic anemia. A1AT deficiency causes panniculitis (rare), panacinar emphysema (lower lobes, worse with smoking), and neonatal cholestasis. MASLD causes cardiovascular disease, the leading cause of death, T2DM, CKD, and extrahepatic cancers (e.g. colorectal).

Diagnosis

Hemochromatosis is evaluated first by increased transferrin saturation (over 45%), then increased ferritin, *HFE* genetic testing (C282Y), and liver MRI or biopsy (Prussian blue) if unclear. Wilson disease is evaluated by decreased serum ceruloplasmin, increased 24-hour urine copper, slit-lamp exam for Kayser-Fleischer rings, and liver biopsy (hepatic copper) or *ATP7B* testing if unclear. A1AT deficiency is evaluated by decreased serum A1AT level, phenotyping or genotyping (PiZZ), liver biopsy showing PAS-positive, diastase-resistant globules, and pulmonary function tests and chest CT for emphysema. MASLD shows mildly increased ALT more than AST (AST:ALT over 2:1 suggests alcohol) and a bright (echogenic) liver on ultrasound; fibrosis is assessed by FIB-4, then elastography, after excluding alcohol, viral hepatitis, and other causes.

Management

Hemochromatosis is treated by therapeutic phlebotomy first, with iron chelation (deferoxamine, deferasirox) if phlebotomy is not possible, and by avoiding iron, vitamin C, alcohol, and raw shellfish. Wilson disease is treated by copper chelation with D-penicillamine or trientine, zinc for maintenance, which blocks intestinal copper absorption, and a low-copper diet that avoids shellfish, liver, nuts, and chocolate. A1AT deficiency is treated by stopping smoking and IV A1AT augmentation therapy for emphysema. MASLD is treated by 7–10% weight loss through diet and exercise first, treating metabolic risk factors (diabetes, lipids, blood pressure), GLP-1 receptor agonists or resmetirom for MASH with fibrosis, and bariatric surgery for severe obesity. Liver transplant treats end-stage disease, and cures the defect in Wilson disease and A1AT deficiency.

Relationships to other conditions

Alcoholic liver disease. Both MASLD and alcohol cause steatosis, but ALT exceeds AST in MASLD and AST:ALT is over 2:1 in alcohol.

Acute liver failure. Wilson disease is a cause of acute liver failure, and all four feed into cirrhosis.

Cholestatic liver disease. A1AT deficiency can cause neonatal cholestasis, and cholestasis itself is the

hallmark of PBC and PSC.

Movement disorders. Wilson disease causes parkinsonism and tremor in young patients.

Iatrogenic and toxic liver injury: postoperative, paracetamol, and alcohol

Drugs, toxins, and complications of medical care injure hepatocytes through direct toxicity or immune reactions. Postoperative causes, paracetamol overdose, alcoholic liver disease, and other hepatotoxic drugs each follow a distinct path. Here, we review their mechanisms, staged presentations, and treatment, and consider how they relate to metabolic and vascular liver disease.

Key points

- The toxic paracetamol dose is over 150 mg/kg, and risk is higher with chronic alcohol use and malnutrition.
- Always check a paracetamol level after an overdose, even if the patient looks well.
- NAC replenishes glutathione and works best within 8 hours.
- Alcoholic liver disease has AST:ALT over 2:1, with AST usually under 300 U/L.

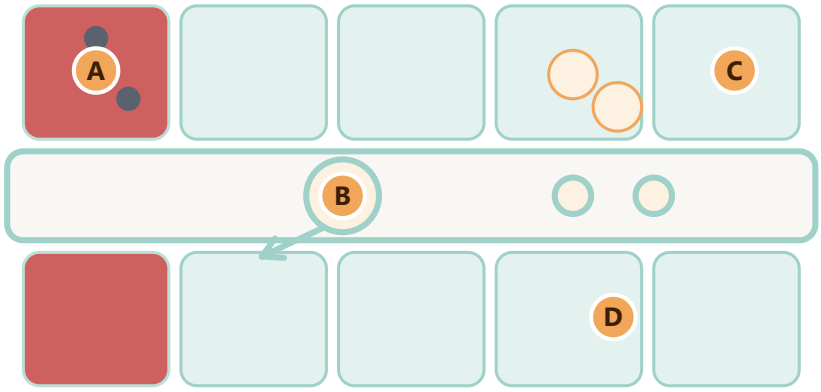


Fig. 1 | Hepatocytes with a toxin damaging one, an immune cell attacking another, a fat-filled hepatocyte, and inflammation from altered proteins. A, Direct toxicity; B, Immune reaction; C, Fatty liver; D, Acetaldehyde adducts.

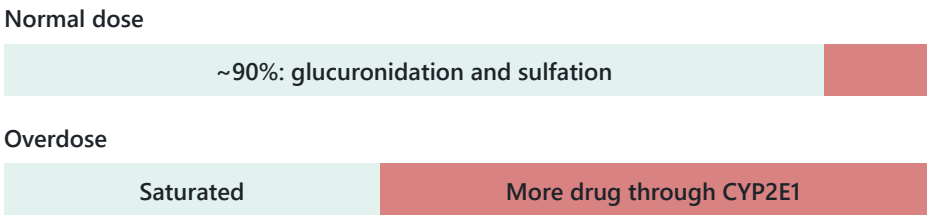


Fig. 2 | Paracetamol metabolism: at a normal dose about 90% is safely conjugated; in overdose conjugation saturates and more becomes toxic NAPQI.

Pathophysiology

Drugs, toxins, and medical care injure hepatocytes directly or through immune reactions. In paracetamol overdose, increased NAPQI uses up glutathione, causing zone 3 necrosis. In halothane hepatitis, altered liver proteins provoke an immune reaction after repeat exposure. In alcohol use, an increased NADH:NAD+ ratio increases fatty acid synthesis, causing steatosis, and acetaldehyde adducts act as new antigens, causing immune activation and alcoholic hepatitis.

Postoperative complications

Intraoperative hypotension causes ischemic hepatitis. Inhaled anesthetics cause halothane hepatitis, which is immune-mediated and follows repeat exposure. Bile duct injury (e.g. cholecystectomy) causes bile leak and bile

duct stricture. Blood transfusions and hematoma resorption cause an increased bilirubin load and jaundice. Sepsis and total parenteral nutrition cause cholestasis.

Paracetamol overdose

At a normal dose, ~90% undergoes glucuronidation and sulfation and is safely excreted, and ~5–10% is converted by CYP2E1 to the toxic metabolite NAPQI, which glutathione neutralizes, so there is no liver injury. In overdose, glucuronidation and sulfation are saturated, more drug goes through CYP2E1, glutathione is used up, and NAPQI binds hepatocyte proteins, causing zone 3 (centrilobular) necrosis. Toxicity is staged: at 0–24 hours, nausea, vomiting, or no symptoms; at 24–72 hours, RUQ pain and rising AST/ALT; at 72–96 hours, peak AST/ALT and liver failure; and at 4 days–2 weeks, recovery or death.

Alcoholic liver disease

Three pathways lead to the same result: ADH (cytosol, main pathway), CYP2E1 (induced by chronic drinking), and catalase (peroxisomes, minor). They produce acetaldehyde with an increased NADH:NAD+ ratio, followed by acetate via aldehyde dehydrogenase. The increased NADH:NAD+ ratio increases fatty acid synthesis and decreases fatty acid breakdown, causing fatty liver (steatosis), which is reversible with abstinence. Acetaldehyde binds proteins, and these acetaldehyde adducts act as new antigens, causing immune activation and free radical damage, leading to alcoholic hepatitis (neutrophils, Mallory-Denk bodies), fibrosis, and cirrhosis. The spectrum runs from fatty liver (usually no symptoms; hepatomegaly) through alcoholic hepatitis (fever, jaundice, tender hepatomegaly) to cirrhosis.

Other hepatotoxic drugs

These include isoniazid, amiodarone, methotrexate, valproate, amoxicillin-clavulanate (cholestatic), and statins (rare). Aspirin given to children during a viral illness, most often influenza or varicella, causes Reye syndrome, an acute encephalopathy with fatty liver failure that begins with vomiting and confusion.

Diagnosis

Paracetamol overdose is evaluated with a 4-hour paracetamol level on the Rumack-Matthew nomogram, with LFTs, INR, creatinine, and glucose. Alcoholic liver disease shows AST:ALT over 2:1, with AST usually under 300 U/L, increased GGT, and macrocytosis, with an alcohol history. Drug-induced liver injury rests on the timing of the drug, exclusion of other causes, and improvement after stopping it.

Management

Paracetamol overdose is treated with the antidote NAC, which replenishes glutathione and is best within 8 hours, activated charcoal if within 1–2 hours of ingestion, and transplant evaluation if acute liver failure develops. Alcoholic liver disease is treated by abstinence, the most important measure, nutrition and thiamine, and prednisolone for severe alcoholic hepatitis if the Maddrey score is 32 or more. Postoperative and drug-induced injury is treated by stopping the offending drug, supportive care, and treating the underlying complication.

Relationships to other conditions

Ischemic hepatitis. Paracetamol and hypoperfusion both injure zone 3, and intraoperative hypotension links this page to vascular injury.

MASLD. Steatosis is shared, and the AST:ALT ratio and alcohol history separate them.

Acute liver failure and cirrhosis. Paracetamol is a leading cause of acute liver failure, and alcohol of cirrhosis.

Pancreatitis. Ethanol and valproate cause pancreatitis as well as liver injury.

Liver tumors: distinguishing benign lesions, primary cancers, and metastases

Benign or malignant tumors replace and compress normal liver tissue, and metastases are the most common liver malignancy. Hemangioma, hepatic adenoma, and focal nodular hyperplasia are benign, whereas HCC, cholangiocarcinoma, and angiosarcoma are primary malignancies. Here, we review their features, complications, imaging, and treatment, and consider how they relate to cirrhosis and PSC.

Key points

- Metastases are the most common liver malignancy, and HCC the most common primary liver cancer.
- Do not biopsy a hemangioma, because of bleeding risk.
- In cirrhosis, arterial-phase enhancement with washout diagnoses HCC without biopsy.
- Hepatic adenoma can rupture, so resect if large (over 5 cm) or bleeding.

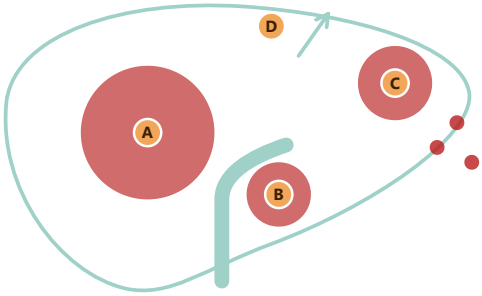


Fig. 1 | A liver with a large tumor replacing tissue, a tumor pressing on a bile duct, a surface tumor that has ruptured, and arrows leaving a tumor to show hormones in the blood. A, Tissue replacement; B, Compression; C, Rupture; D, Paraneoplastic effects.

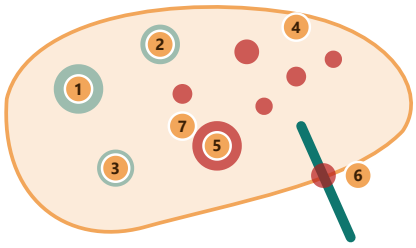


Fig. 2 | Liver containing benign and malignant lesions. Positions are illustrative. 1, Hemangioma; 2, Hepatic adenoma; 3, Focal nodular hyperplasia; 4, Metastases; 5, HCC; 6, Cholangiocarcinoma; 7, Angiosarcoma.

Pathophysiology

Tumors replace normal liver, which can cause liver failure, and press on bile ducts, causing obstructive jaundice. Hepatic adenoma and HCC can rupture, causing hemoperitoneum, and HCC can cause polycythemia, hypoglycemia, and hypercalcemia as paraneoplastic effects.

Etiology

Hemangioma is the most common benign liver tumor and should not be biopsied, because of bleeding risk. Hepatic adenoma occurs in young women on oral contraceptives and anabolic steroid users, and can rupture and bleed. Focal nodular hyperplasia has a central stellate scar and no malignant potential. Metastases are the most common liver malignancy, from the colon, breast, lung, and pancreas. HCC is the most common primary liver cancer, associated with cirrhosis, HBV, HCV, and aflatoxin. Cholangiocarcinoma is bile duct cancer, associated with PSC and liver flukes. Angiosarcoma is rare, associated with vinyl chloride, arsenic, and Thorotrast. A fibrolamellar variant of HCC affects young patients without cirrhosis.

Clinical presentation and complications

Tumors are often found incidentally, or cause RUQ pain or mass, weight loss, and worsening of known cirrhosis (HCC). Complications are rupture and hemoperitoneum (hepatic adenoma, HCC), paraneoplastic syndromes (HCC: polycythemia, hypoglycemia, hypercalcemia), obstructive jaundice (cholangiocarcinoma), and liver failure.

Diagnosis

Any liver mass is evaluated with multiphase contrast CT or MRI. HCC shows arterial-phase enhancement with washout, with no biopsy needed in cirrhosis, and increased AFP. For metastases, the primary tumor is sought (e.g. colonoscopy, CEA). Unclear imaging is biopsied, but not for hemangioma. Hemangiomas show peripheral, nodular arterial enhancement that fills in toward the center on later phases, and focal nodular hyperplasia, mostly in women aged 20–50, has a central scar. Patients with cirrhosis or chronic HBV are surveyed for HCC with ultrasound, with or without AFP, every 6 months.

Management

Hemangioma and FNH usually need no treatment. Hepatic adenoma is treated by stopping oral contraceptives or steroids, with resection if large (over 5 cm) or bleeding. HCC is treated by resection, transplant (Milan criteria), or ablation if early, and TACE or systemic therapy if advanced. Metastases and cholangiocarcinoma depend on the primary tumor and stage, with resection if possible. Adenomas are resected in all men, and in women if over 5 cm or bleeding, because of the risks of rupture and malignant transformation.

Relationships to other conditions

Cirrhosis. Cirrhosis is the main setting for HCC, and HCC screening every 6 months with ultrasound, with or without AFP, is part of its care.

Viral hepatitis. HBV can cause HCC even without cirrhosis.

PSC. PSC is a major risk factor for cholangiocarcinoma.

Budd-Chiari syndrome. HCC is among the malignancies that can cause hepatic vein thrombosis.

Acute fatty liver of pregnancy: fetal LCHAD deficiency and maternal liver failure

Fetal LCHAD deficiency leads to fatty acid buildup in the maternal liver, causing microvesicular steatosis and acute liver failure in the third trimester. Hypoglycemia and coagulopathy distinguish it from HELLP syndrome. Prompt delivery is the only cure. Here, we review its mechanism, recognition, and management.

Key points

- AFLP is a third-trimester, fetal-driven cause of maternal liver failure.
- Hypoglycemia and coagulopathy are common in AFLP and uncommon in HELLP.
- Prompt delivery is the only cure.
- The newborn is tested for LCHAD deficiency.

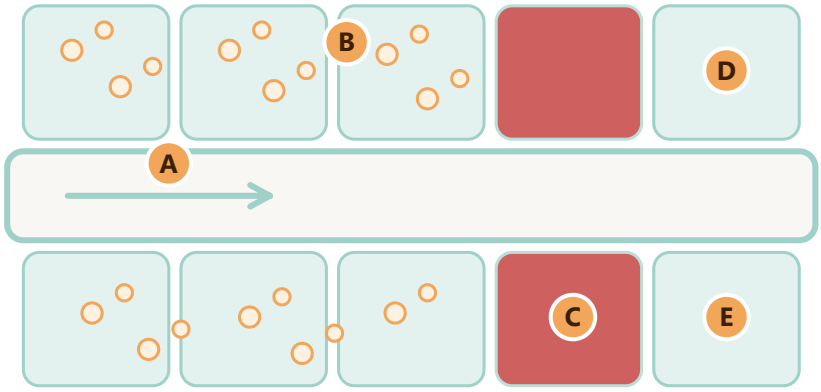


Fig. 1 | A fetus-derived supply of fatty acids reaching maternal hepatocytes, which are filled with tiny fat droplets and failing. A, Fatty acid load; B, Microvesicular steatosis; C, Hypoglycemia; D, Coagulopathy; E, Encephalopathy.

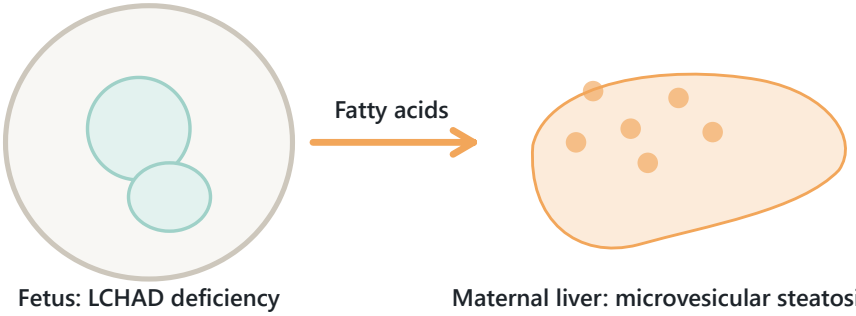


Fig. 2 | Fatty acids passing from the fetus to the mother's liver.

Pathophysiology

In fetal LCHAD deficiency, fetal fatty acids reach the maternal liver and overwhelm its mitochondria, leaving tiny fat droplets in hepatocytes as microvesicular steatosis. Failing hepatocytes cause hypoglycemia, decreased clotting factor synthesis causes increased INR and DIC, and increased ammonia causes encephalopathy, as acute liver failure develops.

Etiology and mechanisms

Fetal LCHAD deficiency, with the mother as a carrier, allows fetal fatty acids to reach the mother. Maternal liver mitochondria are overwhelmed, causing

microvesicular steatosis and acute liver failure. Risk factors are the third trimester (usually 30–38 weeks), first pregnancy, twin pregnancy, male fetus, and pre-eclampsia.

Clinical presentation and complications

Symptoms are nausea and vomiting, abdominal pain, malaise, jaundice, polyuria, and polydipsia. Complications are acute liver failure (hypoglycemia, coagulopathy with DIC, encephalopathy), acute kidney injury, and fetal distress and death.

Diagnosis

Key findings are increased bilirubin, increased AST/ALT (usually under 1000 U/L), decreased glucose, increased INR, decreased fibrinogen, increased ammonia, uric acid, and creatinine, and leukocytosis. Diagnosis is clinical (Swansea criteria), and biopsy is rarely needed.

Management

Prompt delivery is the only cure. Support comprises IV glucose, blood products (FFP, cryoprecipitate), and ICU care. Follow-up includes testing the newborn for LCHAD deficiency and counseling about recurrence.

Relationships to other conditions

HELLP syndrome. Hypoglycemia is common in AFLP and rare in HELLP; coagulopathy is common in AFLP and uncommon in HELLP; hemolysis is uncommon in AFLP and present in HELLP; platelets are normal or mildly low in AFLP and low in HELLP; and hypertension is sometimes present in AFLP and usually present in HELLP. Both are treated with delivery.

Viral hepatitis in pregnancy. HEV and HSV can also cause fulminant hepatitis in the third trimester, and HSV requires empiric IV acyclovir.

Acute liver failure. AFLP is a recognized cause, and shares its big 4 complications.

Acute liver failure and cirrhosis: sudden and chronic loss of hepatic function

Acute liver failure is a sudden loss of function in a previously healthy liver, whereas cirrhosis is chronic scarring in which repeated injury causes fibrosis and regenerating nodules. Their consequences reflect lost liver function and, in cirrhosis, portal hypertension. Child-Pugh scoring grades cirrhosis severity. Here, we review their causes, complications, workup, and treatment, and consider how every cause of liver injury converges on them.

Key points

- Acute liver failure is encephalopathy and INR 1.5 or more within 26 weeks in a previously healthy liver.
- Its big 4 are encephalopathy, coagulopathy, hypoglycemia, and infection.
- Cirrhosis has two arms: portal hypertension and loss of liver function.
- Child-Pugh rows spell ABCDE: albumin, bilirubin, coagulation, distention (ascites), and encephalopathy.

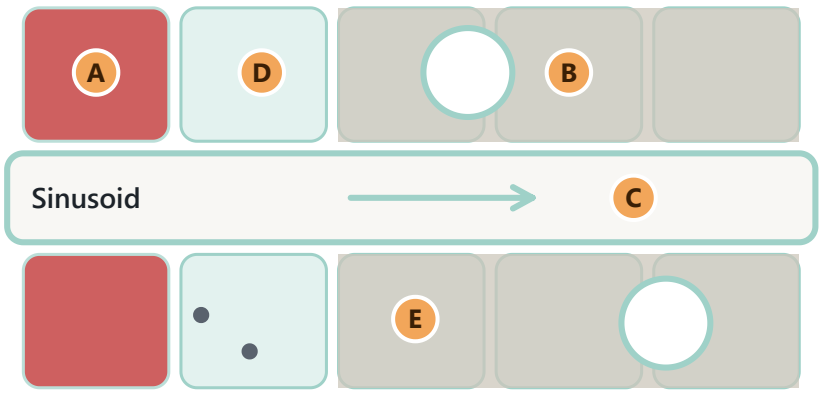


Fig. 1 | Dead hepatocytes on one side, and fibrous bands around regenerating nodules on the other, with pressure rising in the sinusoid and fewer proteins leaving. A, Hepatocyte loss; B, Fibrosis and nodules; C, Portal hypertension; D, Lost synthesis; E, Lost clearance.

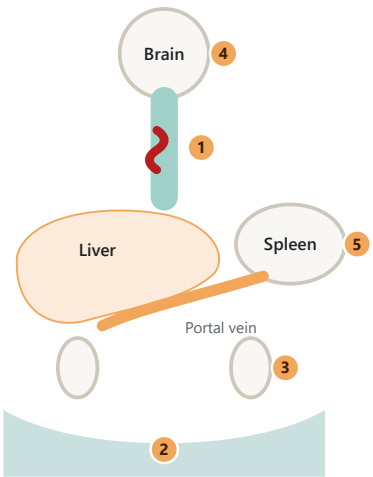


Fig. 2 | Body map of portal hypertension: esophageal varices, ascites, kidneys, brain, and an enlarged spleen. 1, Portosystemic collaterals; 2, Ascites; 3, Hepatorenal syndrome; 4, Hepatic encephalopathy; 5, Splenomegaly.

Pathophysiology

Severe acute injury causes hepatocyte loss in acute liver failure, whereas repeated injury causes fibrosis and regenerating nodules that distort the architecture in cirrhosis. Increased sinusoidal pressure causes portal hypertension, with varices, ascites, and splenomegaly. Decreased albumin and clotting factor synthesis cause ascites, edema, and coagulopathy, and decreased ammonia clearance causes hepatic encephalopathy.

Mechanisms and etiology

Acute liver failure is severe acute injury in a previously healthy liver, with encephalopathy and INR 1.5 or more within 26 weeks; main causes are paracetamol, viral hepatitis, ischemic injury, Wilson disease, and AFLP. In cirrhosis, repeated injury causes fibrosis and regenerating nodules, and distorted architecture raises portal pressure and reduces liver function; main causes are alcohol, MASLD, HBV, HCV, PBC, PSC, and hemochromatosis.

Acute liver failure: the big 4

Encephalopathy causes altered mental status, asterixis, and increased ICP from cerebral edema. Coagulopathy causes increased PT/INR, intracranial hemorrhage, GI bleeding, and mucocutaneous bleeding. Hypoglycemia follows decreased gluconeogenesis and glycogen stores. Infection includes pneumonia, UTI, and sepsis.

Cirrhosis: portal hypertension

Blood diverts into portosystemic collaterals, causing esophageal varices with life-threatening bleeding, caput medusae, and rectal varices. Increased sinusoidal hydrostatic pressure causes transudative ascites (SAAG 1.1 g/dL or more) and SBP (ascitic neutrophils 250/μL or more). Splanchnic vasodilation from nitric oxide decreases effective arterial volume, with RAAS activation and renal vasoconstriction, causing hepatorenal syndrome (functional kidney failure) and sodium and water retention, worsening ascites. Portosystemic shunting lets ammonia bypass the liver, causing astrocyte swelling, cerebral edema, and hepatic encephalopathy. Splenomegaly causes thrombocytopenia as platelets pool in the spleen.

Cirrhosis: loss of liver function

Decreased ammonia clearance causes hepatic encephalopathy. Decreased clotting factor synthesis causes coagulopathy (increased INR) and bleeding. Decreased bilirubin and bile salt excretion causes jaundice and pruritus. Decreased glucose production causes hypoglycemia. Decreased immune function raises infection risk (SBP, pneumonia, sepsis). Decreased vitamin A, D, E, K processing causes fat-soluble vitamin deficiencies (e.g. osteoporosis, bleeding). Damaged hepatocytes release stored B12, increasing serum B12, which is mostly inactive and rises with severity. Decreased estrogen breakdown increases estrogen, causing spider angiomas, palmar erythema, gynecomastia, and testicular atrophy. Decreased albumin synthesis decreases oncotic pressure, causing ascites and edema.

Child-Pugh score

Points of 1, 2, and 3 are given for albumin (over 3.5, 2.8–3.5, under 2.8 g/dL), bilirubin (under 2, 2–3, over 3 mg/dL), INR (under 1.7, 1.7–2.3, over 2.3), ascites (none, mild, moderate to severe), and encephalopathy (none, grade 1–2, grade 3–4). Class A is 5–6 points (1-year survival about 100%), class B is 7–9 points (about 80%), and class C is 10–15 points (about 45%).

Diagnosis

Acute liver failure is evaluated with LFTs (AST/ALT, bilirubin), complications are checked with INR, glucose, ammonia, and cultures, and the cause is sought with paracetamol, viral and autoimmune serology, and Doppler. Cirrhosis is evaluated with bloods (FBC with low platelets, LFTs, INR, albumin) and ultrasound with elastography showing a nodular, stiff liver, with biopsy as the gold standard if unclear. Complications are screened with EGD and diagnostic paracentesis, and HCC screening is every 6 months with ultrasound, with or without AFP. The MELD score, from bilirubin, creatinine, and INR, predicts short-term mortality and ranks patients for transplant. Hepatopulmonary syndrome is hypoxemia from intrapulmonary vasodilation, with dyspnea and platypnea, confirmed by contrast echocardiography and treated by transplant.

Management

Acute liver failure is managed with ICU care, treating the cause (e.g. NAC), managing the big 4, and early transplant referral. Varices are treated with a non-selective beta-blocker or band ligation. Variceal bleeding is treated with octreotide, IV ABX, and endoscopic banding, with TIPS if refractory. Ascites is treated with sodium restriction, spironolactone with furosemide, and large-volume paracentesis with albumin. SBP is treated with a third-generation cephalosporin plus albumin, with long-term prophylaxis afterwards. Hepatic encephalopathy is treated with lactulose, then rifaximin. Hepatorenal syndrome is treated with albumin plus a vasoconstrictor (e.g. terlipressin) and transplant. End-stage disease is treated by liver transplant.

Relationships to other conditions

Toxic injury. Paracetamol leads to acute liver failure, whereas alcohol leads to cirrhosis.

Metabolic disease. Wilson disease can present as acute liver failure, and hemochromatosis and MASLD lead to cirrhosis.

Vascular disease. Budd-Chiari syndrome and congestion also cause portal hypertension, so portal hypertension alone does not mean cirrhosis.

HCC. Cirrhosis is the main setting for HCC.

Cholelithiasis and biliary colic: gallstone formation and its first symptoms

Excess cholesterol or bilirubin in bile, or bile stasis, leads to crystal formation and gallstones. Most stones are asymptomatic, and biliary colic follows when the gallbladder contracts against an impacted stone. Stone location along the biliary tree determines every later condition. Here, we review stone types, risk factors, and management, and consider how stones lead to cholecystitis, choledocholithiasis, cholangitis, and gallstone pancreatitis.

Key points

- Cholesterol stones are ~75% of gallstones.
- Their triad is supersaturation, accelerated nucleation, and gallbladder hypomotility.
- Biliary colic has normal LFTs and no fever; pain over 6 hours or fever suggests cholecystitis.
- Asymptomatic gallstones need no treatment.

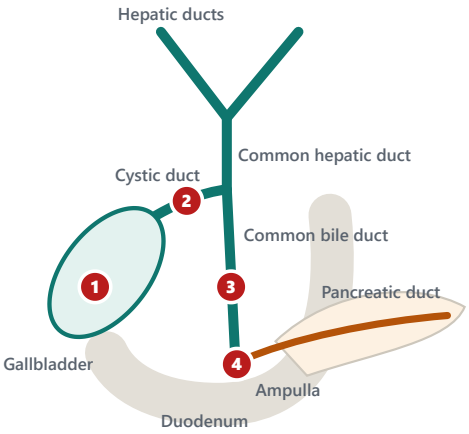


Fig. 1 | The gallbladder and biliary tree.

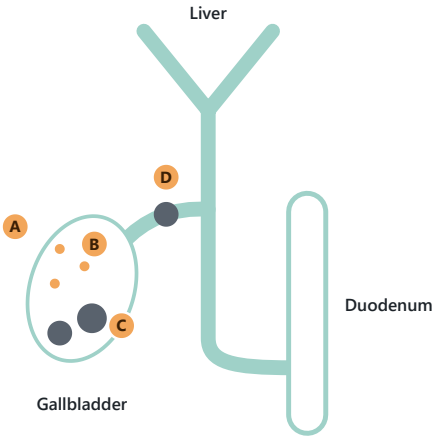


Fig. 2 | A biliary tree with crystals forming in the gallbladder, stones growing in still bile, and a stone blocking the cystic duct. A, Supersaturation; B, Nucleation; C, Hypomotility; D, Cystic duct block.

Pathophysiology

Increased cholesterol, or decreased bile salts or phospholipids, supersaturates bile and forms crystals, excess nucleation factors help them grow, and hypomotility keeps bile in the gallbladder long enough for them to become stones. A stone that briefly blocks the cystic duct causes biliary colic, and a persistent block causes cholecystitis.

Etiology and mechanisms

Cholesterol stones are most common (~75%), arise from cholesterol supersaturation, nucleation, and gallbladder hypomotility, and form in the gallbladder. Black pigment

stones are less common, arise from chronic hemolysis with increased unconjugated bilirubin (e.g. sickle cell, spherocytosis), cirrhosis, or ileal disease such as Crohn disease, and form in the gallbladder as calcium bilirubinate. Brown pigment stones are less common, though more common in East Asia, arise from bile duct infection with bacteria or parasites (*Clonorchis*) or from strictures, and form in the bile ducts. Only ~10% of gallstones show on x-ray, depending on their calcium content. The cholesterol triad comprises supersaturation (increased cholesterol secretion into bile, and decreased bile salts or phospholipids to dissolve it), accelerated nucleation (excess nucleation factors such as mucin, and no nucleation inhibitors to stop crystals forming), and gallbladder hypomotility (stasis, giving crystals time to grow into stones). Risk factors are the 6 Fs: female (estrogen raises cholesterol secretion), fat (obesity), fertile (pregnancy, multiparity), forty (age 40 or older), fair (white or Indigenous American ancestry), and family history. Others are rapid weight loss, bariatric surgery, oral contraceptives, estrogen therapy, fibrates, octreotide, fasting, TPN (hypomotility), Crohn disease, and ileal resection (bile salt loss).

Clinical presentation and complications

Cholelithiasis is mostly asymptomatic and often found by chance on imaging. In biliary colic, a heavy, fatty meal triggers CCK release, the gallbladder contracts against an impacted stone, and epigastric or RUQ pain radiates to the back or right scapula, resolving within 6 hours as the stone dislodges. Stones cause disease by location: in the gallbladder, cholelithiasis; in the cystic duct, biliary colic with a brief block or cholecystitis with a persistent block; in the common bile duct, choledocholithiasis, which can lead to cholangitis; and at the ampulla, gallstone pancreatitis. About 1–2% of people with gallstones develop symptoms each year. Complications are acute cholecystitis from a persistent cystic duct block; chronic cholecystitis from repeated colic, which can calcify the wall as porcelain gallbladder; choledocholithiasis, which can lead to cholangitis; gallstone pancreatitis; gallstone ileus, when a large stone passes through a fistula; and gallbladder carcinoma, whose risk rises with stone size and is highest over 3 cm.

Diagnosis

Gallstones are evaluated by RUQ ultrasound, which is first line and shows mobile, shadowing stones. LFTs and FBC assess complications and are normal in biliary colic. Plain x-ray misses most stones, since only ~10% contain enough calcium to show.

Management

Asymptomatic gallstones need no treatment. Biliary colic is treated with pain relief (e.g. NSAIDs), then elective laparoscopic cholecystectomy. Patients not fit for surgery can have ursodeoxycholic acid, for small cholesterol stones only. Ursodeoxycholic acid dissolves fewer than half of stones.

Relationships to other conditions

Cholecystitis. Both come from a cystic duct stone. A brief block gives colic with normal LFTs and no fever; a persistent block gives cholecystitis, with pain over 6 hours and fever.

Crohn disease. Ileal disease causes gallstones by losing bile salts.

PUD. Epigastric pain overlaps, but colic follows a fatty meal and ultrasound shows stones.

Cholecystitis: persistent cystic duct obstruction and gallbladder inflammation

Persistent cystic duct obstruction leads to gallbladder distention and bacterial colonization, causing inflammation of the gallbladder. Calculous disease accounts for ~90–95%, with Mirizzi syndrome and acalculous cholecystitis as important variants. LFTs are usually normal, so jaundice suggests another process. Here, we review its forms, complications, and management, and consider how it relates to biliary colic and choledocholithiasis.

Key points

- RUQ pain over 6 hours with fever separates cholecystitis from biliary colic.
- RUQ ultrasound is first line, and HIDA scan is used if ultrasound is unclear.
- Jaundice suggests a CBD stone or Mirizzi syndrome.
- Acalculous disease occurs in critically ill patients, with a high risk of gangrene and perforation.

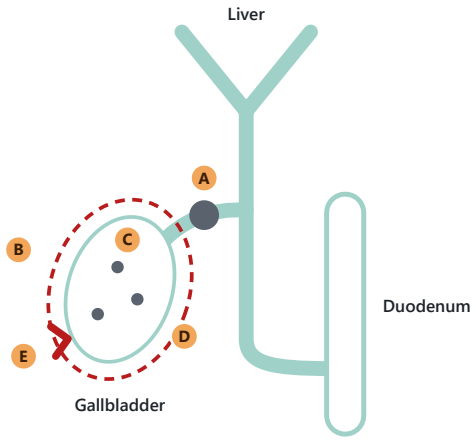


Fig. 1 | A stone blocks the cystic duct. Behind it the gallbladder swells, bacteria multiply, the wall is inflamed, and one part of the wall has died. A, Obstruction; B, Distention; C, Bacterial colonization; D, Wall inflammation; E, Gangrene.

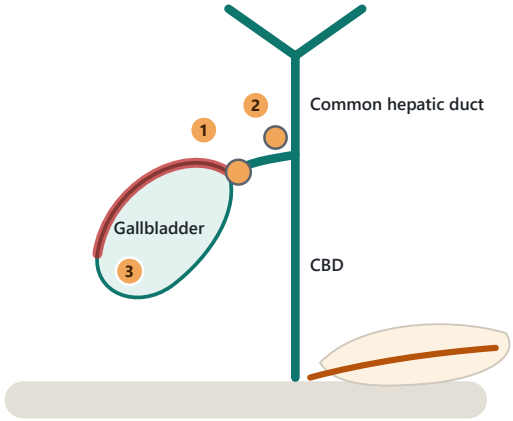


Fig. 2 | Biliary tree with a stone blocking the cystic duct, a stone pressing on the common hepatic duct, and an inflamed gallbladder with no stone. Positions are illustrative. 1, Calculous cholecystitis; 2, Mirizzi syndrome; 3, Acalculous cholecystitis.

Pathophysiology

A stone obstructs the cystic duct, causing RUQ pain over 6 hours, and increased back pressure distends the gallbladder. Bacterial colonization with *E. coli*, *Klebsiella*, and *Enterococcus* causes fever and sepsis, and the inflamed wall causes RUQ tenderness and Murphy sign. Wall necrosis leads to gangrene, with perforation, abscess, and peritonitis.

Etiology and mechanisms

In calculous cholecystitis (~90–95%), a stone obstructs the cystic duct, causing increased back pressure and gallbladder distention, then bacterial colonization (*E.*

coli, *Klebsiella*, *Enterococcus*) and gallbladder wall inflammation. In Mirizzi syndrome, a stone impacted in the cystic duct or gallbladder neck presses on the common hepatic duct from outside, causing cholecystitis plus obstructive jaundice, with cholestatic LFTs unlike simple cholecystitis. Acalculous cholecystitis occurs in critically ill patients (ICU, burns, sepsis, TPN), with gallbladder stasis and ischemia without a stone, causing inflammation with a high risk of gangrene and perforation. Acalculous disease accounts for ~10% of acute cholecystitis.

Clinical presentation and complications

Symptoms are RUQ pain over 6 hours, RUQ tenderness, Murphy sign, fever, and nausea and vomiting. LFTs are usually normal, and jaundice suggests a CBD stone or Mirizzi syndrome. Complications are sepsis, gallbladder empyema (pus-filled gallbladder), gangrenous cholecystitis (leading to perforation, abscess, and peritonitis), emphysematous cholecystitis (gas-forming bacteria, in diabetics), and cholecystoenteric fistula, leading to gallstone ileus. A chronically inflamed gallbladder with a large stone forms a fistula, the stone passes into the bowel, and it lodges at the terminal ileum or ileocecal valve, causing small bowel obstruction (CAVO: colicky pain, abdominal distention, vomiting, obstipation). Emphysematous cholecystitis follows wall ischemia with gas-forming bacteria, especially *Clostridium perfringens*, mostly in men and diabetics, and carries a high mortality. Chronic cholecystitis follows repeated biliary colic, with stones in over 90%, and causes ongoing inflammation and fibrosis of the wall; it gives dull RUQ or epigastric pain after meals for weeks to months, and fever is rare. Rokitansky-Aschoff sinuses, mucosa herniated into the thickened wall, are found in ~90% of specimens. Porcelain gallbladder is a wall calcified by chronic inflammation, often found by chance on x-ray or CT, and its cancer risk is lower than once thought.

Diagnosis

Acute cholecystitis is evaluated by RUQ ultrasound, first line, showing a thick wall, pericholecystic fluid, and sonographic Murphy sign. An unclear ultrasound is followed by HIDA scan, in which the gallbladder does not fill. FBC and LFTs assess inflammation, showing leukocytosis with usually normal LFTs. Mirizzi syndrome is evaluated by MRCP or ERCP, and gallstone ileus by CT, showing Rigler triad. Emphysematous cholecystitis shows gas in the gallbladder wall or lumen on CT. Chronic cholecystitis shows a thick wall and stones on ultrasound, and a HIDA scan with CCK shows an ejection fraction under 35%.

Management

Acute cholecystitis is treated with NPO, IV fluids, pain relief, and IV ABX, then early laparoscopic cholecystectomy within 72 hours. Patients too unwell for surgery, or with acalculous disease, have percutaneous cholecystostomy (drain) with ABX. Gangrene or perforation requires emergency cholecystectomy. Gallstone ileus is treated by enterolithotomy (removing the stone from the bowel), with fistula repair later if needed. Mirizzi syndrome is treated by cholecystectomy (often open), with ERCP stenting if needed. Emphysematous cholecystitis also needs emergency cholecystectomy. Chronic cholecystitis is treated with elective laparoscopic cholecystectomy, or a low-fat diet if the patient is not fit for surgery. Porcelain gallbladder is removed if it causes symptoms or shows selective mucosal calcification; otherwise, observation is discussed with the patient.

Relationships to other conditions

Biliary colic. Both come from a cystic duct stone, but colic is brief, with normal LFTs and no fever.

Choledocholithiasis. Jaundice and a cholestatic pattern move the stone from the cystic duct to the CBD; Mirizzi syndrome mimics this from outside the duct.

Cholangitis. Fever and RUQ pain are shared, but jaundice completes Charcot triad in cholangitis.

Biliary fistulas. Chronic inflammation with a large stone forms the fistula behind gallstone ileus.

Choledocholithiasis and gallstone pancreatitis: obstruction of the common channel

A stone in the CBD blocks bile flow from the liver, causing cholestasis, and a stone at the ampulla blocks both bile and pancreatic ducts, causing gallstone pancreatitis. Workup depends on the likelihood of a stone, and treatment aims to clear the duct and remove the gallbladder. Here, we review these mechanisms and their management, and consider how they relate to cholangitis and pancreatitis.

Key points

- Lipase separates the two: normal in choledocholithiasis and increased 3 or more times in gallstone pancreatitis.
- High likelihood goes to ERCP; intermediate likelihood goes to MRCP or endoscopic ultrasound.
- Cholecystectomy is performed during the same admission.

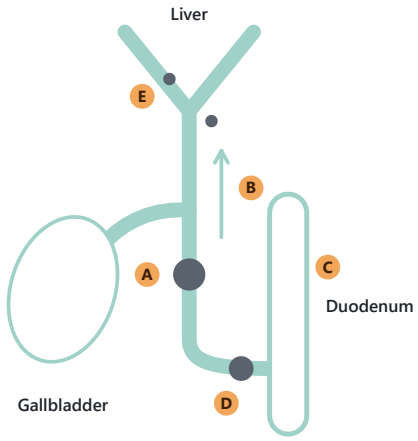


Fig. 1 | A stone blocks the common bile duct. Bile backs up, less bile reaches the duodenum, a second stone sits at the ampulla, and bacteria rise above the block. A, CBD obstruction; B, Back pressure; C, Less bile in the gut; D, Ampulla obstruction; E, Infection.

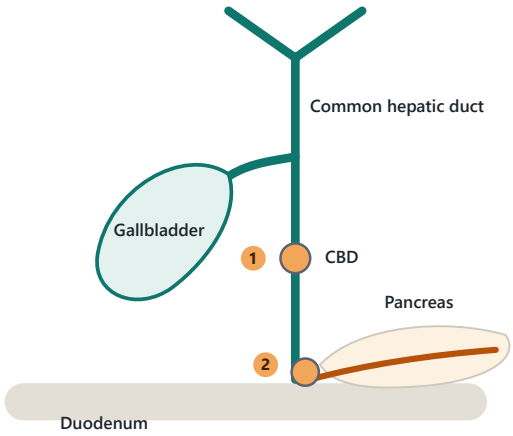


Fig. 2 | Biliary tree with one stone in the common bile duct and another at the ampulla blocking both ducts. 1, Stone in the common bile duct; 2, Stone at the hepatopancreatic ampulla (of Vater).

Pathophysiology

A stone in the CBD stops bile draining into the duodenum. Back pressure increases conjugated bilirubin and bile salts, causing jaundice, pruritus, and dark urine, and decreased bile reaching the gut causes pale stools. A stone at the ampulla blocks both bile and pancreatic ducts, causing gallstone pancreatitis, and bacteria behind the blockage cause ascending cholangitis.

Etiology and mechanisms

A stone in the CBD means bile cannot drain into the duodenum, increasing conjugated bilirubin and bile salts, with jaundice, pruritus, and dark urine, and decreasing bile reaching the gut, with pale stools. A stone

at the hepatopancreatic ampulla (of Vater) blocks both bile and pancreatic ducts, so pancreatic enzymes activate inside the pancreas. Gallstones are the most common cause of pancreatitis. Up to 90% of stones are secondary, forming in the gallbladder and migrating into the CBD, usually as cholesterol stones. Primary stones form in the ducts with bile stasis and infection, as brown pigment stones; they are more common in Asia and cause recurrent cholangitis.

Clinical presentation and complications

Choledocholithiasis causes RUQ pain, jaundice, pruritus, dark urine, and pale stools, with a cholestatic pattern and normal lipase. Gallstone pancreatitis causes severe epigastric pain radiating to the back, nausea, and vomiting, and may also cause jaundice, with a cholestatic pattern and lipase increased 3 or more times normal. Complications are ascending cholangitis (infection behind the blockage), gallstone pancreatitis, and secondary biliary cirrhosis from long-standing obstruction.

Diagnosis

The cholestatic LFT pattern is increased ALP and GGT, increased conjugated bilirubin, and AST/ALT that can spike early, then fall. RUQ ultrasound comes first, showing a dilated CBD (over 6 mm), although the stone itself is often not seen. Intermediate likelihood (e.g. abnormal LFTs or dilated CBD only) is evaluated by MRCP or endoscopic ultrasound. High likelihood (stone seen, cholangitis, or bilirubin over 4 mg/dL with a dilated CBD) proceeds to ERCP, which diagnoses and removes the stone. A dilated CBD is over 6 mm, or over 8 mm after cholecystectomy, and ultrasound shows the stone itself in only 15–40%. Age over 55 is an intermediate predictor, and patients with no predictors proceed to cholecystectomy, with intraoperative cholangiography if needed.

Management

Choledocholithiasis is treated with ERCP with sphincterotomy and stone removal, then cholecystectomy during the same admission. Gallstone pancreatitis is treated with IV fluids, pain relief, and early feeding, with ERCP only if there is cholangitis or ongoing obstruction, and cholecystectomy during the same admission if mild. Alternatively, laparoscopic CBD exploration clears the duct during the cholecystectomy itself.

Relationships to other conditions

Cholangitis. Fever turns an obstructed duct into cholangitis, which needs urgent drainage.

Cholecystitis. Cystic duct stones leave LFTs normal, whereas CBD stones produce a cholestatic pattern.

Malignant obstruction. Stones usually cause painful jaundice, whereas tumors cause painless jaundice with a palpable gallbladder.

Acute pancreatitis. Gallstones are the leading cause; the pancreatitis page covers its systemic and local complications.

Ascending cholangitis: infected obstruction and the need for drainage

Bile duct obstruction lets bacteria ascend from the duodenum, and infected bile under pressure enters the blood, causing sepsis. A CBD stone is the most common cause. Charcot triad progresses to Reynolds pentad as septic shock develops. Here, we review its causes, recognition, and management, and consider how it relates to other biliary disease.

Key points

- Charcot triad is fever, RUQ pain, and jaundice.
- Reynolds pentad adds hypotension and confusion, and means septic shock.
- ERCP with stone removal or stent is definitive.
- The Tokyo Guidelines combine inflammation, cholestasis, and obstruction.

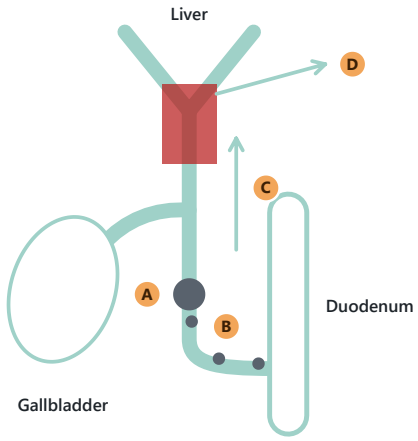


Fig. 1 | A blocked common bile duct with bacteria rising from the duodenum, pressure building in the ducts, and bacteria entering the blood. A, Obstruction and stasis; B, Ascending bacteria; C, Raised pressure; D, Bacteremia.

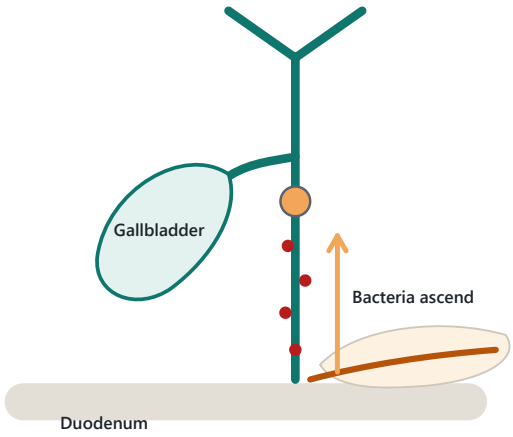


Fig. 2 | Biliary tree with a stone in the common bile duct and bacteria rising from the duodenum.

Pathophysiology

A blocked duct holds bile still, and *E. coli*, *Klebsiella*, and *Enterococcus* ascend from the duodenum into it. Infected bile under pressure causes fever, RUQ pain, and jaundice, and bacteria and toxins that enter the blood cause a cytokine storm and sepsis.

Etiology and mechanisms

Causes are a CBD stone (most common), benign or malignant stricture, a blocked biliary stent after ERCP, and PSC. Bile duct obstruction and stasis allow bacteria to ascend from the duodenum (*E. coli*, *Klebsiella*, *Enterococcus*). Infected bile under pressure allows bacteria and toxins to enter the blood.

Clinical presentation and complications

Charcot triad is fever, RUQ pain, and jaundice. A cytokine storm causes sepsis, leading to Reynolds pentad, which adds hypotension and confusion. Reynolds pentad means septic shock and needs urgent biliary drainage. Charcot triad is specific, but only ~25% of patients have all three features.

Diagnosis

Inflammation is evaluated with FBC (leukocytosis), CRP, and blood cultures; cholestasis with LFTs, showing a cholestatic pattern with increased bilirubin; and obstruction and its cause with ultrasound or CT (dilated bile ducts), with MRCP if unclear. The Tokyo Guidelines combine these three findings to diagnose and grade cholangitis.

Management

Resuscitation uses IV fluids, with vasopressors if in shock. Infection is treated with broad-spectrum IV ABX (e.g. piperacillin-tazobactam). Bile is drained by ERCP with stone removal or stent, which is definitive, with percutaneous drainage if ERCP fails. Recurrence is prevented by cholecystectomy once recovered, if gallstones were the cause. Severe disease, with shock or organ failure, needs urgent drainage, while 80–90% of mild cases respond to ABX and are drained before discharge.

Relationships to other conditions

Choledocholithiasis. An uninfected CBD stone causes jaundice and pain without fever; fever marks the step to cholangitis.

Cholecystitis. RUQ pain and fever without jaundice suggest cholecystitis rather than cholangitis.

PSC. Strictures in PSC cause recurrent bacterial cholangitis as bile stagnates behind them.

Biliary fistulas. Fistulas also let gut bacteria enter the bile ducts.

Biliary fistulas: erosion from the biliary tree into neighboring organs

Chronic inflammation and pressure from a large stone erode into a neighboring organ, creating a fistula. Cholecystoduodenal fistulas are most common. Many patients have no symptoms, but stones can pass through and obstruct the bowel. Here, we review fistula types, complications, and treatment, and consider how they relate to cholecystitis and cholangitis.

Key points

- Cholecystoduodenal fistula is the most common type and the main route for gallstone ileus.
- Pneumobilia on CT suggests a fistula; Rigler triad suggests gallstone ileus.
- Gallstone ileus is treated by enterolithotomy first.

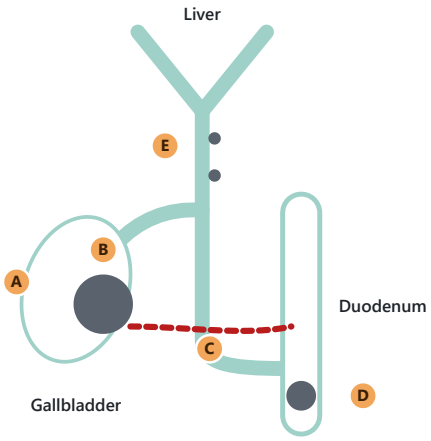


Fig. 1 | A large stone inflames and presses on the gallbladder wall, a tract opens into the duodenum, the stone passes through, and gut bacteria enter the bile ducts. A, Chronic inflammation; B, Pressure from a large stone; C, Fistula; D, Stone passage; E, Gut bacteria.

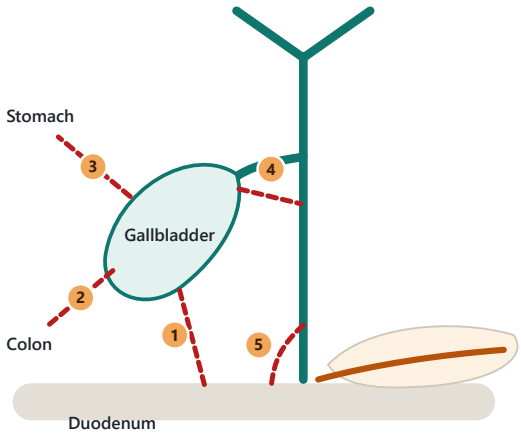


Fig. 2 | Biliary tree with fistulas from the gallbladder to the duodenum, colon, stomach, and bile duct, and from the bile duct to the duodenum. 1, Cholecystoduodenal; 2, Cholecystocolic; 3, Cholecystogastric; 4, Cholecystocholedochal; 5, Choledochoduodenal.

Pathophysiology

Chronic inflammation weakens the gallbladder wall, a large stone presses on it, and a tract opens into a neighboring organ. A stone that enters the bowel can cause gallstone ileus, and gut bacteria that enter the bile ducts cause ascending cholangitis.

Etiology and mechanisms

Cholecystoduodenal fistula connects the gallbladder and duodenum and is the main route for gallstone ileus, with Bouveret syndrome if the stone blocks the pylorus or duodenum. Cholecystocolic fistula connects the gallbladder and colon, causing bile acid diarrhea, and stones may pass in the stool. Cholecystogastric fistula connects the gallbladder and stomach and can cause

gastric outlet obstruction. Cholecystocholedochal fistula connects the gallbladder and common bile duct, as in advanced Mirizzi syndrome. Choledochoduodenal fistula connects the common bile duct and duodenum, from a penetrating duodenal ulcer or CBD stones. Causes are gallstones (most cases, especially large stones), peptic ulcer disease, malignancy, trauma or surgery, and Crohn disease. Of biliary-enteric fistulas, ~68% are cholecystoduodenal, ~17% cholecystocolic, and ~5% cholecystogastric.

Clinical presentation and complications

Fistulas often cause no symptoms, or recurrent cholangitis, and diarrhea in cholecystocolic disease. Complications are gallstone ileus (small bowel obstruction, CAVO), Bouveret syndrome (gastric outlet obstruction with vomiting and epigastric fullness), ascending cholangitis as gut bacteria enter the bile ducts, and GI bleeding. Gallstone ileus usually impacts at the terminal ileum, mostly with stones over 2 cm, and mainly affects older women.

Diagnosis

A suspected fistula is evaluated by CT abdomen, showing pneumobilia. Gallstone ileus is evaluated by CT, showing Rigler triad. The fistula opening is evaluated by ERCP or upper endoscopy. CT detects Rigler triad with a sensitivity of ~93%, and an x-ray may also show it.

Management

A symptomatic fistula is treated by cholecystectomy with fistula repair. Gallstone ileus is treated by enterolithotomy first, with fistula repair later if needed. Bouveret syndrome is treated by endoscopic stone removal, with surgery if this fails. Enterolithotomy is usually open.

Relationships to other conditions

Cholecystitis. Fistulas are a late consequence of chronic inflammation with a large stone.

PUD. A penetrating duodenal ulcer can form a choledochoduodenal fistula, and both can cause gastric outlet obstruction.

Crohn disease. Crohn disease forms fistulas by a different route, through transmural inflammation.

Chronic diarrhea. Bile acid diarrhea from a cholecystocolic fistula is a rare cause to remember in older patients with gallstones.

Biliary tract and periampullary malignancies: painless obstruction of bile flow

Tumors in or around the bile ducts gradually obstruct bile flow, causing painless obstructive jaundice. Pancreatic head adenocarcinoma is the most common periampullary cancer, and cholangiocarcinoma and gallbladder carcinoma arise within the biliary tree. Courvoisier law separates malignant from gallstone obstruction. Here, we review their features, workup, and treatment, and consider how they relate to gallstone disease and PSC.

Key points

- Painless jaundice with a palpable gallbladder suggests malignancy.
- The double duct sign supports a periampullary tumor.
- CA 19-9 and CEA are not for screening.
- Resectable periampullary tumors are treated by pancreaticoduodenectomy.

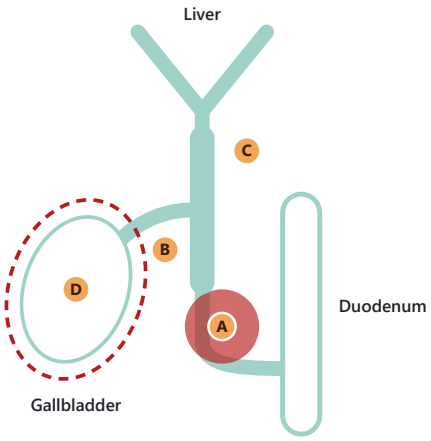


Fig. 1 | A tumor squeezes the lower common bile duct, ducts above it widen, and a large smooth gallbladder is shown, with bile no longer reaching the duodenum. A, Tumor growth; B, Bile duct obstruction; C, Back pressure; D, Courvoisier gallbladder.

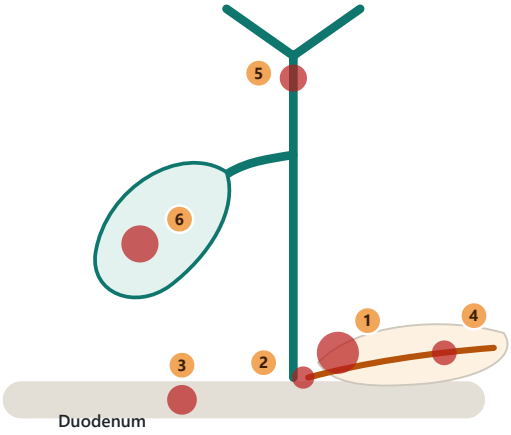


Fig. 2 | Biliary tree with tumors in the pancreatic head, ampulla, duodenum, pancreas, bile duct, and gallbladder. Positions are illustrative. 1, Pancreatic head adenocarcinoma; 2, Ampullary adenocarcinoma; 3, Duodenal adenocarcinoma; 4, Pancreatic neuroendocrine tumors; 5, Bile duct adenocarcinoma (cholangiocarcinoma); 6, Gallbladder carcinoma.

Pathophysiology

A tumor below the cystic duct causes a slow, steady blockage of bile flow. Bile cannot reach the gut, causing pale stools, and builds up behind the block, causing painless jaundice, pruritus, and dark urine. Because the gallbladder wall is healthy and stretchy, it enlarges into a palpable, nontender gallbladder.

Etiology

Pancreatic head adenocarcinoma is the most common periampullary cancer, with risk factors of smoking, chronic pancreatitis, and diabetes. Ampullary

adenocarcinoma blocks bile early, so it is found earlier and has a better prognosis, may cause intermittent jaundice, and is linked to FAP. Duodenal adenocarcinoma is rare, may also cause GI bleeding or obstruction, and is linked to FAP, Lynch syndrome, and celiac disease. Pancreatic neuroendocrine tumors arise from islet cells and may secrete hormones (insulinoma, gastrinoma, vipoma, glucagonoma), with MEN1 as a risk factor. Bile duct adenocarcinoma (cholangiocarcinoma) is intrahepatic, perihilar, or distal, with perihilar (Klatskin) most common, and risk factors of PSC, liver flukes (*Clonorchis*), and choledochal cysts. Gallbladder carcinoma is often found late, or by chance after cholecystectomy, in older women, with risk factors of gallstones (especially large), porcelain gallbladder, polyps over 1 cm, and PSC. Lynch syndrome also raises the risk of ampullary cancer. Of cholangiocarcinomas, ~50% are perihilar, ~40% distal, and ~10% intrahepatic, and hepatitis B or C and cirrhosis also raise the risk. More than half of gallbladder carcinomas are found by chance at cholecystectomy; the risk is highest with stones over 3 cm, and the risk from porcelain gallbladder is lower than once thought.

Clinical presentation and complications

Symptoms are painless jaundice, pruritus, dark urine, pale stools, weight loss, and a palpable, nontender gallbladder. In malignant obstruction, a tumor below the cystic duct (pancreatic head, ampulla, distal CBD) causes slow, steady blockage of a healthy, stretchy gallbladder, producing an enlarged, palpable, nontender gallbladder with painless jaundice. In gallstone obstruction, a stone in the CBD meets a gallbladder already scarred by repeated inflammation, which cannot distend, usually with pain. Courvoisier law states that painless jaundice with a palpable gallbladder is unlikely to be caused by gallstones, so malignancy is suspected. Complications are ascending cholangitis, liver metastases, and gastric outlet obstruction.

Diagnosis

Cholestasis is evaluated by LFTs. Tumor markers are CA 19-9 and CEA, which are not for screening. Imaging is ultrasound, then CT or MRCP, which may show the double duct sign. Tissue diagnosis is by ERCP brushings or endoscopic ultrasound biopsy.

Management

Periampullary tumors are treated by pancreaticoduodenectomy (Whipple procedure) if resectable. Cholangiocarcinoma is treated by resection (bile duct, often with part of the liver), with transplant in selected cases. Gallbladder carcinoma is treated by cholecystectomy with liver resection and lymph node removal. Unresectable disease is treated with a biliary stent to relieve jaundice, and chemotherapy. Gallbladder carcinoma confined to the mucosa needs only cholecystectomy.

Relationships to other conditions

Gallstone obstruction. Pain and a shrunken gallbladder favor stones; painless jaundice with a palpable gallbladder favors malignancy.

PSC. PSC raises the risk of cholangiocarcinoma and gallbladder carcinoma.

Pancreatitis. Chronic pancreatitis is a risk factor for pancreatic head adenocarcinoma.

Hormonal diarrhea. Gastrinomas and vipomas link pancreatic neuroendocrine tumors to chronic secretory diarrhea.

Liver tumors. Intrahepatic cholangiocarcinoma overlaps with primary liver malignancy.

Acute pancreatitis: autodigestion and its local and systemic consequences

Premature activation of trypsin inside the pancreas leads to autodigestion, causing local and systemic inflammation. Gallstones and ethanol are the leading causes in the I GET SMASHED mnemonic. Diagnosis requires 2 of 3 criteria, and severity is graded by organ failure and local complications. Here, we review its causes, complications, and management, and consider how it relates to biliary disease and malabsorption.

Key points

- Gallstones are the most common cause, and ethanol the second.
- Diagnosis requires any 2 of pain, enzymes, and imaging.
- RUQ ultrasound is obtained in every patient.
- CT too early can miss necrosis.
- Prophylactic ABX are not given.

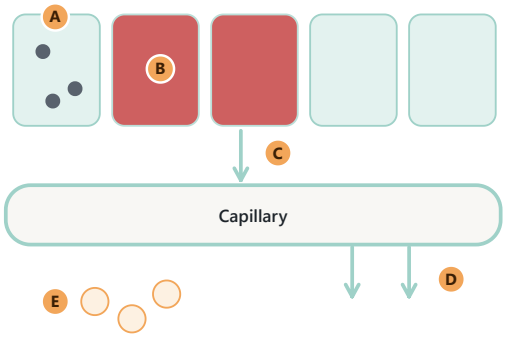


Fig. 1 | Acinar cells with trypsin active inside them, damaged cells, cytokines entering a capillary, fluid leaking from the capillary, and fat necrosis binding calcium. A, Trypsin activation; B, Autodigestion; C, Cytokine release; D, Leaky capillaries; E, Fat saponification.

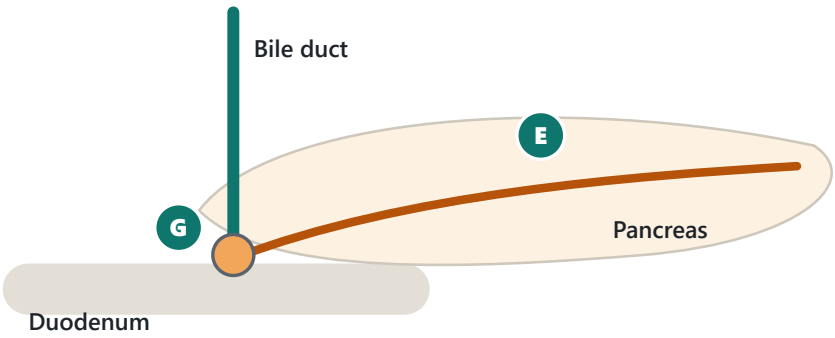


Fig. 2 | Pancreas with its duct joining the bile duct at the ampulla, where a gallstone blocks both, and alcohol injuring the acinar cells.

Pathophysiology

Trypsin is activated inside the pancreas, and the enzymes digest the gland, causing epigastric pain. Increased IL-1, IL-6, and TNF-α cause SIRS. Increased capillary permeability causes third spacing, with hypovolemia and shock, and lipase digests surrounding fat, releasing fatty acids that bind calcium and cause hypocalcemia.

Etiology and mechanisms

I GET SMASHED covers idiopathic disease, with no cause found in about 10–20%; gallstones, the most common cause, in which a stone blocks the ampulla; ethanol, the second most common, through direct acinar toxicity and protein plugs in ducts; trauma, such as blunt abdominal trauma from a bicycle handlebar injury; steroids; mumps and other infections, such as coxsackievirus and CMV; autoimmune causes, including

IgG4-related autoimmune pancreatitis, SLE, and polyarteritis nodosa; scorpion sting (*Tityus trinitatis*, Trinidad); hypercalcemia, hypertriglyceridemia (usually over 1000 mg/dL), and hypothermia; ERCP, of which it is the most common complication; and drugs, including azathioprine, valproate, thiazides, furosemide, sulfonamides, and didanosine.

Clinical presentation and complications

Patients have severe epigastric pain radiating to the back, nausea and vomiting, fever, tachycardia, and epigastric tenderness and guarding. Trypsin activated inside the pancreas causes autodigestion, with epigastric pain and increased IL-1, IL-6, and TNF-α, leading to SIRS and local complications. Systemically, increased capillary permeability causes third spacing, with hypovolemia and fluid in the lungs and abdomen, leading to shock, AKI, ARDS, exudative ascites, and abdominal compartment syndrome. Increased tissue factor causes clotting throughout small vessels, consuming platelets and clotting factors and shearing red cells (schistocytes), leading to DIC. Lipase digests surrounding fat, and fatty acids bind calcium (fat saponification), causing hypocalcemia; islet damage causes hyperglycemia. Locally, interstitial edematous pancreatitis causes pancreatic edema and fluid around the pancreas, forming a pseudocyst after 4 weeks, with no epithelium, which can cause infection, rupture, and compression with small bowel or gastric outlet obstruction. Necrotizing pancreatitis causes tissue necrosis and an acute necrotic collection, becoming walled-off necrosis after 4 weeks; enzymes erode blood vessels, causing retroperitoneal bleeding (Cullen and Grey Turner signs). Repeated attacks can lead to chronic pancreatitis. Chronic pancreatitis is most often caused by alcohol in adults, though under 10% of heavy drinkers develop it, and by smoking; in children, cystic fibrosis and genes such as PRSS1, SPINK1, and CFTR are responsible. Protein plugs in the ducts calcify and form stones, and fibrosis replaces the gland. It causes chronic epigastric pain, often better leaning forward, with pain-free periods; steatorrhea and weight loss once over 90% of the gland is damaged; pancreatogenic (type 3c) diabetes; and a raised risk of pancreatic ductal adenocarcinoma.

Diagnosis

Any 2 of 3 criteria are required: acute, severe epigastric pain, often radiating to the back; lipase (or amylase) at least 3 times the upper limit of normal, with lipase more specific; and typical findings on contrast CT abdomen, MRI, or ultrasound. The cause is sought with RUQ ultrasound for gallstones in every patient, LFTs (ALT over 150 suggests gallstones), triglycerides, calcium, alcohol history, and medications. Contrast CT is obtained if the diagnosis is unclear, or after 48–72 hours if the patient is not improving, as CT too early can miss necrosis. Severity is assessed with FBC, creatinine, BUN, oxygen saturation, and scores such as BISAP. By the revised Atlanta classification, mild disease has no organ failure and no local complications, moderately severe disease has organ failure lasting under 48 hours or local complications, and severe disease has organ failure lasting over 48 hours. Chronic pancreatitis is shown by calcifications, its hallmark, on CT or MRCP, with duct dilation, strictures, or stones, and exocrine function is assessed with fecal elastase (normal over 500 µg/g) or a 72-hour fecal fat over 7 g a day, the most definitive test.

Management

Supportive care comprises goal-directed IV fluids (e.g. Hartmann solution), pain relief, and early oral feeding, with enteral feeding if severe. The cause is treated: cholecystectomy during the same admission for gallstones, stopping drinking for alcohol, insulin infusion for hypertriglyceridemia, and stopping the offending drug. Infected necrosis is treated with ABX (e.g. carbapenem) and delayed drainage or necrosectomy, ideally after 4 weeks, with no prophylactic ABX. Pseudocysts are observed, and drained only if symptomatic or complicated. In chronic pancreatitis, every patient stops alcohol and smoking. Pain is treated with paracetamol, NSAIDs, or pregabalin first and opioids only if these fail, with celiac plexus block, and ERCP or shock wave lithotripsy for obstructing stones or strictures. Steatorrhea is treated with pancreatic enzyme replacement with meals, which does not relieve pain.

Relationships to other conditions

Gallstone disease. A stone at the ampulla blocks both bile and pancreatic ducts. A cholestatic LFT pattern and ALT over 150 favor a gallstone cause.

Malabsorption. Repeated attacks lead to chronic pancreatitis and pancreatic insufficiency.

PUD. Epigastric pain overlaps; radiation to the back and increased lipase favor pancreatitis, and pain that changes with meals favors PUD.

IgG4-related disease. Autoimmune pancreatitis can occur with IgG4-related sclerosing cholangitis.

Pancreatic malignancy. Painless jaundice favors a pancreatic head tumor over pancreatitis.

Malabsorption: separating mucosal injury from failed digestion

Malabsorption syndromes divide into true malabsorption, with mucosal injury and normal enzyme function, and maldigestion, with altered enzyme function and no mucosal injury. Celiac disease, Whipple disease, and tropical sprue injure the mucosa, whereas pancreatic insufficiency, bile salt deficiency, and lactose intolerance impair digestion. Here, we review their mechanisms, nutrient-specific consequences, and treatment, and consider how they relate to diarrhea and IBD.

Key points

- The D-xylose test differentiates malabsorption from maldigestion.
- Celiac disease is confirmed by EGD with biopsy after serology.
- Consequences follow the lost nutrient: carbohydrate, protein, fat, or micronutrient.
- Micronutrient deficiency is treated by replacement and IV fluids.

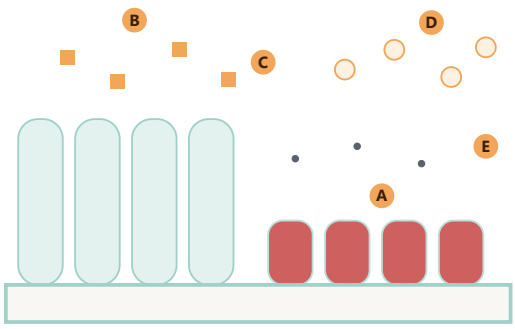


Fig. 1 | Tall villi on the left and blunted villi on the right, with undigested food and fat in the lumen and few nutrients crossing. A, Mucosal injury; B, Failed digestion; C, Undigested carbohydrate; D, Undigested fat; E, Lost micronutrients.

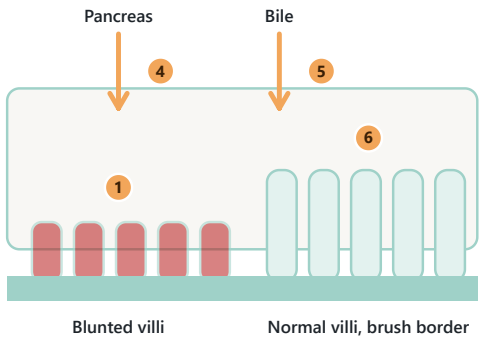


Fig. 2 | Small bowel lumen with blunted villi from mucosal injury, and pancreatic enzymes and bile entering from above. Causes 1–3 injure the mucosa. Causes 4–6 leave food undigested. 1, Celiac disease; 4, Pancreatic insufficiency; 5, Bile salt deficiency; 6, Lactose intolerance.

Pathophysiology

Mucosal injury blunts the villi and decreases absorption in true malabsorption, whereas decreased enzymes or bile salts leave food undigested in maldigestion. Either way, nutrients stay in the lumen. Undigested carbohydrate causes osmotic diarrhea and gas from bacterial digestion, with abdominal pain and flatulence; undigested fat causes steatorrhea and fat-soluble vitamin deficiencies; and iron, B12, and folate deficiency cause anemia.

Etiology and mechanisms

Celiac disease is associated with HLA DQ2/8 and triggered by gliadin from gluten in wheat, rye, and barley. Gliadin binds antigen-presenting cells, which activate T helper cells and then B cells, producing IgA antibodies: tTG, DGP, and EMA. Whipple disease is *Tropheryma whipplei* infection with mucosal damage. Tropical sprue is *E. coli* and *Klebsiella* overgrowth in tropical areas with mucosal injury. Pancreatic insufficiency follows acute or chronic pancreatitis, cystic

fibrosis, and pancreatic cancer or surgery. Bile salt deficiency, a partial maldigestion, follows cirrhosis, hepatocellular injury, biliary tree obstruction, and impaired enterohepatic circulation. Lactose intolerance, also a partial maldigestion, follows genetic mutations and gastroenteritis. Celiac disease is associated with type 1 diabetes and other autoimmune disease, IgA deficiency, and dermatitis herpetiformis. *Tropheryma whipplei* is a gram-positive bacillus that infects macrophages and also involves the joints, heart, and CNS, and tropical sprue follows coliform overgrowth (*E. coli*, *Klebsiella*, *Enterobacter*) after over 1 month in tropical areas. In SIBO, colonic bacteria grow in the small bowel because of stasis, from motility disorders or blind loops, or low acid, and they deconjugate bile salts and damage the mucosa. Abetalipoproteinemia is an autosomal recessive MTTP mutation that prevents apo B lipoproteins from forming, so enterocytes cannot export chylomicrons and fat stays in the mucosa.

Clinical presentation and complications

Carbohydrate malabsorption causes osmotic diarrhea, bowel distention, and increased gas production from bacterial digestion, with abdominal pain and flatulence. Protein malabsorption causes muscle atrophy and weight loss. Fat malabsorption causes steatorrhea and fat-soluble vitamin deficiencies: vitamin A (night blindness), vitamin D (hypocalcemia, osteomalacia, osteoporosis), vitamin E (neuropathy, hemolytic anemia), and vitamin K (coagulopathy). Micronutrient malabsorption causes iron, B12, and folate deficiency, leading to anemia. Celiac disease causes iron deficiency anemia, osteoporosis, fatigue, and arthralgia, and dermatitis herpetiformis, intensely itchy vesicles on extensor surfaces, is pathognomonic. Enteropathy-associated T-cell lymphoma is a rare complication, more likely in refractory disease, and a gluten-free diet lowers the risk. Whipple disease causes weight loss, diarrhea, fever, and migratory large-joint arthralgia, often first, with endocarditis and CNS signs. Tropical sprue causes chronic diarrhea after a traveler's diarrhea-like illness, with B12 and folate deficiency and anemia. SIBO causes bloating, flatulence, and watery diarrhea or steatorrhea, with low B12 and normal or high folate. Abetalipoproteinemia presents in infancy with failure to thrive and steatorrhea, and vitamin E deficiency causes acanthocytes, retinitis pigmentosa, and ataxia.

Diagnosis

The D-xylose test first differentiates malabsorption from maldigestion. Celiac disease is evaluated by serology (tTG, DGP, EMA, HLA), infectious causes by history taking, and confirmation is by EGD with biopsy. Pancreatic function is assessed by fecal elastase, lactose intolerance by hydrogen breath test, and bile salt deficiency by LFTs or biopsy. Celiac serology is done on a gluten diet for 6–8 weeks, measuring tTG-IgA with total IgA, since IgA deficiency gives a false negative and IgG tests are used instead. Duodenal biopsies, 1–2 from the bulb and 4 from the distal duodenum, show villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes. Whipple disease shows PAS-positive foamy macrophages on small bowel biopsy, and tropical sprue is diagnosed from the travel history after excluding other causes. Fecal elastase is unaffected by enzyme replacement. SIBO is shown by a glucose or lactulose breath test, abetalipoproteinemia by acanthocytes with very low cholesterol, LDL, and VLDL, and dermatitis herpetiformis by direct immunofluorescence of skin beside a lesion.

Management

Micronutrient deficiency is treated by replacing the micronutrients and giving IV fluids. Celiac disease is treated with a gluten-free diet; Whipple disease and tropical sprue with ABX; pancreatic insufficiency with enzyme replacement; bile salt deficiency by treating the liver disease or cholestasis; and lactose intolerance with lactase replacement or a lactose-free diet. The gluten-free diet is strict and lifelong, with dapsone for dermatitis herpetiformis. Whipple disease is treated with ceftriaxone or penicillin G for 2 weeks, then TMP-SMX for at least 12 months, and tropical sprue with tetracycline, TMP-SMX, or a quinolone, plus folic acid and iron, for 3–6 months, with parenteral B12. SIBO is treated with rifaximin for 2 weeks, treatment of the cause, and B12 replacement, and abetalipoproteinemia with a low-fat diet and very large doses of oral vitamin E and other fat-soluble vitamins.

Relationships to other conditions

Diarrhea. Every cause here produces chronic osmotic diarrhea, which has a high osmotic gap and stops with fasting, unlike secretory diarrhea.

IBD. Ileal Crohn disease reproduces malabsorption: fat, B12, and bile salts are lost, with oxalate stones and gallstones as consequences.

Cholestatic liver disease. PBC and PSC cause fat-soluble vitamin deficiencies through the same bile salt mechanism.

Pancreatitis. Repeated attacks link pancreatitis to pancreatic insufficiency, which fecal elastase identifies.

Diarrhea: a mechanistic framework of secretory, osmotic, and inflammatory disease

Diarrhea is classified by duration and mechanism. Acute disease is non-inflammatory or inflammatory, and chronic disease is secretory, osmotic, or inflammatory. Inflammatory disease has increased fecal WBC and calprotectin and produces bloody, mucoid stools, whereas secretory and osmotic disease are watery. Here, we review the causes, complications, workup, and treatment of diarrhea, and consider how it relates to malabsorption, IBD, and PUD.

Key points

- Fecal WBC and calprotectin separate non-inflammatory from inflammatory disease.
- Exposure history points to the organism in acute diarrhea.
- HUS follows STEC or *Shigella* and is managed by avoiding ABX and loperamide.
- Loperamide is avoided in HUS, *C. diff*, toxic megacolon, and with fever or bloody stools.

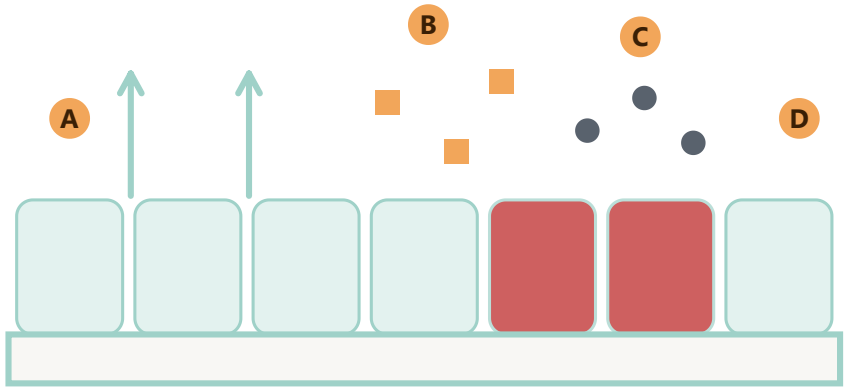


Fig. 1 | Bowel lining with arrows showing water leaving cells, sugars pulling water into the lumen, an inflamed patch with blood, and unabsorbed fluid in the lumen. A, Secretory; B, Osmotic; C, Inflammatory; D, Decreased absorption.

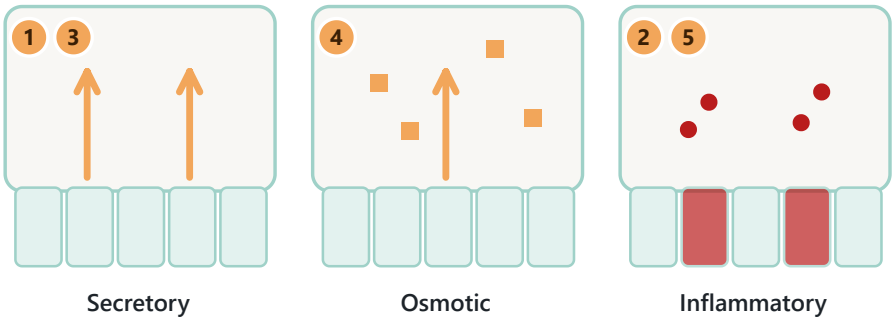


Fig. 2 | Three views of the bowel wall: water secreted into the lumen, sugars pulling water into the lumen, and damaged mucosa bleeding into the lumen. 1, Non-inflammatory; 2, Inflammatory; 3, Secretory; 4, Osmotic; 5, Inflammatory.

Pathophysiology

Fluid reaches the stool in several ways. In secretory diarrhea, channel stimulation increases secretory activity, causing watery stools. In osmotic diarrhea, unabsorbed solutes pull water into the lumen, also causing watery stools. In inflammatory diarrhea, direct mucosal damage and neutrophilic infiltration cause

bloody, mucoid stools. When absorption falls, less water returns from the lumen, causing high-volume stools.

Etiology and mechanisms

Acute non-inflammatory diarrhea has no fecal WBC or calprotectin and produces watery, high-volume stools. Causes are enterotoxic bacteria (*S. aureus* from picnics, *B. cereus* from rice, ETEC and *V. cholerae* from travel), viruses (norovirus on cruises, rotavirus in childcare), and parasites (*Giardia* from river water, *Cryptosporidium* in immunocompromised patients). They decrease absorption and stimulate channels, increasing secretory activity. Acute inflammatory diarrhea has increased fecal WBC and calprotectin and presents as dysentery. Causes are invasive bacteria (*Salmonella* from poultry, *Shigella* from travel and food, *Campylobacter* and EHEC/STEC from food, *C. difficile* from ABX) and parasites (*Entamoeba histolytica* from travel). Neutrophilic infiltration increases goblet cell activity and mucus production, with bleeding. Chronic secretory diarrhea has no direct mucosal damage and arises from hormonal changes (increased VIP from vipomas, gastrin from gastrinomas, and serotonin from carcinoid syndrome) and stimulant laxatives (senna, bisacodyl), through channel stimulation. Chronic osmotic diarrhea has no direct mucosal damage and arises through osmotic effects of malabsorption (celiac disease, Whipple disease, tropical sprue), maldigestion (EPI, bile salt deficiency, lactose intolerance), and osmotic laxatives (lactulose, polyethylene glycol, magnesium citrate). Chronic inflammatory diarrhea has direct mucosal damage, with bloody, mucoid stools, from IBD and chronic infections. *V. cholerae* causes rice-water stools, and *C. difficile* follows ABX, especially clindamycin, cephalosporins, fluoroquinolones, and penicillins. Vipomas cause WDHA: watery diarrhea, hypokalemia, and achlorhydria. Carcinoid syndrome follows liver metastases and causes flushing, wheezing, and tricuspid valve disease. Microscopic colitis is a chronic inflammatory cause alongside IBD.

Clinical presentation and complications

Symptoms are abdominal pain and watery or bloody diarrhea. Acute complications are fluid status disturbances (hypovolemia, hypokalemia, NAGMA), colitis (megacolon, tenesmus, perforation), and HUS from STEC or *Shigella* (translocation-induced coagulopathy), with AKI, thrombocytopenia, and hemolytic anemia. Chronic complications are fluid status disturbances and malabsorption. *C. difficile* ranges from mild diarrhea to pseudomembranous colitis, toxic megacolon, and septic shock, and 3–5% of healthy adults carry it. Microscopic colitis causes chronic watery, nonbloody diarrhea, 4–9 stools a day and often nocturnal, mostly in older women, and is linked to NSAIDs, PPIs, SSRIs, statins, and smoking. In carcinoid syndrome, tryptophan is diverted to serotonin, causing niacin deficiency (pellagra).

Diagnosis

In acute diarrhea, HUS or AKI is evaluated with FBC and BMP, and the initial workup is fecal WBC, fecal calprotectin, stool culture, O&P, and norovirus and rotavirus testing. In chronic diarrhea, the initial workup is fecal WBC, fecal calprotectin, and FOBT. Inflammatory disease is evaluated by checking for infection and colonoscopy, and non-inflammatory disease by stool osmotic gap, laxative use, and hormone levels. *C. difficile* is tested only if there is diarrhea, by 2-step testing with GDH or NAAT, then toxin A and B. A stool osmotic gap under 50 mOsm/kg indicates secretory diarrhea and over 75 osmotic diarrhea, and hormone levels include VIP, gastrin, and urine 5-HIAA for carcinoid. Microscopic colitis needs biopsies, showing a collagen band over 10 μm in collagenous colitis or 20 or more intraepithelial lymphocytes per 100 cells in lymphocytic colitis.

Management

Hypovolemia is treated with IV or PO fluids, and severe colitis or immunocompromise with ABX. HUS is managed by avoiding ABX and loperamide, and by using fluids, blood, or dialysis if needed. Loperamide can be used in most diarrhea, but is avoided in HUS, *C. diff*, toxic megacolon, and with fever or bloody stools. Hypovolemia is treated with ORS, whose salt, water, and glucose are absorbed together by the sodium-glucose cotransporter, or IV fluids. *C. difficile* is treated by stopping unnecessary ABX and giving oral fidaxomicin or oral vancomycin for 10 days, with metronidazole only if neither is available; recurrence is treated with fidaxomicin, bezlotoxumab, or fecal microbiota transplant, and hands are washed with soap, since alcohol gel does not kill spores. STEC is not treated with ABX, which raise the risk of HUS. Cholera is treated with ORS and doxycycline, microscopic colitis by stopping culprit drugs and giving budesonide for 6–8 weeks, and carcinoid syndrome with octreotide or lanreotide.

Relationships to other conditions

Malabsorption. Chronic osmotic diarrhea is the shared endpoint; a high stool osmotic gap and resolution with fasting favor it over secretory disease.

IBD and IBS. Fecal calprotectin and ESR/CRP separate inflammatory disease from IBS. Infectious diarrhea must be excluded before IBD is diagnosed, using stool culture, O&P, and *C. diff*.

PUD. Gastrinomas cause both ulcers and secretory diarrhea.

Intestinal ischemia. Bloody diarrhea with pain in an older patient after hypotension suggests ischemic colitis rather than infection.

Intestinal ischemia: arterial, venous, and low-flow injury to the bowel

Arterial occlusion, venous occlusion, or hypoperfusion causes loss of blood supply to the intestines. Acute mesenteric ischemia presents with out-of-proportion abdominal pain, chronic mesenteric ischemia with postprandial pain, and ischemic colitis with LLQ pain and hematochezia. Here, we review these three patterns, their workup, and their treatment, and consider how they relate to other causes of abdominal pain and bloody stools.

Key points

- The SMA is the most commonly affected vessel in acute mesenteric ischemia.
- Chronic mesenteric ischemia requires atherosclerosis of 2 of the celiac trunk, SMA, and IMA.
- CT angiogram evaluates mesenteric ischemia; CT AP and colonoscopy evaluate ischemic colitis.
- Treatment combines bowel decompression, ABX, heparin, and revascularization.

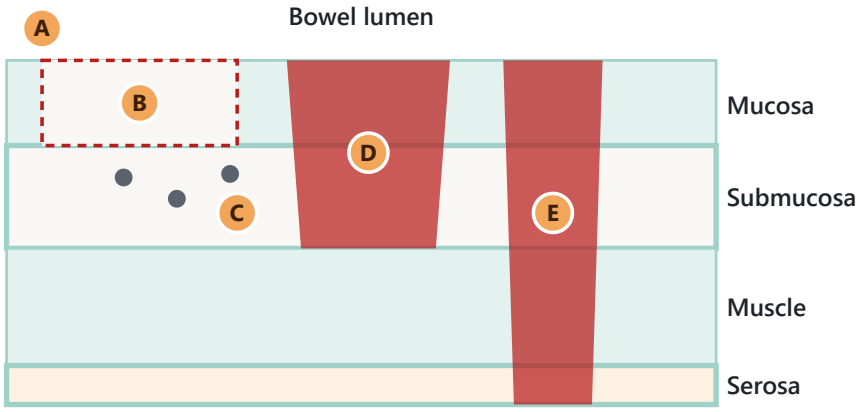


Fig. 1 | A bowel wall in section with a narrowed artery below it. Oxygen falls, the mucosa dies first, vessels leak, and necrosis reaches through the wall in one area. A, Decreased oxygen supply; B, Anaerobic respiration; C, Leaky vessels; D, Infarction; E, Perforation.

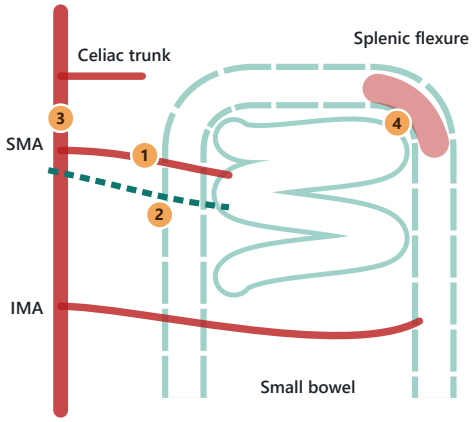


Fig. 2 | Aorta with the celiac trunk, SMA, and IMA supplying the bowel, with the sites of arterial and venous occlusion and the splenic flexure. Positions are illustrative. 1, Arterial; 2, Venous; 3, Atherosclerosis; 4, Hypoperfusion of the IMA.

Pathophysiology

Arterial occlusion, venous occlusion, or hypoperfusion decreases oxygen supply to the bowel, causing abdominal pain, and increased demand after eating makes this worse in chronic disease. Anaerobic respiration causes AGMA, and increased vascular permeability causes hematochezia. Prolonged ischemia causes infarction and necrosis, with out-of-proportion abdominal pain, and

full-thickness necrosis perforates, causing pneumoperitoneum and peritonitis.

Etiology and mechanisms

Acute arterial ischemia follows thrombosis (atherosclerosis), embolism (history of A-fib or AAA), or hypoperfusion (shock), which decrease perfusion or block the SMA, most commonly, or IMA, causing infarction and necrosis with out-of-proportion abdominal pain. Acute venous ischemia follows mesenteric thrombosis (hypercoagulation) or portal thrombosis (cirrhosis), with venous occlusion of the portal vein, SMV, or IMV, venous congestion, and decreased oxygen supply. Chronic mesenteric ischemia follows atherosclerosis of 2 of the celiac trunk, SMA, and IMA; decreased oxygen supply cannot meet increased oxygen demand when eating, causing postprandial abdominal pain. Ischemic colitis follows shock or a history of vasopressor usage, with hypoperfusion of the IMA and decreased oxygen supply to the splenic flexure, descending colon, and rectosigmoid junction, causing LLQ pain and hematochezia. Of acute mesenteric ischemia, embolism, usually cardiac, accounts for 40–50%, thrombosis for 25–30%, and nonocclusive mesenteric ischemia (NOMI) for ~20%, from vasospasm in critically ill patients on vasopressors. Venous thrombosis follows malignancy, oral contraceptives, pregnancy, factor V Leiden, or protein C or S deficiency. In chronic mesenteric ischemia, pain begins 15–30 minutes after meals and lasts up to 4 hours, causing fear of eating and weight loss. Ischemic colitis mostly affects the watershed areas, the splenic flexure (Griffiths point) and rectosigmoid junction (Sudeck point), which account for ~70%.

Clinical presentation and complications

Symptoms are abdominal pain and hematochezia. Complications are AGMA, from increased vascular permeability and anaerobic respiration, and perforation (pneumoperitoneum, peritonitis). Acute ischemia causes pain out of proportion to examination, and chronic ischemia postprandial pain and weight loss. Mortality is 60–80% in acute mesenteric ischemia, and under 5% in nongangrenous ischemic colitis.

Diagnosis

Acute or chronic mesenteric ischemia is evaluated by CT angiogram, and ischemic colitis by CT AP and colonoscopy. CT angiography is preferred even with renal impairment, and lactate and LDH rise only late. Ischemic colitis is confirmed by colonoscopy with biopsy within 48 hours.

Management

Bowel decompression uses NPO, IV fluids, and NGT. ABX prevent sepsis, heparin prevents clotting, and revascularization uses endovascular procedures and surgery. Vasopressors and alpha-agonists are avoided, since they cause vasospasm. NOMI is treated by correcting the low-flow state, with catheter-directed papaverine, and venous thrombosis with anticoagulation. Mild ischemic colitis is managed supportively, moderate disease with broad-spectrum ABX and surgical review, and severe disease with surgery.

Relationships to other conditions

Vascular liver injury. The same low-flow logic applies: shock injures the watershed splenic flexure in the colon and zone 3 in the liver.

Diverticular disease. Both cause LLQ pain and hematochezia. Painless, large-volume bleeding favors diverticulosis, whereas ischemic colitis causes pain with bleeding.

IBD *and infectious colitis*. Bloody diarrhea is shared, but a recent episode of hypotension or vasopressor use points to ischemia.

Embolic disease. A-fib embolizes to both the SMA and the brain, so the same source explains both.

Diverticular diseases: from pressure-driven outpouching to inflammation

Increased sigmoid colon intraluminal pressure pushes the mucosa and submucosa out through the muscularis externa, forming diverticula. Diverticulitis follows when a fecalith obstructs a diverticulum. LLQ pain is the main symptom, and complications include LGIB, abscess, enterovesical fistula, and perforation. Here, we review their mechanisms and management, and consider how they relate to ischemic colitis, IBD, and IBS.

Key points

- Low-fiber diet, constipation, obesity, and older age raise sigmoid pressure.
- Suspected diverticulitis is evaluated with CT AP with contrast.
- Avoid colonoscopy during acute diverticulitis; perform it 6–8 weeks after recovery.
- Treatment follows the complication, and severity is monitored for potential colectomy.

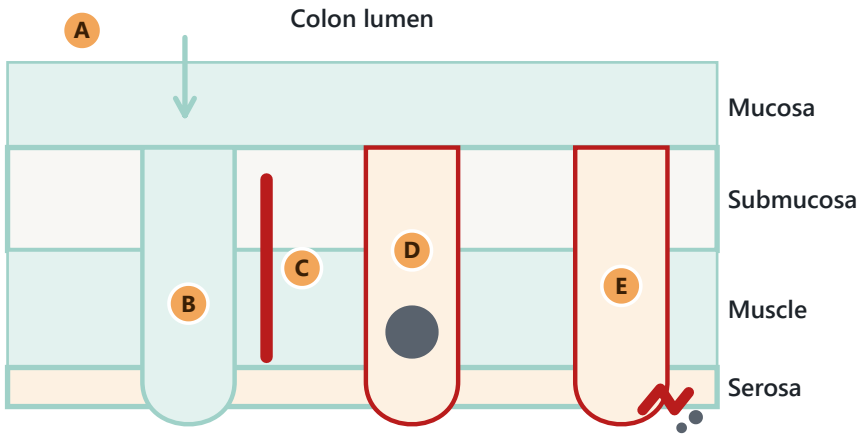


Fig. 1 | A colon wall in section with mucosa and submucosa pushed through the muscle to form a pouch beside a vessel. A second pouch holds a fecalith and is inflamed, and a third has perforated. A, Raised pressure; B, False diverticulum; C, Bleeding; D, Diverticulitis; E, Perforation.

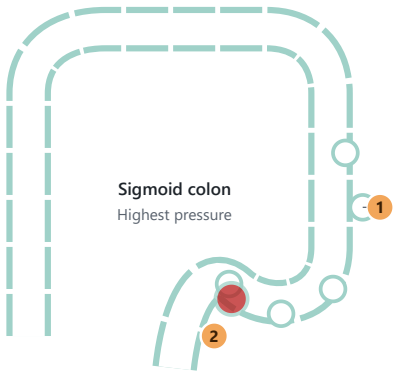


Fig. 2 | Colon with outpouchings along the sigmoid, one of them inflamed. 1, Diverticulosis; 2, Diverticulitis.

Pathophysiology

Low fiber and constipation increase sigmoid intraluminal pressure, which pushes mucosa and submucosa through the muscularis externa where the vasa recta enter, forming false diverticula. A vessel beside a diverticulum can bleed, causing LGIB with hematochezia. A fecalith can obstruct a diverticulum, and the back pressure and bacterial colonization that follow cause diverticulitis with

LLQ pain. An inflamed diverticulum can rupture, causing abscess, pneumoperitoneum, and peritonitis.

Etiology and mechanisms

Diverticulosis follows a low-fiber diet, constipation, obesity, and older age, which increase sigmoid colon intraluminal pressure and form diverticula. In diverticulitis, a fecalith obstructs a diverticulum, causing increased back pressure and bacterial colonization, with LLQ pain and inflammation. Diverticulosis affects over half of people over 60 and is left-sided in up to 90% in Western countries, but only ~4% develop diverticulitis. Meckel diverticulum forms when the vitelline (omphalomesenteric) duct fails to close and is the most common congenital GI anomaly. Its rule of 2s is 2% of people, 2% symptomatic, under age 2, males twice as often, 2 feet from the ileocecal valve, up to 2 inches long, and 2 types of ectopic tissue. Acid from ectopic gastric mucosa causes painless rectal bleeding, currant jelly or brick-colored in children, and accounts for ~50% of LGIB under age 2; in adults, small bowel obstruction is the most common complication, and intussusception also occurs.

Clinical presentation and complications

The main symptom is LLQ pain. Complications are LGIB (hematochezia), abscess (persistent pain and fever refractory to treatment), enterovesical fistula (pneumaturia, fecaluria, UTI), and perforation (pneumoperitoneum, peritonitis). Diverticulosis is usually asymptomatic, while diverticulitis causes LLQ pain, fever, and leukocytosis. Diverticular bleeding is painless and comes from the right colon in 50–90%. Pylephlebitis, a septic portal or mesenteric vein clot, carries a mortality of up to 25%, and up to 30% have a recurrence after a first episode.

Diagnosis

Perforation is evaluated by abdominal x-ray and CT AP, bleeding by colonoscopy, and suspected diverticulitis by CT AP with contrast. Colonoscopy is avoided during acute diverticulitis and performed 6–8 weeks after recovery. Diverticulitis shows leukocytosis and raised CRP. Meckel diverticulum is shown by a Meckel scan, in which technetium-99m is taken up by ectopic gastric mucosa.

Management

Perforation is treated by laparotomy, bleeding by colonoscopy, abscess by ABX, percutaneous drainage, and surgery, fistula by surgery, and infection by ABX. Otherwise, fiber intake is increased. Severity is monitored for potential colectomy. Uncomplicated diverticulitis is treated as an outpatient with a liquid diet and paracetamol rather than NSAIDs, with ABX (amoxicillin-clavulanate, or a fluoroquinolone with metronidazole, for 7–10 days) if the patient is immunocompromised, comorbid, or has a CRP over 140. Fever, inability to keep hydrated, immunocompromise, or comorbidity calls for admission and IV ABX. Perforation is treated by laparotomy with a Hartmann procedure or primary anastomosis. Nuts and seeds are safe, and a symptomatic Meckel diverticulum is excised.

Relationships to other conditions

Ischemic colitis. Both cause LLQ pain and hematochezia, but diverticular bleeding is typically painless, whereas ischemic colitis causes pain with bleeding after hypoperfusion.

Crohn disease. Enterovesical fistula with pneumaturia, fecaluria, and UTI occurs in both.

Appendicitis. The mechanisms mirror each other: a fecalith obstructs a blind pouch, and pressure and infection follow. Diverticulitis causes LLQ pain, and appendicitis RLQ pain.

IBS. Both can cause pain and altered bowel habit, but IBS has no structural damage.

Inflammatory bowel disease: contrasting Crohn disease and ulcerative colitis

IBD comprises Crohn disease and ulcerative colitis, which differ in location, depth, and lesion pattern. Crohn disease occurs anywhere in the GIT with transmural skip lesions, whereas ulcerative colitis is continuous from the rectum and confined to the colon. Their complications therefore differ, while extraintestinal features overlap. Here, we review their pathology, etiology, complications, and stepwise treatment, and consider how they relate to IBS, malabsorption, and PSC.

Key points

- Crohn disease is transmural with skip lesions; ulcerative colitis is continuous from the rectum.
- Smoking is a risk factor for Crohn disease and decreases the risk of ulcerative colitis.
- Fecal calprotectin and ESR/CRP rule out IBS.
- Primary sclerosing cholangitis occurs only in UC among the extraintestinal features listed.

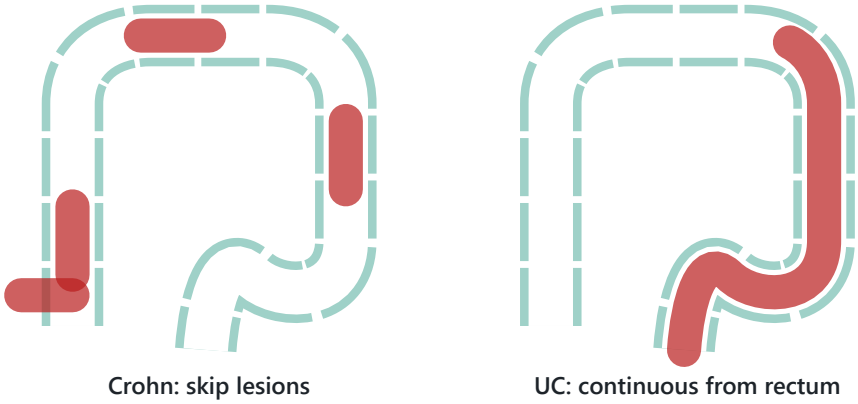


Fig. 1 | Two colons: Crohn disease with skip lesions from the ileum, sparing the rectum; ulcerative colitis with continuous disease starting at the rectum.

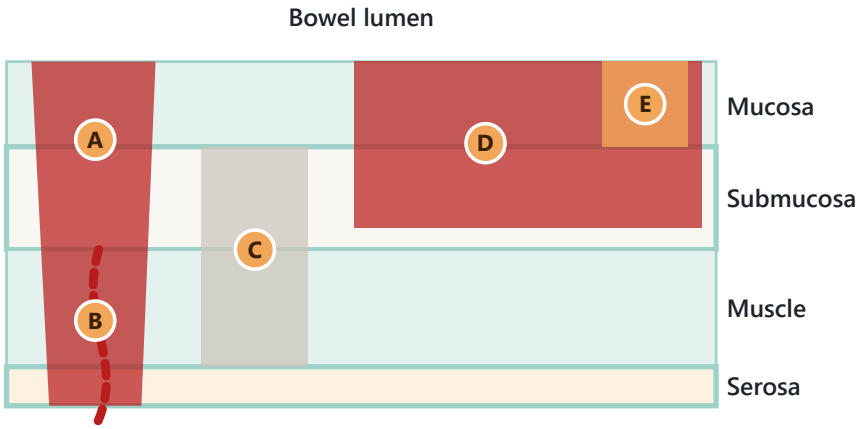


Fig. 2 | A bowel wall in section. On the left, inflammation runs through every layer with a fistula tract and a narrowing. On the right, continuous inflammation stays in the mucosa and submucosa, with a small area of dysplasia. A, Transmural inflammation; B, Fistulas and abscesses; C, Strictures; D, Continuous inflammation; E, Dysplasia.

Pathophysiology

Genes and environment cause immune dysregulation, with Th1 predominating in Crohn disease and Th2 in ulcerative colitis, and the result is chronic inflammation. Crohn disease involves every layer of the wall, in skip lesions, so tracts through the wall form enterovesical, enteroenteral, and enterocutaneous fistulas, and scarring causes small bowel obstruction. Ulcerative colitis spreads continuously from the rectum, causing bloody diarrhea. Chronic inflammation raises the risk of colorectal cancer.

Pathology

Crohn disease occurs anywhere in the GIT, most commonly in the ileum, is rectum-sparing, and causes RLQ pain. Lesions skip, diarrhea is watery and bloody in flare-ups, ulcers are transmural, granulomas may be present, and there is Th1 dysregulation. Ulcerative colitis involves the colon only, starts at the rectum, and causes LLQ pain. Lesions are continuous, diarrhea is bloody (hematochezia), ulcers are submucosal, there are no granulomas, and there is Th2 dysregulation.

Etiology

Crohn disease is associated with family history, NOD-2 mutations, Western diet, and smoking. Ulcerative colitis is associated with family history, HLA DRB1, and Western diet, and smoking decreases risk.

Clinical presentation and complications

In Crohn disease, malabsorption is especially marked when the ileum is involved. Decreased fat absorption causes steatorrhea, fat-soluble vitamin deficiencies, and increased free oxalate levels with nephrolithiasis from oxalate. Decreased B12 absorption causes anemia. Decreased enterohepatic circulation decreases bile salt recycling and increases cholesterol, causing cholelithiasis. Fistulas and abscesses include enterovesical fistula (pneumaturia, fecaluria, UTI), enteroenteral fistula (diarrhea), enterocutaneous fistula (increased drainage), and perianal abscesses. Strictures cause small bowel obstruction with cramping, abdominal pain and distention, vomiting, and obstipation. In ulcerative colitis, colitis causes pain, fever, and bloody stools, and decreased myenteric plexus activity leads to megacolon, tenesmus, and perforation; chronic inflammation leads to colorectal cancer. Extraintestinal features are arthralgia, uveitis, pyoderma gangrenosum, erythema nodosum, hypercoagulability (VTE, DVT, PE), and primary sclerosing cholangitis. Crohn disease also causes oral aphthous ulcers, and at least 5% of patients with ulcerative colitis develop colorectal cancer over time. Toxic megacolon follows severe colitis, usually ulcerative colitis or *C. difficile*, and can be precipitated by opioids, anticholinergics, and antimotility drugs. It requires a colon over 6 cm on x-ray, at least 3 of fever over 38 °C, heart rate over 120, neutrophilia, or anemia, and 1 of dehydration, confusion, electrolyte disturbance, or hypotension; it is the leading cause of death in ulcerative colitis.

Diagnosis

Perforation and small bowel obstruction are evaluated by abdominal x-ray, and nephrolithiasis or cholelithiasis by ultrasound. IBS is ruled out with fecal calprotectin and ESR/CRP, and infectious diarrhea with stool culture, O&P, and *C. diff*. Otherwise, ileocolonoscopy or colonoscopy with biopsy is obtained. In Crohn disease, colonoscopy shows skip lesions and cobblestone mucosa, noncaseating granulomas are found in up to 33%, small bowel follow-through shows a string sign from stricture or spasm, and ASCA is more common. In ulcerative colitis, colonoscopy shows continuous inflammation from the rectum, friable mucosa, and pseudopolyps, biopsy shows cryptitis, crypt distortion, and crypt abscesses, a barium enema in longstanding disease shows a stove-pipe, or lead-pipe, colon, and p-ANCA is positive in 60–70%. Colonoscopy surveillance for colorectal cancer is done every 1–2 years in ulcerative colitis.

Management

Treatment moves up a pyramid with severity. 5-ASA and sulfasalazine are maintenance for UC; corticosteroids (prednisone, budesonide) are for induction and flare-ups; immunomodulators (AZA, 6-MP, MTX) are maintenance for Crohn disease; biological therapy comprises certolizumab, infliximab, adalimumab, ustekinumab, and vedolizumab; and surgery is the top tier. Colectomy cures ulcerative colitis and is used for failed medical therapy, fulminant colitis, toxic megacolon, perforation, uncontrolled bleeding, or dysplasia. Toxic megacolon is treated with an IV corticosteroid for 3 days, then infliximab or cyclosporine for 3 days, and subtotal colectomy with end ileostomy if this fails. Crohn disease is treated surgically by resection or stricturoplasty for complications, and recurs at the anastomosis in ~90%.

Relationships to other conditions

IBS. Both cause pain and altered bowel habit, but IBS has no structural damage and normal inflammatory markers.

Malabsorption. Ileal Crohn disease produces the same fat, B12, and bile salt losses.

PSC. Most PSC patients have IBD, usually ulcerative colitis, and the combination raises colorectal cancer risk.

Infectious colitis. Bloody diarrhea overlaps, so infection is excluded before IBD is diagnosed.

Gallstones. Ileal disease leads to cholesterol gallstones through lost bile salts.

Irritable bowel syndrome: a disorder of gut-brain interaction without structural damage

IBS is a type of DGBI that stems from visceral hypersensitivity, altered microbiome, and hypermotility, with a lack of structural damage. Patients have recurrent abdominal pain with a predominant bowel pattern of diarrhea, constipation, or both. Diagnosis uses the Rome V criteria. Here, we review its mechanisms, diagnosis, and symptom-directed treatment, and consider how it is distinguished from IBD.

Key points

- IBS has no structural damage.
- Rome V requires recurrent pain for at least 3 days per month with 2 of 3 criteria.
- Treatment is directed at the dominant symptom.

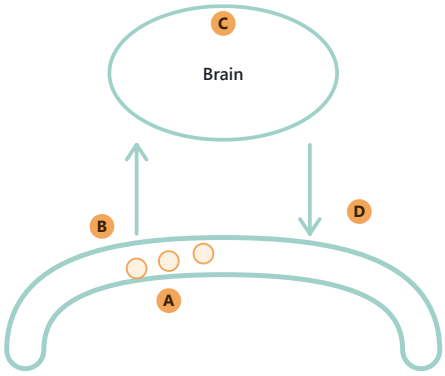


Fig. 1 | A brain above a healthy colon. Gas stretches the bowel, a signal rises to the brain and is amplified, and contractions pass along the bowel, with no structural damage. A, Gas and osmotic load; B, Stretch receptors; C, CNS amplification; D, Hypermotility.

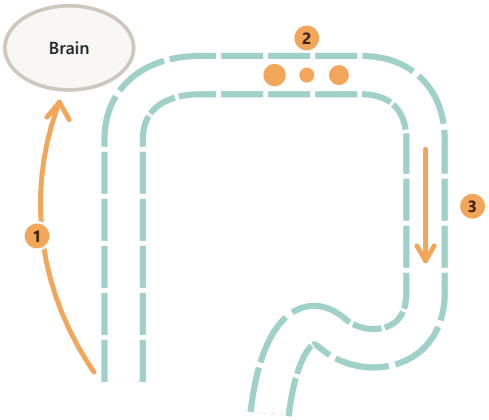


Fig. 2 | Gut-brain loop: stretch signals from the colon are amplified in the brain, with gas in the bowel and increased motility. There is no structural damage to show. 1, Visceral hypersensitivity; 2, Altered microbiome; 3, Hypermotility.

Pathophysiology

There is no structural damage. An altered microbiome turns carbohydrates into gas, and gas and osmotic effects stimulate stretch receptors, causing reflex contraction. Visceral hypersensitivity means the CNS amplifies these signals, causing abdominal pain, and hypermotility causes diarrhea.

Etiology and mechanisms

In visceral hypersensitivity, psychiatric disorders, fibromyalgia, depression, and anxiety increase CNS amplification of stretch receptor stimulation from osmotic effects of carbohydrates, causing abdominal pain. In altered microbiome, the gut microbiome metabolizes carbohydrates into gas, which stimulates stretch receptors and reflex contraction, causing

abdominal pain. In hypermotility, increased gut motility causes diarrhea.

Clinical presentation

The predominant bowel pattern is abdominal pain with diarrhea, constipation, or mixed. Alarm features that point to another cause are onset at 50 or older, blood in the stool, fever or night sweats, nocturnal diarrhea, weight loss, iron deficiency anemia, raised CRP or fecal calprotectin, a family history of colorectal cancer, IBD, or celiac disease, and recent ABX.

Diagnosis

The Rome V criteria require recurrent but not continuous abdominal pain or discomfort for at least 3 days per month, and 2 of the following: related to defecation, changes in stool frequency, and changes in stool appearance. Perforation is evaluated by abdominal x-ray and CT AP. Diarrhea is evaluated by fecal calprotectin, ESR/CRP, O&P, *C. diff*, and colonoscopy with biopsy. Diarrhea also calls for celiac serology with total IgA, and alarm features call for colonoscopy with random biopsies of the left and right colon.

Management

Triggers are reduced with a low FODMAP diet. Abdominal pain is relieved with TCAs and peppermint oil, constipation with laxatives, and diarrhea with loperamide. Abdominal pain is treated with antispasmodics such as dicyclomine first, then peppermint oil, then TCAs, with rifaximin if these fail. Constipation is treated with PEG first, then lubiprostone, linaclotide, or plecanitide. Diarrhea is treated with loperamide, then rifaximin or eluxadoline, which is avoided after cholecystectomy, and alosetron for women with severe disease, which carries a risk of ischemic colitis. Soluble fiber such as psyllium, not insoluble fiber, improves global symptoms, and psychological therapy helps refractory disease.

Relationships to other conditions

IBD. Inflammatory markers separate them: fecal calprotectin and ESR/CRP are normal in IBS.

Malabsorption. Carbohydrates drive symptoms in both, through osmotic effects and gas. Celiac disease must be excluded when symptoms suggest IBS.

Chronic diarrhea. Alarm features such as iron-deficiency anemia, rectal bleeding, weight loss, and nocturnal symptoms call for further investigation.

Appendicitis: luminal obstruction, migrating pain, and perforation

Obstruction of the appendix lumen leads to increased pressure, infection, and ischemia, causing inflammation and eventually perforation. Pain classically migrates from the periumbilical region to the RLQ as inflammation reaches the parietal peritoneum. Here, we review its causes, examination signs, age-specific imaging, and treatment, and consider how it relates to diverticulitis and other causes of abdominal pain.

Key points

- Fecalith is the most common cause in adults, and lymphoid hyperplasia in children.
- Pain migrates from visceral (T8–T10) to parietal over 12–24 hours.
- Imaging depends on the patient: CT AP in adults, and ultrasound first in children and pregnancy.
- Diagnosis is often clinical.

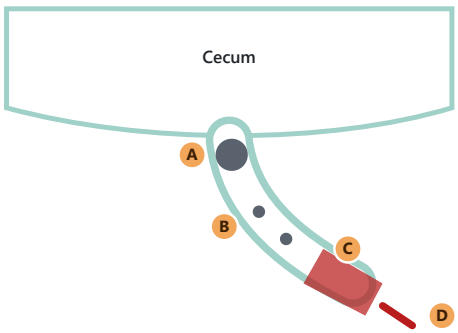


Fig. 1 | An appendix hanging from the cecum, blocked at its base by a fecalith. Beyond the block it swells with bacteria, its tip turns pale, and the tip has perforated. A, Obstruction; B, Distention and infection; C, Ischemia; D, Perforation.

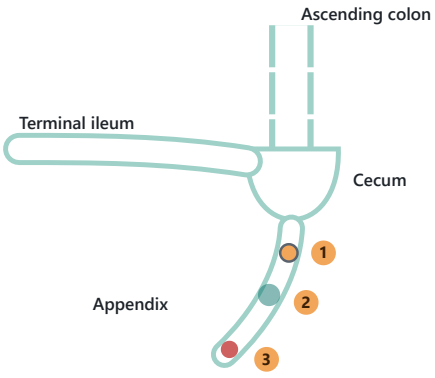


Fig. 2 | Cecum and terminal ileum with the appendix hanging below, showing a fecalith at its base, lymphoid swelling, and a tumor at the tip. 1, Fecalith; 2, Lymphoid hyperplasia; 3, Neoplasm.

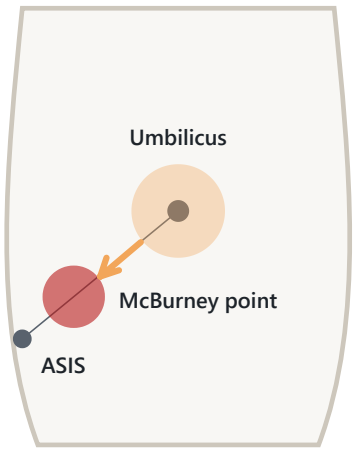


Fig. 3 | Abdomen with pain moving from the umbilicus to McBurney point in the right lower quadrant. The patient's right is on the left.

Pathophysiology

A fecalith or lymphoid tissue blocks the lumen, and increased intraluminal pressure causes distention and infection, with periumbilical visceral pain. Pressure then compresses the wall, causing ischemia and infarction, inflammation reaches the parietal peritoneum, causing sharp RLQ pain, and the dead wall ruptures, causing abscess and peritonitis.

Etiology and mechanisms

Obstruction arises from a fecalith, most common in adults; lymphoid hyperplasia, in children and after viral infection; or a neoplasm, such as carcinoid, in older adults. Increased intraluminal pressure causes distention and infection, then ischemia and infarction, then perforation and abscess.

Clinical presentation and complications

Periumbilical visceral pain (T10) moves to sharp, localized RLQ pain as inflammation reaches the parietal peritoneum over 12–24 hours, with anorexia, nausea and vomiting after the pain, and low-grade fever. McBurney point is tenderness one-third of the way from the right ASIS to the umbilicus and suggests typical appendix position. Rovsing sign is pain in the RLQ on pressing the LLQ, suggesting peritoneal irritation in the RLQ. Psoas sign is pain on extending the right hip with the patient lying on the left side, suggesting a retrocecal appendix irritating the iliopsoas. Obturator sign is pain on internally rotating the flexed right hip, suggesting a pelvic appendix irritating the obturator internus. Rebound tenderness and guarding, with pain when pressure is released and a tense abdominal wall, suggest peritonitis. Complications are perforation after 48–72 hours, particularly in young children and older adults; abscess or phlegmon; and generalized peritonitis and sepsis.

Diagnosis

Bloods show leukocytosis with left shift and CRP, β-hCG is obtained in anyone who could be pregnant to rule out ectopic pregnancy, and urinalysis is performed. Adults have CT AP with contrast, showing an appendix over 6 mm and fat stranding. Children and pregnant patients have ultrasound first to avoid radiation, with MRI if unclear in pregnancy. Diagnosis is often clinical (e.g. Alvarado score).

Management

Uncomplicated appendicitis is treated by laparoscopic appendectomy with perioperative ABX, or ABX alone in selected cases. Perforation with generalized peritonitis requires emergency appendectomy with IV ABX. Perforation with a contained abscess is treated with IV ABX and percutaneous drainage, with appendectomy later if needed.

Relationships to other conditions

Diverticulitis. Both are blind pouches obstructed by a fecalith, with abscess and perforation as complications. Appendicitis causes RLQ pain, and diverticulitis LLQ pain.

Crohn disease. RLQ pain is shared; a longer history with diarrhea and weight loss favors Crohn disease.

Carcinoid syndrome. Carcinoid is a cause of appendiceal obstruction in older adults, and can cause secretory diarrhea.

Ectopic pregnancy. This is the mimic to exclude, so β-hCG is obtained in anyone who could be pregnant.

Localizing abdominal pain: from where it hurts to the test that confirms it

Abdominal pain is localized by its site, its character and triggers, the signs it brings beyond the abdomen, and specific examination findings, and is confirmed by targeted bloods, stool tests, imaging, and endoscopy. Dangerous causes outside the gut must be excluded early. Here, we review how each step narrows the differential and which test answers each question.

Key points

- Visceral pain is dull and poorly localized; somatoparietal pain from the parietal peritoneum is sharp and localized.
- Pain out of proportion to the examination suggests acute mesenteric ischemia.
- Myocardial infarction, ruptured AAA, aortic dissection, ectopic pregnancy, and torsion must be excluded early.
- An upright CXR is the first test for suspected perforation, and RUQ ultrasound for RUQ pain.

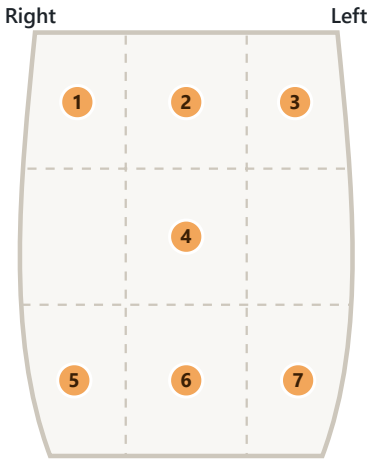


Fig. 1 | Abdomen divided into 9 regions, numbered 1 to 7: right upper quadrant, epigastrium, left upper quadrant, periumbilical region, right lower quadrant, suprapubic region, and left lower quadrant. The patient's right is on the left. Diffuse pain (8) has no single region. 1, Right upper quadrant; 2, Epigastric; 3, Left upper quadrant; 4, Periumbilical; 5, Right lower quadrant; 6, Suprapubic and pelvic; 7, Left lower quadrant.

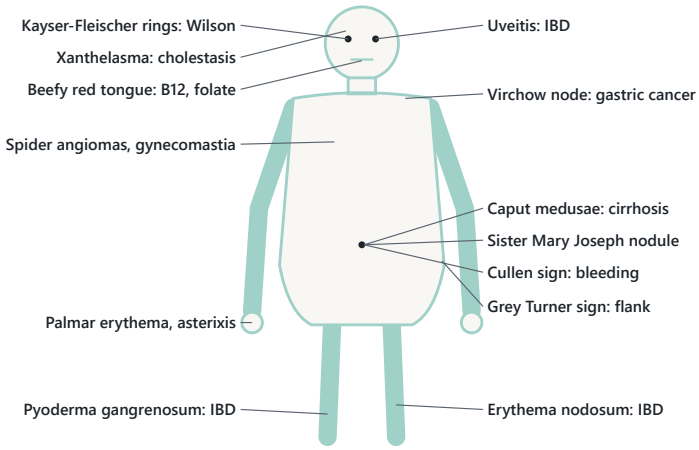


Fig. 2 | Signs beyond the abdomen placed on the body: Kayser-Fleischer rings in Wilson disease, xanthelasma in cholestasis, and uveitis in IBD at the eyes; a beefy red tongue in B12 or folate deficiency; a Virchow node above the left clavicle in gastric cancer; spider angiomas and gynecomastia on the chest; palmar erythema and asterixis at the hands; caput medusae, a Sister Mary Joseph nodule, and the Cullen sign at the umbilicus; the Grey Turner sign at the flank; and pyoderma gangrenosum and erythema nodosum on the legs in IBD.

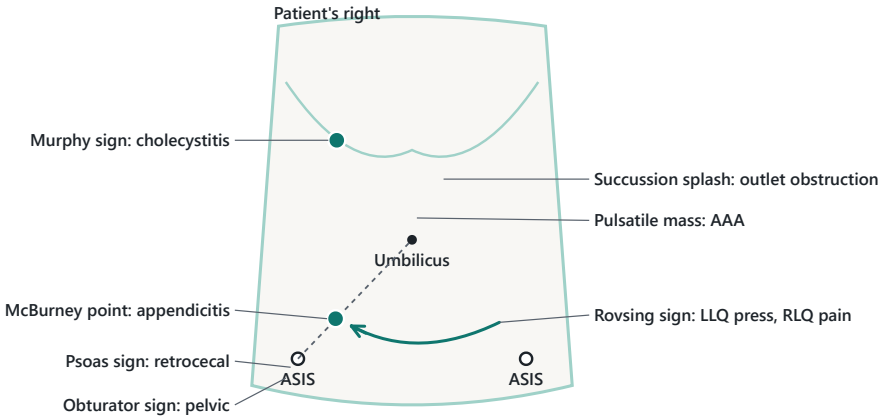


Fig. 3 | Examination signs on the abdomen, with the patient's right on the left: the Murphy sign under the right costal margin in cholecystitis; McBurney point a third of the way from the right ASIS to the umbilicus in appendicitis; the Rovsing sign, pressure in the LLQ causing RLQ pain; the psoas and obturator signs at the right hip for retrocecal and pelvic appendicitis; a succussion splash over the stomach in gastric outlet obstruction; and a pulsatile mass above the umbilicus in AAA.

Approach

The location of pain narrows the organs involved, its character and triggers narrow the process, signs beyond the abdomen and specific examination findings point to particular conditions, and tests confirm the leading diagnosis. Visceral pain arises from stretch of hollow organs, with receptors in the serosa, mesentery, muscle, and mucosa, and is dull and poorly localized. Somatoparietal pain arises from inflamed or stretched parietal peritoneum and is sharp and localized, as when appendicitis reaches the RLQ. Referred pain occurs when visceral and somatic nerves share a spinal level, as in left shoulder pain with splenic rupture.

Location

Right upper quadrant pain suggests biliary colic, acute cholecystitis, cholangitis, acute hepatitis and liver abscess, and Budd-Chiari syndrome and portal vein thrombosis. Epigastric pain suggests peptic ulcer disease, gastritis, GERD, and acute and chronic pancreatitis; exclude myocardial infarction. Left upper quadrant pain suggests splenic infarct, abscess, or rupture (Kehr sign: left shoulder pain); pancreatitis. Periumbilical pain suggests early appendicitis, before it localizes; embolism to the proximal SMA (mesenteric ischemia). Right lower quadrant pain suggests appendicitis at McBurney point, terminal ileal Crohn disease, and right-sided diverticulitis; ectopic pregnancy and ovarian torsion. Suprapubic and pelvic pain suggests ectopic pregnancy, pelvic inflammatory disease, ovarian or testicular torsion, urinary tract infection or obstruction, and incarcerated hernia. Left lower quadrant pain suggests diverticulitis, ischemic colitis, and ulcerative colitis. Diffuse pain suggests acute mesenteric ischemia, bowel obstruction, perforation with peritonitis, and ruptured AAA; pneumonia or PE can cause diffuse upper pain.

Character of pain

Colicky RUQ or epigastric pain after a fatty meal, gone within 6 hours; normal LFTs, no fever points to biliary colic. RUQ pain over 6 hours, fever, and Murphy sign points to acute cholecystitis. Fever, RUQ pain, and jaundice (Charcot triad), with hypotension and confusion in Reynolds pentad points to ascending cholangitis. Burning epigastric pain, worse with eating points to gastric ulcer. Burning epigastric pain, better with eating, worse fasting and at night points to duodenal ulcer. Heartburn and regurgitation points to GERD. Severe epigastric or LUQ pain radiating to the back, with vomiting points to acute pancreatitis. Chronic epigastric pain, better leaning forward, with steatorrhea points to chronic pancreatitis. Periumbilical pain moving to the RLQ over 12–24 hours, anorexia, then vomiting points to appendicitis. Sudden, severe pain out of proportion to the examination; the patient writhes points to acute mesenteric ischemia. Pain 15–30 minutes after meals, lasting up to 4 hours, with fear of eating and weight loss points to chronic mesenteric ischemia. LLQ pain, fever, and leukocytosis points to diverticulitis. LLQ pain and hematochezia after shock or vasopressors points to ischemic colitis. RLQ pain, chronic diarrhea, weight loss, and perianal disease points to crohn disease. Bloody diarrhea with tenesmus points to ulcerative colitis. Recurrent pain related to defecation, with changes in stool frequency or appearance points to IBS. Pain, vomiting, obstipation, and distention points to bowel obstruction, including gallstone ileus. Sudden pain, then a rigid abdomen; the patient lies still points to perforation with peritonitis, such as a perforated ulcer. Painless rectal bleeding points to diverticular bleeding; under age 2, Meckel diverticulum.

Causes outside the gut

Epigastric pain with cardiac risk factors, dyspnea, sweating, or exertional symptoms suggests myocardial infarction, evaluated with ECG and troponin. Abdominal, back, or flank pain with hypotension and a pulsatile mass suggests ruptured AAA, evaluated with bedside ultrasound if unstable; CT if stable. Sharp, tearing pain with pulse deficits suggests aortic dissection, evaluated with CT angiogram of the chest, abdomen, and pelvis. Pain and vaginal bleeding 6–8 weeks after the last period suggests ectopic pregnancy, evaluated with β-hCG, then transvaginal ultrasound. Severe pelvic pain and vomiting with an adnexal mass suggests ovarian torsion, evaluated with pelvic ultrasound. Severe testicular and lower abdominal pain in a young male suggests testicular torsion, evaluated with testicular examination; scrotal ultrasound. LUQ pain after trauma, referred to the left shoulder suggests splenic rupture (Kehr sign), evaluated with imaging, with resuscitation.

Signs beyond the abdomen

Jaundice points to biliary obstruction, hepatitis, or hemolysis; see Reading Liver Function Tests. Painless jaundice with a palpable, nontender gallbladder (Courvoisier law) points to periampullary or biliary malignancy. Dark urine, pale stools, and pruritus points to cholestasis, such as a bile duct stone. Pyoderma gangrenosum, erythema nodosum, uveitis, arthralgia, or oral aphthous ulcers points to IBD; pyoderma gangrenosum accompanies ulcerative colitis in 5–12% and Crohn disease in 1–2%. Dermatitis herpetiformis points to celiac disease. Angular cheilitis or a smooth, pale tongue; a beefy, red tongue points to iron deficiency; B12 or folate deficiency, as in malabsorption or autoimmune gastritis. Virchow node, Sister Mary Joseph nodule, or Krukenberg tumor points to metastatic gastric cancer. Acanthosis nigricans or Trousseau syndrome points to gastric cancer, as a paraneoplastic sign. Cullen sign (periumbilical bruising) or Grey Turner sign (flank bruising) points to intra-abdominal or retroperitoneal bleeding, as in necrotizing pancreatitis. Asterixis, fetor hepaticus, spider angiomas, palmar erythema, gynecomastia, ascites, or caput medusae points to cirrhosis and liver failure. Kayser-Fleischer rings points to wilson disease. Bronze skin with diabetes points to hemochromatosis. Xanthelasma points to chronic cholestasis, such as PBC. Flushing, wheezing, and tricuspid valve disease points to carcinoid syndrome. Migratory large-joint arthralgia with weight loss and diarrhea points to whipple disease. Pneumaturia and fecaluria points to enterovesical fistula, from diverticulitis or Crohn disease.

Examination

Distention and visible peristalsis suggests bowel obstruction. Dilated veins draining away from the umbilicus suggests portal hypertension. Pulsatile mass suggests AAA. High-pitched, tinkling bowel sounds suggests ileus or early small bowel obstruction. Absent bowel sounds suggests late obstruction, intestinal ischemia, ileus, or peritonitis. Shifting dullness suggests ascites. Murphy sign: inspiration stops as the gallbladder meets the hand suggests acute cholecystitis. Tenderness at McBurney point, a third of the way from the ASIS to the umbilicus suggests appendicitis. Rovsing sign: LLQ pressure causes RLQ pain suggests appendicitis. Psoas sign: pain on passive extension of the right hip suggests retrocecal appendicitis. Obturator sign: pain on internally rotating the flexed right hip; sensitivity only 8% suggests pelvic appendicitis or abscess. Rebound tenderness, involuntary guarding, or rigidity suggests peritonitis. Succussion splash suggests delayed gastric emptying, as in gastric outlet obstruction. Rectal examination suggests GI bleeding; adds little in appendicitis. Pelvic or testicular examination suggests pelvic disease or torsion in lower abdominal pain.

Investigations

For any significant abdominal pain, obtain FBC, electrolytes, creatinine, glucose, calcium, LFTs, and lipase; β-hCG if pregnancy is possible; urinalysis. For peritonitis, sepsis, or suspected perforation, obtain upright CXR first, for free air; blood cultures; then CT AP with contrast. For RUQ pain, obtain LFTs and lipase; RUQ ultrasound first; HIDA if cholecystitis is unclear; MRCP or endoscopic ultrasound for a bile duct stone. For epigastric pain, obtain ECG and troponin if cardiac risk; lipase; EGD for suspected ulcer; urea breath test or stool antigen for *H. pylori*. For suspected appendicitis, diverticulitis, or obstruction, obtain CT AP with contrast in adults; ultrasound first in children and pregnancy. For suspected mesenteric ischemia, obtain CT angiogram; lactate rises only late (Intestinal Ischemia). For chronic diarrhea or suspected IBD, obtain fecal calprotectin and CRP; stool culture, O&P, and *C. difficile*; colonoscopy with biopsy (IBD). For suspected malabsorption, obtain tTG-IgA with total IgA, then duodenal biopsy; fecal elastase; hydrogen breath test (Malabsorption). For painless rectal bleeding in a young child, obtain meckel scan (Diverticular Diseases). EGD is used for persistent upper abdominal pain or pain with alarm features; persistent vomiting; dysphagia or odynophagia; refractory GERD and Barrett surveillance; upper GI bleeding; chronic diarrhea or malabsorption; iron deficiency anemia with a suspected upper source. Colonoscopy is used for rectal bleeding, changed bowel habit, weight loss, iron deficiency anemia, positive FOBT,

and chronic diarrhea (with biopsies); screening from age 45–50. Not needed for uncomplicated abdominal pain or diarrhea under 4 weeks; avoided in peritonitis and acute diverticulitis. ERCP is used for mainly therapeutic: removing bile duct stones, stenting strictures, and draining cholangitis.

Relationships to other conditions

Reading liver function tests. Jaundice and RUQ pain lead to the LFT pattern, which separates hepatocellular, cholestatic, and hemolytic causes.

Appendicitis. Its migrating pain shows the shift from visceral to somatoparietal pain.

Intestinal ischemia. Pain out of proportion to a soft abdomen distinguishes it from peritonitis, where the abdomen is rigid.

Reading liver function tests: patterns of injury, cholestasis, and failing function

The liver panel mostly measures injury and cholestasis rather than function: AST and ALT reflect hepatocyte injury, ALP and GGT reflect bile flow, and bilirubin, albumin, and INR reflect synthetic reserve. The pattern, the degree of rise, and the trend, read with the clinical picture, point to the cause. Here, we review each test, the main patterns, the bilirubin pathway, and how combinations of results and signs narrow the differential.

Key points

- ALT is the most liver-specific enzyme; AST also comes from muscle, heart, and red cells.
- A high ALP with a normal GGT suggests bone rather than liver.
- Massive rises over 10000 IU/L suggest paracetamol toxicity or profound ischemia.
- Falling AST and ALT with rising bilirubin and INR mean a failing liver, not recovery.

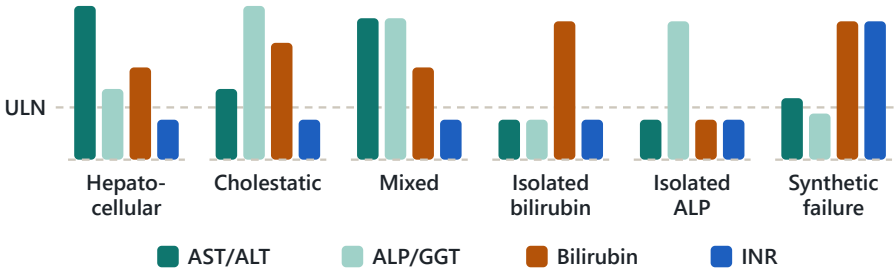


Fig. 1 | LFT patterns as bars against the upper limit of normal. Hepatocellular: AST and ALT rise much more than ALP. Cholestatic: ALP and GGT rise, often with bilirubin, more than AST and ALT. Mixed: both rise. Isolated bilirubin: bilirubin alone. Isolated ALP: ALP alone. Synthetic failure: bilirubin and INR rise while AST and ALT may already be falling.

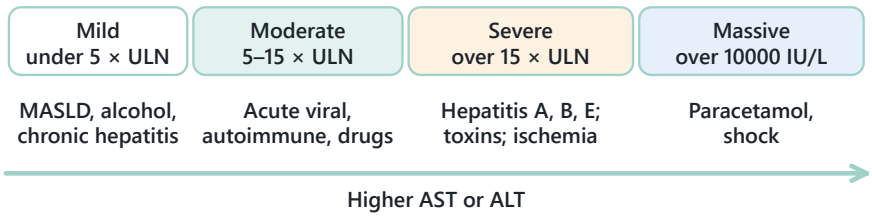


Fig. 2 | How high AST and ALT rise: mild, under 5 times the upper limit of normal, in MASLD, alcohol, and chronic hepatitis; moderate, 5 to 15 times, in acute viral and autoimmune hepatitis and drug injury; severe, over 15 times, in hepatitis A, B, or E, toxins, and ischemic hepatitis; and massive, over 10000 IU/L, in paracetamol toxicity and shock.

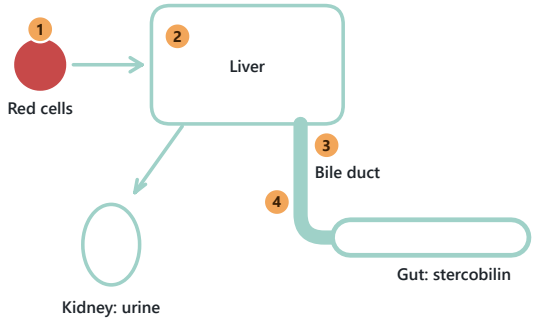


Fig. 3 | Bilirubin moves from red cells to the liver, where it is conjugated, then through the bile duct to the gut, with some reaching the kidney. Numbered sites: 1 red cells, 2 uptake and conjugation in the liver, 3 excretion into bile, 4 the bile duct. Only conjugated bilirubin is water soluble, so only it reaches the urine. 1, Too much heme; 2, Uptake and conjugation; 3, Hepatocyte injury and excretion into bile; 4, Bile duct obstruction.

Approach

Most of the panel measures injury or cholestasis rather than function: AST and ALT reflect hepatocyte injury, ALP and GGT reflect bile flow, and bilirubin, albumin, and INR reflect what the liver can still do. The pattern matters more than any single value, and trends more than a single result. Normal LFTs do not exclude liver disease, since MASLD, fibrosis, and compensated cirrhosis can all have normal enzymes.

The tests

ALT comes from hepatocyte cytosol; the most liver-specific, changes in hepatocellular injury, and has a normal range of 4–36 IU/L. AST comes from liver, muscle, heart, kidney, brain, and red cells; cytosol and mitochondria, changes in hepatocellular injury, alcohol, muscle injury, and hemolysis, and has a normal range of 5–30 IU/L. ALP comes from canalicular membrane and bile duct epithelium; also bone, placenta, and intestine, changes in cholestasis; bone disease, pregnancy, and growth, and has a normal range of 30–120 IU/L. GGT comes from bile ducts; induced by alcohol and drugs, changes in confirms that a high ALP comes from the liver, and has a normal range of 6–50 IU/L. Bilirubin comes from heme breakdown; conjugated by UGT1A1 in hepatocytes, changes in hemolysis, conjugation defects, liver injury, or obstruction, and has a normal range of 2–17 µmol/L; direct 0–6. Albumin comes from made only by the liver; half-life 14–21 days, changes in falls slowly in chronic liver disease, and has a normal range of 35–50 g/L. PT/INR comes from clotting factors made by the liver; II, VII, IX, and X need vitamin K, changes in rises quickly when synthesis fails, and has a normal range of PT 10.9–12.5 s.

Patterns

In the hepatocellular pattern, AST and ALT rise much more than ALP; causes include ALT higher: viral hepatitis, MASLD, hemochromatosis, Wilson disease, α1-antitrypsin deficiency, ischemic hepatitis and acute Budd-Chiari syndrome, autoimmune hepatitis, drugs and toxins, and celiac disease. AST higher: alcohol, cirrhosis, and nonhepatic causes (hemolysis, myopathy, thyroid disease, exercise). In the cholestatic pattern, ALP and GGT rise, often with bilirubin, more than AST and ALT; causes include bile duct obstruction, PBC and PSC, drugs, infiltration (sarcoidosis, amyloidosis, lymphoma), metastases, and cystic fibrosis. In the mixed pattern, both rise; causes include overlapping processes, or a pattern changing over time. In the isolated bilirubin pattern, bilirubin alone; fractionate it; causes include unconjugated: hemolysis or Gilbert syndrome. Conjugated: Dubin-Johnson or Rotor syndrome. In the isolated ALP pattern, ALP alone; check GGT; causes include normal GGT: bone, pregnancy, or growth. High GGT: a hepatobiliary source. In the synthetic failure pattern, low albumin, high INR, and rising bilirubin; causes include acute liver failure or decompensated cirrhosis; AST and ALT may already be falling. Falling AST and ALT with rising bilirubin and INR mean a failing liver, not recovery.

Degree of rise

A mild rise, under 5 × ULN suggests MASLD; alcohol, often with AST:ALT of 2:1 or more; chronic hepatitis B or C; drugs such as statins, NSAIDs, and antiepileptics; early hemochromatosis; muscle and thyroid disease. A moderate rise, 5–15 × ULN suggests acute viral hepatitis, autoimmune hepatitis, and drug-induced injury; alcoholic hepatitis rarely exceeds 300–500 IU/L. A severe rise, over 15 × ULN suggests acute hepatitis A, B, or E; severe drug injury; toxins such as *Amanita phalloides*; ischemic hepatitis. A massive rise, over 10000 IU/L suggests paracetamol toxicity, profound ischemia after cardiac arrest or shock, and, rarely, HSV hepatitis in pregnancy or immunocompromise.

Bilirubin

Bilirubin moves from red cells to the liver, where UGT1A1 conjugates it, then into bile and the gut, where bacteria form urobilinogen and stercobilin. Excess heme from hemolysis, a large hematoma, or ineffective erythropoiesis raises unconjugated bilirubin, rarely above 4 mg/dL. Defective uptake and conjugation also raise unconjugated bilirubin, as in Gilbert syndrome, with UGT1A1 at 30–50% and jaundice after fasting,

illness, menstruation, or exertion that needs no treatment, in Crigler-Najjar syndrome, and with drugs that block UGT1A1. Hepatocyte injury and defective excretion raise conjugated, often mixed, bilirubin, as in hepatitis, cirrhosis, drugs, and PBC, and in Dubin-Johnson syndrome (MRP2, with a black liver) and Rotor syndrome (without pigment). Bile duct obstruction by stones, strictures, PSC, or tumors raises conjugated bilirubin with ALP and GGT, causing pale stools, dark urine, and pruritus. Conjugated bilirubin is water soluble and darkens urine, whereas unconjugated bilirubin is bound to albumin and does not, so bilirubin in urine means a conjugated rise.

Synthetic function

Albumin: half-life 14–21 days, so it falls in chronic, not acute, liver disease; also low in malnutrition, inflammation, nephrotic syndrome, burns, sepsis, and protein-losing enteropathy. PT/INR: clotting factors have short half-lives, so it rises quickly; also prolonged by vitamin K deficiency, warfarin, DIC, and antiphospholipid antibodies. With a vitamin K test, INR that corrects with parenteral vitamin K points to deficiency, as in malabsorption; failure to correct means severe hepatocellular dysfunction. Bilirubin, albumin, and INR feed the Child-Pugh and MELD scores.

Combining results with the picture

Cholestatic pattern with fever, RUQ pain, and jaundice suggests ascending cholangitis, and calls for blood cultures, ultrasound, then ERCP. Cholestatic pattern with RUQ pain and a dilated CBD suggests choledocholithiasis, and calls for MRCP or endoscopic ultrasound; ERCP if likelihood is high. Cholestatic pattern with painless jaundice, weight loss, and a palpable gallbladder suggests periampullary or biliary malignancy, and calls for CT or MRCP; CA 19-9. Cholestatic pattern with itch and fatigue in a middle-aged woman suggests PBC, and calls for AMA, positive in over 90%. Cholestatic pattern with ulcerative colitis suggests PSC, and calls for MRCP: beaded ducts; p-ANCA. AST:ALT of 2:1 or more with raised GGT and heavy drinking suggests alcohol-related liver disease, and calls for maddrey score if alcoholic hepatitis. Very high ALT after paracetamol ingestion suggests paracetamol toxicity, and calls for paracetamol level; NAC. ALT over 1000 after shock, cardiac arrest, or heart failure suggests ischemic hepatitis, and calls for treat the low-flow state. ALT-predominant rise with viral risk factors suggests viral hepatitis, and calls for HAV IgM, HBsAg, anti-HBc IgM, and HCV antibody. ALT-predominant rise in a young woman with high IgG suggests autoimmune hepatitis, and calls for ANA and ASMA (type 1); anti-LKM1 (type 2). Mild rise with obesity, diabetes, or dyslipidemia suggests MASLD, and calls for ultrasound; fibrosis score such as FIB-4. Hepatocellular rise with diabetes, bronze skin, or arthropathy suggests hemochromatosis, and calls for transferrin saturation over 45%, then ferritin and HFE testing. Hepatocellular rise in a young patient with tremor, psychiatric change, or Kayser-Fleischer rings suggests wilson disease, and calls for ceruloplasmin. Hepatocellular rise with early lower-lobe emphysema suggests α1-antitrypsin deficiency, and calls for α1-antitrypsin level. Raised transaminases and hypoglycemia in the third trimester suggests acute fatty liver of pregnancy, and calls for swansea criteria. Isolated unconjugated bilirubin after fasting or illness, with normal enzymes and FBC suggests gilbert syndrome, and calls for no treatment needed. Unconjugated bilirubin with anemia suggests hemolysis, and calls for FBC and blood film. High INR and encephalopathy within 26 weeks, with no prior liver disease suggests acute liver failure, and calls for find the cause; transplant referral. Low albumin, high INR, ascites, and varices suggests cirrhosis, and calls for child-Pugh and MELD scores.

Relationships to other conditions

Localizing abdominal pain. RUQ pain and jaundice lead here, and the LFT pattern decides between imaging the ducts and testing for hepatitis.

Acute liver failure and cirrhosis. Bilirubin, albumin, and INR measure the failing liver and feed the Child-Pugh and MELD scores.

Choledocholithiasis. A cholestatic pattern with a dilated duct is its signature.

Abbreviations

25-OH 25-hydroxy

5-ASA 5-aminosalicylic acid

5-HIAA 5-hydroxyindoleacetic acid, a serotonin breakdown product

5-HT 5-hydroxytryptamine (serotonin)

5-HT3 Serotonin receptor 3, an ion channel

5-HT4 Serotonin receptor 4

6-MP 6-Mercaptopurine

A1AT Alpha-1 antitrypsin

AAA Abdominal aortic aneurysm

AAST American Association for the Surgery of Trauma

ABCG5/8 ATP-binding cassette transporters G5 and G8

ABX Antibiotics

ACh Acetylcholine

AChE Acetylcholinesterase

ADH Alcohol dehydrogenase

AFLP Acute fatty liver of pregnancy

AFP Alpha-fetoprotein

AGMA Anion gap metabolic acidosis

AIDS Acquired immunodeficiency syndrome

AKI Acute kidney injury

ALP Alkaline phosphatase

ALT Alanine aminotransferase

AMA Anti-mitochondrial antibody

ANA Antinuclear antibody

anti-HAV Antibody to hepatitis A virus

anti-HBc Antibody to hepatitis B core antigen

Anti-HCV Antibody to hepatitis C virus

anti-HDV Antibody to hepatitis D virus

anti-HEV Antibody to hepatitis E virus

anti-LKM1 Anti-liver-kidney microsomal antibody type 1

ApoB-100 Apolipoprotein B-100

ApoB-48 Apolipoprotein B-48

APRI AST to platelet ratio index

AQP4 Aquaporin-4

ARDS Acute respiratory distress syndrome

ASCA Anti-Saccharomyces cerevisiae antibodies

ASIS Anterior superior iliac spine

ASMA Anti-smooth muscle antibody

AST Aspartate aminotransferase

AT-III Antithrombin III

ATP7A Copper-transporting ATPase A

ATP7B Copper-transporting ATPase B

AZA Azathioprine

BB2 Bombesin receptor 2, the GRP receptor

BBB Blood-brain barrier

BCRP Breast cancer resistance protein

BISAP Bedside Index of Severity in Acute Pancreatitis

BMP Basic metabolic panel

BNP B-type natriuretic peptide

BSEP Bile salt export pump

BUN Blood urea nitrogen

CA 19-9 Cancer antigen 19-9

CagA Cytotoxin-associated gene A

cAMP Cyclic adenosine monophosphate

CaSR Calcium-sensing receptor

CAVO Colicky pain, abdominal distention, vomiting, and obstipation

CBD Common bile duct

CCB Calcium channel blocker

cccDNA Covalently closed circular DNA: a stable template that persists in the nucleus

CCK Cholecystokinin: released by the duodenum in response to fat; makes the gallbladder contract

CCK-2 Cholecystokinin receptor 2

CEA Carcinoembryonic antigen

CF Cystic fibrosis

CFTR Cystic fibrosis transmembrane conductance regulator gene

CINV Chemotherapy-induced nausea and vomiting

CIPO Chronic intestinal pseudo-obstruction

CKD Chronic kidney disease

CMV Cytomegalovirus

CN Cranial nerve

CN X Cranial nerve X (the vagus nerve)

CNS Central nervous system

COX Cyclooxygenase

CPT1 Carnitine palmitoyltransferase 1

CREST Calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia

CRP C-reactive protein

CSF Cerebrospinal fluid

CT Computed tomography

CT AP Computed tomography of the abdomen and pelvis

CTR1 Copper transporter 1

CTZ Chemoreceptor trigger zone

CVP Central venous pressure

CXR Chest x-ray

CYP Cytochrome P450

CYP1A2 Cytochrome P450 1A2

CYP2C19 Cytochrome P450 2C19

CYP2C9 Cytochrome P450 2C9

CYP2D6 Cytochrome P450 2D6

CYP2E1 Cytochrome P450 2E1

CYP3A4 Cytochrome P450 3A4

CYP7A1 Cholesterol 7α-hydroxylase

CYP8B1 Sterol 12α-hydroxylase

D2 Dopamine receptor 2

DGBI Disorders of gut-brain interaction

DGP Deamidated gliadin peptide

DIC Disseminated intravascular coagulation

DMT1 Divalent metal transporter 1

DMV Dorsal motor nucleus of the vagus

DMX Dorsal motor nucleus of the vagus

DNA Deoxyribonucleic acid

DVT Deep vein thrombosis

EBV Epstein-Barr virus

ECG Electrocardiogram

ECL Enterochromaffin-like

EGD Esophagogastroduodenoscopy

EGF Epidermal growth factor

EGFR Epidermal growth factor receptor

EHEC Enterohemorrhagic E. coli

EMA Endomysial antibody

ENS Enteric nervous system

EP Prostaglandin E receptor

EP3 Prostaglandin E receptor 3

EPI Exocrine pancreatic insufficiency

ER Endoplasmic reticulum

ERCP Endoscopic retrograde cholangiopancreatography

ESR Erythrocyte sedimentation rate

ETEC Enterotoxigenic E. coli

FAP Familial adenomatous polyposis

FAST Focused assessment with sonography in trauma

FBC Full blood count

FFP Fresh frozen plasma

FIB-4 Fibrosis-4 index

FNH Focal nodular hyperplasia

FOBT Fecal occult blood test

FODMAP Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols

GALT Gut-associated lymphoid tissue

GDH Glutamate dehydrogenase

GERD Gastroesophageal reflux disease

GGT Gamma-glutamyl transferase

GI Gastrointestinal

GIP Glucose-dependent insulinotropic peptide

GIT Gastrointestinal tract

GLP-1 Glucagon-like peptide 1

GLP-2 Glucagon-like peptide 2

GLUT2 Glucose transporter 2

GLUT5 Glucose transporter 5

Grey Turner Flank bruising

GRP Gastrin-releasing peptide

GSH Reduced glutathione

GSSG Oxidized glutathione

GST Glutathione S-transferase

H1 Histamine receptor 1

H2 Histamine receptor 2

H2R Histamine H2 receptor

H2RA Histamine-2 receptor antagonist

HAV Hepatitis A virus

HBeAg Hepatitis B e antigen: active replication, high infectivity

HBIG Hepatitis B immune globulin

HBsAg Hepatitis B surface antigen: current infection

HBV Hepatitis B virus

HBV DNA Viral load: active replication

HCC Hepatocellular carcinoma

HCl Hydrochloric acid

HCO_3^- Bicarbonate

HCV Hepatitis C virus

HDL High-density lipoprotein

HDV Hepatitis D (delta) virus

HELLP Hemolysis, elevated liver enzymes, low platelets

HEV Hepatitis E virus

HFE Hereditary hemochromatosis gene

HIDA Hepatobiliary iminodiacetic acid scan (cholescintigraphy)

HIV Human immunodeficiency virus

HLA Human leukocyte antigen

HMG-CoA 3-hydroxy-3-methylglutaryl coenzyme A

HSV Herpes simplex virus

HUS Hemolytic uremic syndrome

IBD Inflammatory bowel disease

IBS Irritable bowel syndrome

ICC Interstitial cells of Cajal

ICP Intracranial pressure

ICU Intensive care unit

IDL Intermediate-density lipoprotein

IF Intrinsic factor

IgA Immunoglobulin A

IgE Immunoglobulin E

IGF-1 Insulin-like growth factor 1

IgG Immunoglobulin G

IgG4 Immunoglobulin G4

IgM Immunoglobulin M

IL-1 Interleukin 1

IL-10 Interleukin 10

IL-13 Interleukin 13

IL-25 Interleukin 25

IL-4 Interleukin 4

IL-5 Interleukin 5

IL-6 Interleukin 6

IM Intramuscular

IMA Inferior mesenteric artery

IMV Inferior mesenteric vein

INR International normalized ratio

IRP Integrated relaxation pressure

IV Intravenous

IVC Inferior vena cava

JAK Janus kinase

JAK2 Janus kinase 2 (V617F mutation)

JAK2 V617F Janus kinase 2 V617F mutation, found in most myeloproliferative neoplasms

JVD Jugular venous distention

KCl Potassium chloride

LCHAD Long-chain 3-hydroxyacyl-CoA dehydrogenase, an enzyme of fatty acid beta-oxidation

LDH Lactate dehydrogenase

LDL Low-density lipoprotein

LES Lower esophageal sphincter

LFT Liver function test

LFTs Liver function tests

LGIB Lower gastrointestinal bleed

LGR5 Leucine-rich repeat-containing G protein-coupled receptor 5

LLQ Left lower quadrant

LPS Lipopolysaccharide

LUQ Left upper quadrant

M3 Muscarinic acetylcholine receptor 3

MAdCAM Mucosal addressin cell adhesion molecule

MALT Mucosa-associated lymphoid tissue

MASH Metabolic dysfunction-associated steatohepatitis (formerly NASH)

MASLD Metabolic dysfunction-associated steatotic liver disease (formerly NAFLD)

MCP Metacarpophalangeal

MDR1 Multidrug resistance protein 1

MDR3 Multidrug resistance protein 3

MELD Model for End-Stage Liver Disease

MEN1 Multiple endocrine neoplasia type 1

MMC Migrating motor complex

MPGN Membranoproliferative glomerulonephritis

MR Magnetic resonance

MRCP Magnetic resonance cholangiopancreatography

MRI Magnetic resonance imaging

MRP2 Multidrug resistance-associated protein 2

MRP4/6 Multidrug resistance-associated proteins 4 and 6

MTTP Microsomal triglyceride transfer protein gene

MTX Methotrexate

NAAT Nucleic acid amplification test

NAC N-acetylcysteine

NAD Nicotinamide adenine dinucleotide

NADH Reduced nicotinamide adenine dinucleotide

NADPH Reduced nicotinamide adenine dinucleotide phosphate

NAFLD Non-alcoholic fatty liver disease

NAGMA Non-anion gap metabolic acidosis

NANC Non-adrenergic, non-cholinergic

NAPQI N-acetyl-p-benzoquinone imine

NAT N-acetyltransferase

NF-κB Nuclear factor kappa B

NG Nasogastric

NGT Nasogastric tube

NK1 Neurokinin-1 receptor

NMOSD Neuromyelitis optica spectrum disorder

NO Nitric oxide

NOD-2 Nucleotide-binding oligomerization domain 2, a gene involved in immune responses to gut bacteria

NOMI Nonocclusive mesenteric ischemia

NPO Nil per os: nothing by mouth

NS3/4A Nonstructural protein 3/4A

NS5A Nonstructural protein 5A

NS5B Nonstructural protein 5B

NSAID Nonsteroidal anti-inflammatory drug

NSAIDs Nonsteroidal anti-inflammatory drugs

NTCP Sodium taurocholate co-transporting polypeptide, a bile salt transporter on hepatocytes

NTS Nucleus tractus solitarius

O&P Ova and parasites

OATP Organic anion transporting polypeptide

OCPs Oral contraceptive pills

OCT1 Organic cation transporter 1

ORS Oral rehydration solution

p-ANCA Perinuclear anti-neutrophil cytoplasmic antibody

PAF Platelet-activating factors

PAIR Puncture, aspiration, injection, and reaspiration

PAS Periodic acid-Schiff

PBC Primary biliary cholangitis (formerly primary biliary cirrhosis)

PCR Polymerase chain reaction

PE Pulmonary embolism

PEG Polyethylene glycol

PEP Phosphoenolpyruvate

PEPCK Phosphoenolpyruvate carboxykinase

PepT1 Peptide transporter 1

PFK-1 Phosphofructokinase 1

PG Prostaglandins

PGE2 Prostaglandin E2

pIgR Polymeric immunoglobulin receptor

PNH Paroxysmal nocturnal hemoglobinuria

PO By mouth (oral)

POEM Peroral endoscopic myotomy

PONV Postoperative nausea and vomiting

PPI Proton pump inhibitor

PPIs Proton pump inhibitors

PRSS1 Cationic trypsinogen gene

PSC Primary sclerosing cholangitis

PT Prothrombin time

PTH Parathyroid hormone

PUD Peptic ulcer disease

PYY Peptide YY

RAAS Renin-angiotensin-aldosterone system

RBCs Red blood cells

RdRp RNA-dependent RNA polymerase

RET Rearranged during transfection proto-oncogene

RHF Right heart failure

RLQ Right lower quadrant

RNA Ribonucleic acid

RUQ Right upper quadrant

SAAG Serum-ascites albumin gradient

SBP Spontaneous bacterial peritonitis: infection of ascitic fluid without a surgical source

SGLT1 Sodium-glucose cotransporter 1

SIBO Small intestinal bacterial overgrowth

SIgA Secretory immunoglobulin A

SIRS Systemic inflammatory response syndrome

SLE Systemic lupus erythematosus

SMA Superior mesenteric artery

SMV Superior mesenteric vein

SPINK1 Serine protease inhibitor Kazal type 1 gene

SSRIs Selective serotonin reuptake inhibitors

SSTR2 Somatostatin receptor 2

STEC Shiga toxin-producing E. coli

SULT Sulfotransferase

T2DM Type 2 diabetes mellitus

TACE Transarterial chemoembolization

TCAs Tricyclic antidepressants

TEF Tracheoesophageal fistula

TG Triglycerides

TGF Transforming growth factor

Th1 T helper type 1 cells

Th2 T helper type 2 cells

TIPS Transjugular intrahepatic portosystemic shunt

TMP-SMX Trimethoprim-sulfamethoxazole

TNF Tumor necrosis factor

TPN Total parenteral nutrition

TPO Thrombopoietin

tTG Tissue transglutaminase

UC Ulcerative colitis

UDCA Ursodeoxycholic acid

UDP-glucose Uridine diphosphate glucose

UGT UDP-glucuronosyltransferase

UGT1A1 Bilirubin UDP-glucuronosyltransferase

ULN Upper limit of normal

UTI Urinary tract infection

VacA Vacuolating cytotoxin A

VIP Vasoactive intestinal peptide

VKOR Vitamin K epoxide reductase

VLDL Very low-density lipoprotein

VTE Venous thromboembolism

WBC White blood cells

WDHA Watery diarrhea, hypokalemia, and achlorhydria

References

Embryology of the gut and its defects: separation, rotation, migration, and closure

1. Bhatia A, Shatanof RA, Bordoni B. Embryology, Gastrointestinal. 2023. In: *StatPearls*. StatPearls Publishing. NBK537172
2. Malone JC, Arbor TC, Shah AB. Embryology, Midgut. 2023. In: *StatPearls*. StatPearls Publishing. NBK553156
3. Ehrhardt JD, Gomez F. Embryology, Pancreas. 2023. In: *StatPearls*. StatPearls Publishing. NBK545243
4. Salik I, Paul M. Tracheoesophageal Fistula. 2024. In: *StatPearls*. StatPearls Publishing. NBK535376
5. Baldwin DL, Yadav D. Esophageal Atresia. 2023. In: *StatPearls*. StatPearls Publishing. NBK560848
6. Garfield K, Sergent SR. Pyloric Stenosis. 2023. In: *StatPearls*. StatPearls Publishing. NBK555931
7. Sigmon DF, Eovaldi BJ, Cohen HL. Duodenal Atresia and Stenosis. 2023. In: *StatPearls*. StatPearls Publishing. NBK470548
8. Aleem A, Desai H, Shah H. Annular Pancreas. 2025. In: *StatPearls*. StatPearls Publishing. NBK559236
9. Holt AC, Anand S, Nada H, Ahmad H. Midgut Volvulus. 2022. In: *StatPearls*. StatPearls Publishing. NBK441962
10. An J, Zabbo CP. Meckel Diverticulum. 2023. In: *StatPearls*. StatPearls Publishing. NBK499960
11. Gunasekaran V, Hussain N, Vadakekut ES. Gastroschisis. 2026. In: *StatPearls*. StatPearls Publishing. NBK557894
12. Zahouani T, Pattnaik P. Omphalocele. 2026. In: *StatPearls*. StatPearls Publishing. NBK519010
13. Siddiqui AI, Desai H, Ahmad T. Biliary Atresia. 2026. In: *StatPearls*. StatPearls Publishing. NBK537262
14. Smith CA, Avansino J. Anorectal Malformations. 2023. In: *StatPearls*. StatPearls Publishing. NBK542275
15. Anand S, Kulkarni KK, Sodhani S. Hirschsprung Disease. 2026. In: *StatPearls*. StatPearls Publishing. NBK562142

Gastric secretion: neural, endocrine, and local control of acid and its protection

1. Do C, Vasquez PC, Soleimani M. Metabolic alkalosis pathogenesis, diagnosis, and treatment: core curriculum 2022. *Am J Kidney Dis*. 2022;80(4):536–551. PMC10947768

The emetic center: inputs, the vomiting reflex, and antiemetic targets

1. MacDougall MR, Sharma S. Physiology, Chemoreceptor Trigger Zone. 2023. In: *StatPearls*. StatPearls Publishing. NBK537133
2. Mirza M, Das JM. Neuroanatomy, Area Postrema. 2023. In: *StatPearls*. StatPearls Publishing. NBK544249
3. Hauser JM, Azzam JS, Kasi A. Antiemetic Medications. 2022. In: *StatPearls*. StatPearls Publishing. NBK532303
4. Theriot J, Wermuth HR, Ashurst JV. Antiemetics, Selective 5-HT3 Antagonists. 2024. In: *StatPearls*. StatPearls Publishing. NBK513318
5. Ibrahim MA, Preuss CV. Antiemetic Neurokinin-1 Receptor Blockers. 2024. In: *StatPearls*. StatPearls Publishing. NBK470394
6. Takov V, Tadi P. Motion Sickness. 2023. In: *StatPearls*. StatPearls Publishing. NBK539706
7. Sizemore DC, Singh A, Dua A, Singh K, Grose BW. Postoperative Nausea. 2022. In: *StatPearls*. StatPearls Publishing. NBK500029
8. Cue L, Chu F, Cascella M. Cannabinoid Hyperemesis Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK549915
9. Parajuli SR, Goosenberg E. Cyclic Vomiting Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK500018
10. Vadakekut ES, Mahdy H. Hyperemesis Gravidarum. 2025. In: *StatPearls*. StatPearls Publishing. NBK532917

Liver function: structure, metabolism, bile, and detoxification

1. Kalra A, Yetiskul E, Wehrle CJ, Tuma F. Physiology, Liver. 2023. In: *StatPearls*. StatPearls Publishing. NBK535438
2. Chen I, Cassaro S. Physiology, Bile Acids. 2023. In: *StatPearls*. StatPearls Publishing. NBK549765
3. Kalakonda A, Jenkins BA, John S. Physiology, Bilirubin. 2022. In: *StatPearls*. StatPearls Publishing. NBK470290

Digestion and absorption: enzyme activation, nutrient transport, and the sites that fail

1. Patricia JJ, Dhamoon AS. Physiology, Digestion. 2022. In: *StatPearls*. StatPearls Publishing. NBK544242
2. Zuvarox T, Goosenberg E, Belletieri C. Malabsorption Syndromes. 2025. In: *StatPearls*. StatPearls Publishing. NBK553106
3. Sorathia SJ, Chippa V, Rivas JM. Small Intestinal Bacterial Overgrowth. 2023. In: *StatPearls*. StatPearls Publishing. NBK546634
4. Junaid SZS, Patel K. Abetalipoproteinemia. 2023. In: *StatPearls*. StatPearls Publishing. NBK513355

The immune system of the gut: sampling, secretory IgA, and tolerance

1. Panneerselvam D, Vaqar S. Peyer Patches. 2023. In: *StatPearls*. StatPearls Publishing. NBK557457
2. Sauls RS, Taylor BN. Histology, M Cell. 2022. In: *StatPearls*. StatPearls Publishing. NBK534232
3. Patel A, Jialal I. Biochemistry, Immunoglobulin A. 2023. In: *StatPearls*. StatPearls Publishing. NBK551516
4. Killeen RB, Joseph NI. Selective IgA Deficiency. 2023. In: *StatPearls*. StatPearls Publishing. NBK538205
5. Daley SF, Lopez CM. Food Allergies. 2026. In: *StatPearls*. StatPearls Publishing. NBK482187
6. Dao DPD, Le PH. Histology, Goblet Cells. 2023. In: *StatPearls*. StatPearls Publishing. NBK553208
7. Yu LE, Yang WC, Liang YC. Crosstalk within the intestinal epithelium: aspects of intestinal absorption, homeostasis, and immunity. *Biomedicines*. 2024;12(12):2771. PMC11673925
8. Wehkamp J, Stange EF. An update review on the Paneth cell as key to ileal Crohn's disease. *Front Immunol*. 2020;11:646. PMC7174711
9. Shang K, Qi X, Tian T, Shi H, et al. Regulation of the tuft cell-ILC2 circuit in intestinal mucosal immunity. *Front Immunol*. 2025;16:1568062. PMC12066627

The enteric nervous system: plexuses, pacemakers, and motility

1. Tobias A, Sadiq NM. Physiology, Gastrointestinal Nervous Control. 2022. In: *StatPearls*. StatPearls Publishing. NBK545268
2. Patel KS, Thavamani A. Physiology, Peristalsis. 2023. In: *StatPearls*. StatPearls Publishing. NBK556137
3. Malone JC, Thavamani A. Physiology, Gastrocolic Reflex. 2023. In: *StatPearls*. StatPearls Publishing. NBK549888
4. Mawer S, Alhawaj AF. Physiology, Defecation. 2023. In: *StatPearls*. StatPearls Publishing. NBK539732
5. Parikh A, Thevenin C. Physiology, Gastrointestinal Hormonal Control. 2023. In: *StatPearls*. StatPearls Publishing. NBK537284
6. Anand S, Kulkarni KK, Sodhani S. Hirschsprung Disease. 2026. In: *StatPearls*. StatPearls Publishing. NBK562142
7. Puri R, Tian C. Achalasia. 2026. In: *StatPearls*. StatPearls Publishing. NBK519515

8. Gandhi MH, Goosenberg E. Gastroparesis. 2026. In: *StatPearls*. StatPearls Publishing. NBK551528
9. Menon G, Collier SA. Intestinal Pseudo-Obstruction. 2026. In: *StatPearls*. StatPearls Publishing. NBK560669

Esophagitis: routes of mucosal injury and relationships to other foregut disease

1. Antunes C, Sharma A. Esophagitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK442012
2. Sharma M, Sharma A, Goosenberg E. Pill-Induced Esophagitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK549818
3. Bielecki JE, Chen RJ, Gupta V. Caustic Ingestions. 2024. In: *StatPearls*. StatPearls Publishing. NBK557442
4. Roussel JM, Pandit S. Eosinophilic Esophagitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK459297
5. Robertson KD, Nagra N, Mehta D. Esophageal Candidiasis. 2023. In: *StatPearls*. StatPearls Publishing. NBK537268

Esophageal motility disorders: neuromuscular failure of the LES and esophageal body

1. Goyal M, Nagalli S. Esophageal Motility Disorders. 2022. In: *StatPearls*. StatPearls Publishing. NBK562155
2. Puri R, Tian C. Achalasia. 2026. In: *StatPearls*. StatPearls Publishing. NBK519515
3. Goel S, Nookala V. Diffuse Esophageal Spasm. 2023. In: *StatPearls*. StatPearls Publishing. NBK541106
4. Adigun R, Goyal A, Hariz A. Systemic Sclerosis (Scleroderma). 2024. In: *StatPearls*. StatPearls Publishing. NBK430875
5. Nesheiwat Z, Antunes C. Zenker Diverticulum. 2023. In: *StatPearls*. StatPearls Publishing. NBK499996
6. Kassama Z, Goosenberg E. Plummer-Vinson Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK538306
7. Ghazaleh S, Patel K. Esophageal Webs and Rings. 2023. In: *StatPearls*. StatPearls Publishing. NBK539771

Gastroesophageal reflux disease: failure of the antireflux barrier and its consequences

1. Goosenberg E, Vadakekut ES. Gastroesophageal Reflux Disease (GERD). 2025. In: *StatPearls*. StatPearls Publishing. NBK554462
2. Puri R, Sharma S. Hiatal Hernia. 2026. In: *StatPearls*. StatPearls Publishing. NBK562200
3. Khieu M, Goosenberg E. Barrett Esophagus. 2025. In: *StatPearls*. StatPearls Publishing. NBK430979

Gastritis: superficial mucosal injury and the boundary with peptic ulcer disease

1. Azer SA, Awosika AO, Akhondi H. Gastritis. 2024. In: *StatPearls*. StatPearls Publishing. NBK544250
2. Raza M, Bhatt H. Atrophic Gastritis. 2023. In: *StatPearls*. StatPearls Publishing. NBK563275
3. Megha R, Farooq U, Lopez PP. Stress-Induced Gastritis. 2023. In: *StatPearls*. StatPearls Publishing. NBK499926
4. Vaqar S, Shackelford KB. Pernicious Anemia. 2023. In: *StatPearls*. StatPearls Publishing. NBK540989
5. Menon G, El-Nakeep S, Babiker HM. Gastric Cancer. 2024. In: *StatPearls*. StatPearls Publishing. NBK459142
6. Farooq U, Chakinala RC. MALToma. 2024. In: *StatPearls*. StatPearls Publishing. NBK553122
7. Gore M, Bansal K, Goosenberg E. Menetrier Disease. 2026. In: *StatPearls*. StatPearls Publishing. NBK563154

Peptic ulcer disease: *H. pylori*, NSAIDs, and the balance of acid and mucosal defense

1. Malik TF, Gnanapandithan K, Singh K. Peptic Ulcer Disease. 2023. In: *StatPearls*. StatPearls Publishing. NBK534792
2. Helbing A, Menon G, Karanchi H. Gastrinoma. 2025. In: *StatPearls*. StatPearls Publishing. NBK441842
3. Garg DR, Nassereddin A, Arbor TC. Anatomy, Abdomen and Pelvis: Celiac Trunk. 2025. In: *StatPearls*. StatPearls Publishing. NBK459241
4. Fish EM, Shumway KR, Burns B. Physiology, Small Bowel. 2024. In: *StatPearls*. StatPearls Publishing. NBK532263

Vascular liver injury: inflow failure, outflow congestion, and zone 3 hypoxia

1. Hitawala AA, Goosenberg E. Budd-Chiari Syndrome. 2025. In: *StatPearls*. StatPearls Publishing. NBK558941
2. Rodriguez Ziccardi M, Pendela VS, Singhal M. Cardiac Cirrhosis. 2023. In: *StatPearls*. StatPearls Publishing. NBK431053
3. Samant H, Asafo-Agyei KO, Kimyaghalam A, Garfield K. Portal Vein Thrombosis. 2024. In: *StatPearls*. StatPearls Publishing. NBK534157
4. Taylor RM, Tujios S, Jinjuvadia K, Davern T, et al. Short and long-term outcomes in patients with acute liver failure due to ischemic hepatitis. *Dig Dis Sci*. 2012;57(3):777–785. PMC5154383
5. Yoon JH, Choi CW, Won JH. Hepatic sinusoidal obstruction syndrome/veno-occlusive disease after hematopoietic cell transplantation: historical and current considerations in Korea. *Korean J Intern Med*. 2021;36(6):1261–1280. PMC8588980

Infectious liver injury: the hepatitis viruses and their clinical divergence

1. Girish V, Grant LM, John S. Hepatitis A. 2024. In: *StatPearls*. StatPearls Publishing. NBK459290
2. Tripathi N, Mousa OY. Hepatitis B. 2023. In: *StatPearls*. StatPearls Publishing. NBK555945
3. Basit H, Koirala J. Hepatitis C. 2023. In: *StatPearls*. StatPearls Publishing. NBK430897
4. Sachdeva K, Girish V, John S. Hepatitis D. 2025. In: *StatPearls*. StatPearls Publishing. NBK470436
5. Girish V, Grant LM, Sharma B, Koirala J. Hepatitis E. 2025. In: *StatPearls*. StatPearls Publishing. NBK532278
6. Akhondi H, Sabih DE. Liver Abscess. 2023. In: *StatPearls*. StatPearls Publishing. NBK538230
7. Jackson-Akers JY, Goosenberg E, Oliver TI. Amebic Liver Abscess. 2025. In: *StatPearls*. StatPearls Publishing. NBK430832
8. Bento D, John S. Echinococcus Granulosus. 2026. In: *StatPearls*. StatPearls Publishing. NBK539751
9. Peine W, Jenks JD, Horrall S. Schistosomiasis. 2026. In: *StatPearls*. StatPearls Publishing. NBK554434

Traumatic liver injury: hemodynamic stability as the decision point

1. Smoker S, Taghavi S, Patel PJ. Liver Trauma. 2026. In: *StatPearls*. StatPearls Publishing. NBK513236

Autoimmune liver disease: bile duct injury in PBC and PSC, and autoimmune hepatitis

1. Pandit S, Samant H. Primary Biliary Cholangitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK459209
2. Malik S, Dbouk N, Grant LM, Samant H. Primary Sclerosing Cholangitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK537181
3. Linzay CD, Sharma B, Pandit S. Autoimmune Hepatitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK459186

Metabolic liver disease: iron, copper, misfolded protein, and fat

1. Killeen RB, Vadakekut ES. Hemochromatosis. 2026. In: *StatPearls*. StatPearls Publishing. NBK430862
2. Immergluck J, Grant LM, Anilkumar AC. Wilson Disease. 2025. In: *StatPearls*. StatPearls Publishing. NBK441990
3. Meseeha M, Sankari A, Attia M. Alpha-1 Antitrypsin Deficiency. 2024. In: *StatPearls*. StatPearls Publishing. NBK442030

4. Girish V, John S. Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD). 2025. In: *StatPearls*. StatPearls Publishing. NBK541033

Iatrogenic and toxic liver injury: postoperative, paracetamol, and alcohol

1. Schaffer DH, Murray BP, Khazaeni B. Acetaminophen Toxicity. 2026. In: *StatPearls*. StatPearls Publishing. NBK441917
2. Patel R, Mueller M. Alcohol-Associated Liver Disease. 2023. In: *StatPearls*. StatPearls Publishing. NBK546632
3. John S, Sachdeva K, Grant LM. Alcohol-Associated Hepatitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK470217
4. Francis P, Navarro VJ. Drug-Induced Liver Injury. 2024. In: *StatPearls*. StatPearls Publishing. NBK557535
5. Chapman J, Arnold JK. Reye Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK526101

Liver tumors: distinguishing benign lesions, primary cancers, and metastases

1. Asafo-Agyei KO, Samant H. Hepatocellular Carcinoma. 2023. In: *StatPearls*. StatPearls Publishing. NBK559177
2. Smith M, Tomboc PJ, Markovich B. Fibrolamellar Hepatocellular Carcinoma. 2022. In: *StatPearls*. StatPearls Publishing. NBK553113
3. Evans J, Willyard CE, Sabih DE. Cavernous Hepatic Hemangioma. 2022. In: *StatPearls*. StatPearls Publishing. NBK470283
4. Hamad S, Willyard CE, Mukherjee S. Focal Nodular Hyperplasia. 2022. In: *StatPearls*. StatPearls Publishing. NBK532244
5. Shreenath AP, Grant LM, Kahloon A. Hepatocellular Adenoma. 2024. In: *StatPearls*. StatPearls Publishing. NBK513264
6. Griscom JT, Wolf PS. Liver Metastasis. 2023. In: *StatPearls*. StatPearls Publishing. NBK553118

Acute fatty liver of pregnancy: fetal LCHAD deficiency and maternal liver failure

1. Hadi Y, Kupec J. Acute Fatty Liver of Pregnancy. 2023. In: *StatPearls*. StatPearls Publishing. NBK545315

Acute liver failure and cirrhosis: sudden and chronic loss of hepatic function

1. Girish V, Royer A, John S. Acute Liver Failure. 2025. In: *StatPearls*. StatPearls Publishing. NBK482374
2. Sharma B, John S. Hepatic Cirrhosis. 2022. In: *StatPearls*. StatPearls Publishing. NBK482419
3. Mandiga P, Kommu S, Bollu PC. Hepatic Encephalopathy. 2025. In: *StatPearls*. StatPearls Publishing. NBK430869
4. Girish V, Ranasinghe IR, Rout P. Hepatorenal Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK430856
5. Goosenberg E, Kudaravalli P, Samant H. Ascites. 2025. In: *StatPearls*. StatPearls Publishing. NBK470482
6. Meseeha M, Tian C, Attia M. Esophageal Varices. 2026. In: *StatPearls*. StatPearls Publishing. NBK448078
7. Bansal K, Gore M, Grant LM, Mittal S. Hepatopulmonary Syndrome. 2025. In: *StatPearls*. StatPearls Publishing. NBK562169

Cholelithiasis and biliary colic: gallstone formation and its first symptoms

1. Jones MW, Weir CB, Marietta M. Gallstones (Cholelithiasis). 2026. In: *StatPearls*. StatPearls Publishing. NBK459370
2. Jones MW, Kashyap S, Ferguson T. Gallbladder Imaging. 2022. In: *StatPearls*. StatPearls Publishing. NBK470366
3. Jones MW, Weir CB, Del Toro D, Patel PJ. Porcelain Gallbladder. 2026. In: *StatPearls*. StatPearls Publishing. NBK518979

Cholecystitis: persistent cystic duct obstruction and gallbladder inflammation

1. Jones MW, Santos G, Patel PJ. Acute Cholecystitis. 2025. In: *StatPearls*. StatPearls Publishing. NBK459171
2. Jones MW, Santos G, Patel PJ. Acalculous Cholecystitis. 2025. In: *StatPearls*. StatPearls Publishing. NBK459182
3. Menon G, Pfeifer C. Emphysematous Cholecystitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK448099
4. Jones MW, Sall AR, Panneerselvam D, Patel PJ. Chronic Cholecystitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK470236
5. Jones MW, Weir CB, Del Toro D, Patel PJ. Porcelain Gallbladder. 2026. In: *StatPearls*. StatPearls Publishing. NBK518979
6. Jones MW, Ferguson T. Mirizzi Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK482491
7. Turner AR, Sharma B, Mukherjee S. Gallstone Ileus. 2022. In: *StatPearls*. StatPearls Publishing. NBK430834

Choledocholithiasis and gallstone pancreatitis: obstruction of the common channel

1. LaGattuta N, Puri R, St Hill CR. Choledocholithiasis. 2026. In: *StatPearls*. StatPearls Publishing. NBK441961
2. ASGE Standards of Practice Committee, Buxbaum JL, Abbas Fehmi SM, Sultan S, et al. ASGE guideline on the role of endoscopy in the evaluation and management of choledocholithiasis. *Gastrointest Endosc*. 2019;89(6):1075–1105.e15. PMC8594622

Ascending cholangitis: infected obstruction and the need for drainage

1. Virgile J, Marathi R. Cholangitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK558946

Biliary fistulas: erosion from the biliary tree into neighboring organs

1. Turner AR, Kudaravalli P, Al-Musawi JH, Ahmad H. Bouveret Syndrome (Bilioduodenal Fistula). 2023. In: *StatPearls*. StatPearls Publishing. NBK430738
2. Turner AR, Sharma B, Mukherjee S. Gallstone Ileus. 2022. In: *StatPearls*. StatPearls Publishing. NBK430834

Biliary tract and periampullary malignancies: painless obstruction of bile flow

1. Paluri RK, Kasi A. Periampullary Tumors. 2022. In: *StatPearls*. StatPearls Publishing. NBK555958
2. Menon G, Garikipati SC, Roy P. Cholangiocarcinoma. 2024. In: *StatPearls*. StatPearls Publishing. NBK560708
3. Menon G, Babiker HM. Gallbladder Carcinoma. 2024. In: *StatPearls*. StatPearls Publishing. NBK442002

Acute pancreatitis: autodigestion and its local and systemic consequences

1. Wang CF, Tariq A, Chandra S. Acute Pancreatitis. 2025. In: *StatPearls*. StatPearls Publishing. NBK482468
2. Goosenberg E, Lappin SL. Chronic Pancreatitis. 2025. In: *StatPearls*. StatPearls Publishing. NBK482325
3. Al Zureikat Q, Goosenberg E, Sood T. Pancreatic Pseudocyst. 2026. In: *StatPearls*. StatPearls Publishing. NBK557594

Malabsorption: separating mucosal injury from failed digestion

1. Zuvarox T, Goosenberg E, Belletieri C. Malabsorption Syndromes. 2025. In: *StatPearls*. StatPearls Publishing. NBK553106
2. Daley SF, Gillespie E. Celiac Disease. 2026. In: *StatPearls*. StatPearls Publishing. NBK441900
3. Mirza HA, Gharbi A, Bhutta BS. Dermatitis Herpetiformis. 2023. In: *StatPearls*. StatPearls Publishing. NBK493163
4. Antunes C, Singhal M. Whipple Disease. 2023. In: *StatPearls*. StatPearls Publishing. NBK441937
5. Goosenberg E, Aloysius MM, Afzal M. Tropical Sprue. 2025. In: *StatPearls*. StatPearls Publishing. NBK567742

6. Sorathia SJ, Chippa V, Rivas JM. Small Intestinal Bacterial Overgrowth. 2023. In: *StatPearls*. StatPearls Publishing. NBK546634
7. Junaid SZS, Patel K. Abetalipoproteinemia. 2023. In: *StatPearls*. StatPearls Publishing. NBK513355
8. Tian C, Ghodeif AO, Arshad S, Gillespie E. Exocrine Pancreatic Insufficiency. 2025. In: *StatPearls*. StatPearls Publishing. NBK555926

Diarrhea: a mechanistic framework of secretory, osmotic, and inflammatory disease

1. Descoteaux-Friday GJ, Shrimanker I. Chronic Diarrhea. 2023. In: *StatPearls*. StatPearls Publishing. NBK544337
2. Mada PK, Alam MU. Clostridioides difficile infection. 2024. In: *StatPearls*. StatPearls Publishing. NBK431054
3. Hempel KA, Sharma AV. Collagenous and Lymphocytic Colitis. 2022. In: *StatPearls*. StatPearls Publishing. NBK541100
4. Menon G, Pandit S, Bhusal K. Carcinoid Syndrome. 2025. In: *StatPearls*. StatPearls Publishing. NBK448096
5. Helbing A, Menon G, Sandhu S, Jialal I. VIPoma. 2025. In: *StatPearls*. StatPearls Publishing. NBK507698
6. Ojeda Rodriguez JA, Kahwaji CI. Cholera (Vibrio cholerae Infection). 2024. In: *StatPearls*. StatPearls Publishing. NBK526099
7. Mueller M, Rausch-Phung EA, Tainter CR. Escherichia coli Infection. 2025. In: *StatPearls*. StatPearls Publishing. NBK564298
8. Stuempfig ND, Tobin EH, Seroy J. Viral Gastroenteritis. 2025. In: *StatPearls*. StatPearls Publishing. NBK518995

Intestinal ischemia: arterial, venous, and low-flow injury to the bowel

1. Zafer S, Lopez RA, Kimyaghalam A. Acute Mesenteric Ischemia. 2026. In: *StatPearls*. StatPearls Publishing. NBK431068
2. Kimyaghalam A, Costanza M. Chronic Mesenteric Ischemia. 2026. In: *StatPearls*. StatPearls Publishing. NBK430748
3. Sulger E, Dhaliwal HS, Oropallo A, Gonzalez L. Mesenteric Venous Thrombosis. 2024. In: *StatPearls*. StatPearls Publishing. NBK459184
4. Amini A, Nagalli S. Bowel Ischemia. 2023. In: *StatPearls*. StatPearls Publishing. NBK554527

Diverticular diseases: from pressure-driven outpouching to inflammation

1. Bach K, Menon G, Goosenberg E. Diverticular Disease of the Colon. 2026. In: *StatPearls*. StatPearls Publishing. NBK541110
2. Nallapeta NS, Farooq U, Patel K. Diverticulosis. 2023. In: *StatPearls*. StatPearls Publishing. NBK430771
3. An J, Zabbo CP. Meckel Diverticulum. 2023. In: *StatPearls*. StatPearls Publishing. NBK499960

Inflammatory bowel disease: contrasting Crohn disease and ulcerative colitis

1. Tian C, Ranasinghe IR, Hsu R. Crohn Disease. 2025. In: *StatPearls*. StatPearls Publishing. NBK436021
2. Lynch WD, Hsu R. Ulcerative Colitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK459282
3. Goosenberg E, Pico J. Toxic Megacolon. 2026. In: *StatPearls*. StatPearls Publishing. NBK547679
4. Boyle B, Collins MH, Wang Z, Mack D, et al. Histologic correlates of clinical and endoscopic severity in children newly diagnosed with ulcerative colitis. *Am J Surg Pathol*. 2017;41(11):1491–1498. PMC5636675
5. Massinha P, Portela F, Campos S, Duque G, et al. Ulcerative colitis: are we neglecting its progressive character. *GE Port J Gastroenterol*. 2018;25(2):74–79. PMC5892362

Irritable bowel syndrome: a disorder of gut-brain interaction without structural damage

1. Nathani RR, Sodhani S, Goosenberg E. Irritable Bowel Syndrome. 2025. In: *StatPearls*. StatPearls Publishing. NBK534810

Appendicitis: luminal obstruction, migrating pain, and perforation

1. Lotfollahzadeh S, Lopez RA, Deppen JG. Appendicitis. 2024. In: *StatPearls*. StatPearls Publishing. NBK493193

Localizing abdominal pain: from where it hurts to the test that confirms it

1. Kopitnik NL, Kashyap S, Dominique E. Acute Abdomen. 2025. In: *StatPearls*. StatPearls Publishing. NBK459328
2. Goldin J, Sodhani S. Abdominal Examination. 2025. In: *StatPearls*. StatPearls Publishing. NBK459220
3. Schmieder SJ, Krishnamurthy K. Pyoderma Gangrenosum. 2023. In: *StatPearls*. StatPearls Publishing. NBK482223
4. Hafsi W, Badri T. Erythema Nodosum. 2022. In: *StatPearls*. StatPearls Publishing. NBK470369
5. Hudgi A, Parthasarathy S, Mirza A, Woodcock A, Goosenberg E. Esophagogastroduodenoscopy (EGD). 2026. In: *StatPearls*. StatPearls Publishing. NBK532268
6. Dsouza R, Menon G, Pfeifer C. Colonoscopy. 2025. In: *StatPearls*. StatPearls Publishing. NBK559274

Reading liver function tests: patterns of injury, cholestasis, and failing function

1. Jogu P, Zubair M, Minter DA. Liver Function Tests. 2026. In: *StatPearls*. StatPearls Publishing. NBK482489
2. Singh A, Koritala T, Jialal I. Unconjugated Hyperbilirubinemia. 2023. In: *StatPearls*. StatPearls Publishing. NBK549796
3. Tripathi N, Jialal I. Conjugated Hyperbilirubinemia. 2023. In: *StatPearls*. StatPearls Publishing. NBK562172
4. Grant LM, Faust TW, Thoguluva Chandrasekar V, John S. Gilbert Syndrome. 2024. In: *StatPearls*. StatPearls Publishing. NBK470200
5. Gounden V, Easton-Carr R, Jialal I. Hypoalbuminemia. 2026. In: *StatPearls*. StatPearls Publishing. NBK526080
6. Zubair M, Yang R, Moosavi L. Prothrombin Time. 2026. In: *StatPearls*. StatPearls Publishing. NBK544269